

# EXHALE: User Manual

**Scope.** This manual describes how to set up, run, and analyze models with EXHALE, the in-house extension of the ATES (ATmospheric EScape) code to trace metals, metal-line cooling, a runtime opacity dispatcher, the Roche tidal potential, and excited-hydrogen / Ly $\alpha$  physics. It documents every input parameter (with a brief physical meaning), every output file (column by column, with units), the analysis tools, and a set of worked examples. The detailed physics derivations live in the companion notes (Update\_EXHALE\_early\_phase.tex, transmission\_spectrum.tex, Update\_EXHALE.tex, cooling\_formulas.tex (the complete reference for every analytic cooling coefficient), EXHALE\_BC\_and\_IC.tex (the boundary- and initial-condition reference), and wind\_ae\_solver.tex, the Wind-AE steady-wind solver used by IC mode: windae); this document is the practical user guide.

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## 1. Introduction

### 1.1 What the code does

EXHALE solves the time-dependent, spherically-symmetric (1-D) radiation–hydrodynamics of an irradiated planetary upper atmosphere and the hydrodynamic wind it drives (“atmospheric escape”). It integrates the Euler equations for mass, momentum, and energy of a single fluid,

$$\partial_t \rho + \frac{1}{r^2} \partial_r (r^2 \rho v) = 0, \quad \partial_t (\rho v) + \frac{1}{r^2} \partial_r (r^2 \rho v^2) + \partial_r p = -\rho \partial_r \Phi,$$

$$\partial_t E + \frac{1}{r^2} \partial_r (r^2 v (E + p)) = -\rho v \partial_r \Phi + \mathcal{H} - \mathcal{C},$$

to a steady state, where  $\Phi$  is the gravitational (optionally Roche) potential (§1.3),  $\mathcal{H}$  is the XUV photoionization heating and  $\mathcal{C}$  the radiative cooling. Coupled to the hydrodynamics, at every step it solves the *ionization equilibrium* of H, He, and the trace metals (C, N, O, Mg, Si, Ca, Na, K, S, Fe) with a Newton (MINPACK) solver, including photoionization, collisional ionization, radiative + dielectronic recombination, and charge exchange. The photoelectron energy budget carries secondary ionization (a fast photoelectron drives further H I / He I / H<sub>2</sub> ionizations instead of thermalizing fully) on the Shull & van Steenberg (1985) ionization budget with the Dalgarno, Yan & Liu (1999) heating fraction and molecular channels, the He II  $\rightarrow$  He I recombination radiation is coupled to H I ionization on the spot (Draine 2011 y/z), and the He  $\leftrightarrow$  H charge-exchange pair (Koskinen 2013) acts in every helium-bearing ionization system. When the metastable He 2<sup>3</sup>S triplet is included the code adds its explicit collisional-excitation cooling (the 10830 Å channel) and the He 2<sup>3</sup>S + H / H<sub>2</sub> Penning ionization that removes the metastable. The converged run yields the radial profiles of density, velocity, temperature, and the ionization state, plus the steady-state *mass-loss rate*  $\dot{M}$ . A post-processing pass adds advection to the ionization balance, and the Python tool EXHALE\_transit.py turns a converged model into transit *transmission spectra* (He 10830, Ly $\alpha$ , H $\alpha$ /H $\beta$ , and the metal resonance lines Mg II, Ca II, Na I).

## 1.2 The lower boundary, radii, and normalization

The computational domain starts at a **lower boundary** placed at the 1  $\mu$ bar pressure level, which defines  $r = 1$  and the base radius  $R_p$  (the input Planet radius). The base number density  $n_0$  and temperature  $T_0$  (the equilibrium temperature) are set there. Internally the code is non-dimensional: lengths in  $R_p$ , velocities in  $v_0 = \sqrt{k_B T_0 / \mu}$ , densities in  $n_0$ , etc. *All output files are written back in cgs* (see §4. for the exact normalization of each column). The key dimensionless number is the *Jeans / escape parameter*  $b_0 = GM_p \mu / (k_B T_0 R_p)$ , the depth of the gravitational well in units of the thermal energy; small  $b_0$  means an easily-escaping (or tidally unbound) atmosphere.

## 1.3 The gravitational potential

The potential  $\Phi$  in the momentum and energy equations is selected by Domain mode (Table 3). The code writes it in the dimensionless units introduced above (lengths in  $R_p$ , potential in  $v_0^2 = k_B T_0 / \mu$ ), so it is fixed by just three numbers: the Jeans parameter  $b_0 = GM_p \mu / (k_B T_0 R_p)$ , the stellar-to-planet mass ratio  $q = M_\star / M_p$ , and the orbital separation  $\tilde{a} = a / R_p$ .

**Spherical mode (Domain mode: Spherical).** Only the planet's own gravity is kept, so the domain can extend far past the Hill radius:

$$\phi(r) = -\frac{b_0}{r}.$$

**Roche mode (default).** The full co-rotating Roche potential, evaluated along the star–planet (substellar) line  $x = r$ , adds the stellar tidal attraction and the centrifugal term of the synchronously-rotating frame:

$$\phi(r) = -\frac{b_0}{r} - \frac{b_0 q}{\tilde{a} - r} - \frac{b_0(1+q)}{2\tilde{a}^3} \left( \frac{q\tilde{a}}{1+q} - r \right)^2,$$

where the second term is the star's gravity at distance  $\tilde{a} - r$  and the third is the centrifugal potential about the center of mass at  $x_{\text{cm}} = q\tilde{a} / (1+q)$  with orbital frequency  $\Omega^2 = G(M_p + M_\star) / a^3$  (synchronous rotation). The effective gravity used

by the hydrodynamics is  $g(r) = -d\phi/dr$ , with

$$\frac{d\phi}{dr} = \frac{b_0}{r^2} - \frac{b_0 q}{(\tilde{a}-r)^2} + \frac{b_0(1+q)}{\tilde{a}^3} \left( \frac{q\tilde{a}}{1+q} - r \right).$$

The inner Lagrange point  $L_1$  is the root of  $d\phi/dr = 0$ , where planetary gravity, the stellar tide, and the centrifugal force cancel; in Roche mode the outer boundary is placed at this  $L_1$  radius by default, beyond which gas is no longer bound to the planet.

**Roche mode with an extended outer boundary (Outer radius in Roche mode).** By default Roche mode truncates at  $L_1$ . Supplying an explicit Outer radius [R\_p]:  $> 1$  *without* Domain mode: Spherical now keeps the full tidal Roche potential above *and* pushes the outer boundary out to the requested radius (e.g.  $10R_p$ ), reproducing the extended-domain treatment of Yan et al. (2022) / Huang et al. (2023), whose escape models retain the tidal force but integrate well past  $L_1$ . This differs from Domain mode: Spherical, which also extends the domain but *drops* the tidal and centrifugal terms (pure  $-b_0/r$ ). Two caveats apply. First, beyond  $L_1$  the Roche potential is past its maximum, so the net force is outward and the 1-D radial flow there is a spherical approximation to what is physically an  $L_1$  funnel: the same approximation Yan/Huang make. Second, the cold hydrostatic IC is invalid past  $L_1$ ; use IC mode: auto (or transonic) so the solver seeds a transonic wind through the interior sonic point. The tidal singularity of the  $-b_0 q/(\tilde{a}-r)$  term sits at the star ( $r = \tilde{a} \gg 10$ ), so a  $10R_p$  domain is numerically safe.

**Full 3-D form (transmission post-processing).** The 3-D Roche-equipotential reconstruction in EXHALE\_transit.py / roche\_recon.py (§5.2) needs the potential off-axis as well. With the substellar direction  $\hat{x}$  pointing toward the star and  $(y, z)$  the sky plane,

$$\phi(x, y, z) = -\frac{b_0}{\sqrt{x^2 + y^2 + z^2}} - \frac{b_0 q}{\sqrt{(\tilde{a}-x)^2 + y^2 + z^2}} - \frac{b_0(1+q)}{2\tilde{a}^3} \left[ (x_{\text{cm}} - x)^2 + y^2 \right],$$

which reduces to the substellar  $\phi(r)$  above on the  $+x$  axis. The centrifugal term involves only the rotation-plane coordinates  $(x, y)$ , not  $z$ , so surfaces of constant  $\phi$  are triaxial (most extended toward  $L_1$ , least along the spin axis  $z$ ).

## 1.4 Workflow

A model run consists of:

1. Prepare input.inp (planet/star parameters, numerics, physics switches), by hand or via the GUI; optionally metals.inp (trace metal abundances) and opacity.inp (opacity model).
2. Build and run ./EXHALE.x; it reads the input files from the current directory and writes profiles to output/.
3. Inspect with EXHALE\_plots.py (live profiles) and the example notebook; compute transmission spectra with EXHALE\_transit.py.

## 2. Installation and running

### 2.1 Requirements

A Fortran compiler (gfortran by default; Intel ifort/ix supported), make, and (for the GUI and analysis) Python 3 with tkinter, numpy, scipy, matplotlib, and astropy (the last is used by EXHALE\_transit.py for astropy.convolution).

### 2.2 Build

EXHALE uses an incremental Makefile; module dependencies are auto-generated, so only edited sources and their dependents recompile.

```
make                # build ./EXHALE.x with gfortran
make FC=ifort       # Intel ifort (or FC=ifx for the LLVM compiler)
make wind_ae_ic     # standalone Wind-AE IC generator -> ./wind_ae_ic.x
make check          # byte-identical regression against the stored goldens
make clean          # remove .o/.mod (keeps EXHALE.x)
make distclean      # wipe build/ and EXHALE.x
```

make check rebuilds, re-runs each regression case single-threaded (OMP\_NUM\_THREADS=1, so the result is deterministic) and *bitwise*-compares output/Hydro\_ioniz.txt and output/Ion\_species.txt against the stored reference outputs. The standing rule it encodes: a refactor with no intended physics change must be byte-identical, and a new physics option must default to off. The harness and its reference outputs live in backup/regression/; the target reports that it found no harness, without failing, in a checkout that does not carry it. make wind\_ae\_ic builds the out-of-process IC generator described in §2.10.

### 2.3 Running

The driver script run\_EXHALE.sh (1) launches a Python/Tk GUI that writes input.inp, (2) builds, and (3) runs ./EXHALE.x:

```
chmod +x run_EXHALE.sh
./run_EXHALE.sh          # gfortran (GUI -> build -> run)
./run_EXHALE.sh --ifort  # Intel ifort (--ifx for ifx)
```

The GUI writes the legacy core block only: it offers Load from file, Power-law and Monochromatic as spectrum types, and every optional key of Table 3 (Spectrum type: Planck and the two stellar quantities it needs among them) is added to the written file by hand or inherited by copying an examples/ configuration.

You may equally skip the GUI: place a hand-written input.inp (and optional metals.inp/opacity.inp) in a run directory, ensure an output/ subdirectory exists, and run ./EXHALE.x there. Each modeled planet typically lives in its own subdirectory holding its input.inp, its EXHALE.x, and its output\*/ folders. Progress (iteration counter, the

mass-flux spread  $du$ , and the time-derivative norm  $dtu$ ) is printed to the terminal; the run stops when  $du$  or  $dtu$  falls below its convergence threshold.

## 2.4 Convergence control

The time loop stops when *any* of the following criteria is met (defaults in `src/modules/init/parameters.f90`; most are runtime-settable, Table 3):

| Parameter                           | Default            | Stop condition                                   |
|-------------------------------------|--------------------|--------------------------------------------------|
| <code>du_th</code>                  | $1 \times 10^{-3}$ | mass-flux spread $du < du\_th$                   |
| <code>dtu_th</code>                 | $1 \times 10^{-8}$ | time-derivative $\infty$ -norm $dtu < dtu\_th$   |
| <code>stall_tol, N_stall</code>     | $10^{-6}$ , 2000   | $du$ on a plateau for <code>N_stall</code> steps |
| <code>count_max</code> (Max steps:) | $10^6$             | hard iteration cap                               |

$du$  is the radial spread of the mass flux  $F = \rho v r^2$  over the convergence window  $r \geq \text{Escape radius}$  (cells  $j_{\min} \dots N$ ),

$$du = \frac{\max_j F_j - \min_j F_j}{|\bar{F}|}, \quad \bar{F} = \frac{1}{N - j_{\min} + 1} \sum_{j=j_{\min}}^N F_j,$$

with  $F$  taken *signed*: no absolute value is applied before the extrema, so a window whose flux reverses (inflow at one radius and outflow at another) is measured as the far-from-steady state it is rather than read as flat, and the normalization is the window's own mean level, floored at  $10^{-30}$  so that a vanishing mean gives a large number rather than an infinity. A steady spherical wind carries one mass flux at every radius, which is what this measures. It is evaluated once per step by a single function (`mass_flux_spread` in `EXHALE_main.f90`) that the  $du$  stop, the arming of the  $du$  triggers, the PLM  $\rightarrow$  WENO3 stage flip, the secondary-ionization flip and the JFNK hand-off all read, so every one of them speaks about the same number; the value at the final step is written to `run.log`. It is *not* a velocity change or a step-to-step variation: that is  $dtu$ .

The stall and `count_max` backstops guarantee termination even when the strict  $du/dtu$  thresholds are physically unreachable for a given setup (e.g. a transonic wind whose  $du$  floors at  $\sim 1\%$ ). Each iteration prints `count`, `du`, `dtu` (and the windowed flux-level change when `Level_tol` is set), and the run reports which criterion stopped it. The optional `Force_start` forces the first 1000 iterations regardless (useful when starting near a fixed point).

Two of these triggers are *armed* before they can fire. A generated initial condition (cold hydrostatic, transonic, warm-seed, Wind-AE) is an analytic profile the marching loop has not touched, so its  $du$  measures the smoothness of the formula that built it, not relaxation of the wind, and it can start orders of magnitude below every threshold. The  $du < du\_th$  stop and the Newton hand-off (together with the secondary-ionization flip that shares its  $du$  comparison) therefore require  $du$  to have been seen at or above their own threshold first, so only a *descending* crossing stops the run; both arm immediately when the run started from `Load IC? True`, whose state was relaxed by the run that wrote it. Arming is reported in the log. The PLM  $\rightarrow$  WENO3 switch is deliberately not guarded. The rationale and history of these settings are in `Update_EXHALE_solver`.

**Caveat:  $du$  can stop a run prematurely.**  $du$  measures only the *spatial spread* of  $\rho v r^2$  in the wind window; it is blind to a uniform drift of the flux level (the mean it normalizes by drifts with it), and on strongly irradiated planets it *oscil-*

lates, dipping below  $10^{-3}$  transiently long before the wind settles. A WASP-121b reference stopped this way was low by  $\sim 0.08$  dex in  $\dot{M}$  with the heating/cooling grossly out of balance (energy residual  $\sim 30$ ). For production answers use the residual-based stop (`Resid tol:`) or the Newton finish (`Solver: Newton`, §2.7); details in `steady_solver_memo`.

**Staged secondary ionization.** With `Secondary_ionization: True` (the default) the photoelectron coupling is *not* applied from step 0. From a cold initial condition it would deepen the base cooling valley and amplify the base acoustic mode into a NaN runaway on high-gravity cases. Instead the wind first relaxes without the coupling; the moment that first relaxation converges (or stalls) the coupling switches on (the terminal prints `secondary ionization activated at step N`), the convergence logicals and flux-level history are reset, and stops are held for `N_stall` steps while the base adjustment wave forms, after which the run re-converges with the full physics. Expect roughly double the step count on high-gravity cases. `Secondary_ionization: False` disables the coupling entirely; `Immediate` applies it from step 0 (for A/B tests only: it reproduces the cold-start crash on high-gravity cases).

**A consequence worth knowing before reading a short run.** Staged is not a synonym for on. A run that never reaches its first convergence (one stopped at a fixed step count, or one left at a du plateau) never switches the coupling on, and its answer carries no secondary ionization at all. The startup report (`EXHALE_setup.out`) says which of the three states the run is in, and the terminal prints the activation line only when it happens.

## 2.5 Base grid resolution (Base grid [dr,cells])

The Mixed grid stacks cells uniform cells of size `dr` (in  $R_p$ ) on the lower boundary and fills the rest of the domain with `N - cells` geometrically stretched cells; the total cell count `N` is set by the optional key `Grid cells:` (default 500). The two numbers share one input line because they are not independent: their product is the radial extent of the uniform region ( $0.01 R_p$  with the default  $2.0 \times 10^{-4}$  50).

The quantity that matters physically is how many uniform cells span the base density scale height,  $H/dr$  with  $H = kT/(\mu g)$ . Where  $H/dr$  is of order a few cells the discretization supports a *stationary*  $2dr$  entropy (contact) mode that nothing in the scheme removes: HLLC resolves a contact of zero speed exactly and so adds no dissipation to it, the gravity source is cell-local, the WENO3 pressure gradient sees only interface pressures, and there is no conduction or physical viscosity unless they are switched on. The measured dependence is an alternating  $\ln \rho$  amplitude of  $\sim 10^{-2}$  for  $H/dr < 5$  and  $\sim 10^{-4}$  for  $H/dr > 100$ . High-gravity planets have the least margin: at  $T_{eq}$  on the default grid HD 189733 b spans only  $\sim 27$  cells per  $H$ , against  $\sim 60$ – $100$  for HD 209458 b, WASP-52 b and WASP-121 b, and a cool, shielded base shrinks  $H$  further. The value in effect is echoed in `EXHALE_setup.out` as `Basescale-heightresolution:H(T_eq)/dr`, with a warning below 10 cells.

Refine at fixed extent, i.e. divide `dr` and multiply `cells` by the same factor: `Base grid [dr,cells]: 5.0e-5 200` is the  $4\times$  refinement. Two costs follow. The CFL time step is set by the smallest cell, so a  $4\times$  finer base means  $\sim 4\times$  more steps for the same physical time; and at fixed `N` the stretched region loses those cells and coarsens (for a  $4.43 R_p$  domain, the cell size at  $1.5 R_p$  grows from  $6.1 \times 10^{-3}$  to  $1.3 \times 10^{-2} R_p$ ). Raising `Grid cells:` alongside the refinement keeps the upper stretch fixed instead: the pairs  $1.0 \times 10^{-4}$  100/500,  $5.0 \times 10^{-5}$  200/658,  $2.5 \times 10^{-5}$  400/916 and  $1.25 \times 10^{-5}$  800/1375 hold the stretch ratio at its default-grid value, at the price of proportionally more work in every sweep. In practice the  $2\times$  step ( $1.0 \times 10^{-4}$  100) is the one to try first: it is where the mode goes, and  $4\times$  is not reachable

to convergence on HD 189733 b within the step cap (Max steps:) because its time step is  $4\times$  smaller. That  $2\times$  setting is what HD189733b/input.inp carries; the other three production planets run on the default. The full investigation is docs/hd189\_base\_checkerboard.md.

## 2.6 The lower boundary: a characteristic condition at the face

The lower boundary is imposed at the first cell *face*  $r_{\text{edg}}(0)$ , not in a ghost cell. At that face the characteristic speeds are  $v-c$ ,  $v$  and  $v+c$ , and a wave enters the domain when its speed is positive; in EXHALE the radius increases outward, so gas entering the base moves outward and inflow is  $v > 0$ . For the subsonic inflow a wind rests on, two characteristics enter and one leaves, so the lower atmosphere states *two* conditions and the interior supplies *one*.

The two the lower atmosphere states are its **pressure** and its **specific entropy** at the base level  $r = 1$ , carried as the isentrope through  $(p_0, T_0)$  at the base composition. The one the interior supplies is the outgoing acoustic invariant of the first cell,

$$p_b - \rho_i c_i v_b = p_i - \rho_i c_i v_i, \quad (1)$$

where the subscript  $i$  is cell 1 continued to the face along its own hydrostatic isentrope and  $c_i$  uses cell 1's own  $\gamma_{\text{eff}}$ . The closed-form invariant  $v - 2c/(\gamma - 1)$  is deliberately not used: the caloric equation of state gives every cell its own  $\gamma_{\text{eff}}(x_{\text{H}_2}, T)$ , and that invariant assumes one constant  $\gamma$ .

The ghost cells are then the volume averages of the same hydrostatic isentrope continued *below* the face, with the velocity carrying a constant  $\rho v r^2$ . They are not copies of the face state: the two differ by half a cell of stratification, and that half cell is what the previous ghost-cell closure left as a resolution-independent momentum imbalance of 25% of the weight of cell 1.

Where the flow reverses ( $-c < v < 0$ ) only one characteristic enters, so the reservoir states the pressure alone and the entropy at the face is the interior's, advected out. The handover between the two is a smooth blend of width  $10^{-6}$  in face Mach number, far below the  $\sim 10^{-5}$  a wind's own base carries. Supersonic outflow prescribes nothing. Supersonic *inflow* would need three reservoir conditions and a  $(p, s)$  reservoir has two, so it is reported as a configuration outside the model rather than repaired.

Nothing in this is selectable. There is no ghost velocity, pressure or temperature closure to choose, and the four keys that used to choose them (Valve eps, Hydrostatic base, Base ghost temperature, Base velocity) are refused at startup with a message naming the replacement, rather than ignored. The one key that remains is Base BC:, which states *where* the boundary is, not what it does.

## 2.7 The steady-state Newton solver (Solver: Newton)

The finite-volume steady state can be solved directly ( $F(Y) = 0$  for the hydro state, with the ionization equilibrium eliminated locally inside the residual) by a preconditioned Jacobian-free Newton–Krylov method (banded LAPACK preconditioner, frozen WENO weights, diagonal scaling, non-monotone line search; the boundary is differentiable of its own accord, the branch handover being a blend rather than a kink, so nothing has to be smoothed for it). Each unknown is scaled by its own cell's state  $(\rho, \rho(|v| + c_s), E)$ , so that no cell sets another's scale; the momentum scale uses the fastest signal speed because  $\rho v$  itself vanishes wherever the flow reverses. This reaches residual levels unattainable

by marching at a small fraction of the cost: WASP-121b converges in 8 Newton iterations ( $\sim 200$  marching-steps of cost, vs.  $>35,000$  insufficient marching steps), HD189733b in 33 s where marching spent  $>3$  h without converging.

Two ways to use it:

- **Production:** add Solver: Newton to input.inp. The run marches (including the two-stage PLM $\rightarrow$ WENO3 when Reconstruction scheme: PLM+WENO3 is set) until the flux metric  $du$  (the radial spread of  $\rho v r^2$ ) drops below the hand-off threshold newton\_du\_switch (default  $10^{-2}$ ; the  $du$ -based convergence stops are suspended), then the Newton solve finishes to Resid tol (default  $10^{-5}$ ), and the standard outputs and post-processing follow. The hand-off point is adjustable through an optional third token on the key, Solver: Newton <du\_switch> (e.g. Solver: Newton 5e-3 hands off later); a bare Solver: Newton keeps the  $10^{-2}$  default.
- **From an existing state** (recommended for re-converging old runs): put the state in output/\*\_IC.txt, set Load IC? True and Resid tol: 1.0e-5, and run with the environment hook

```
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x
```

which solves from the loaded state, writes output/, and stops.

The warm start matters: Newton has a finite basin, so the hand-off must wait until the marching warm-up has flattened the wind. The default  $du < 10^{-2}$  is calibrated to land inside the basin: handing off earlier (a larger newton\_du\_switch) risks starting Newton outside it and stalling, whereas a cold-IC start (no warm-up at all) stalls outright. Earlier revisions keyed the hand-off on the residual itself ( $\|R\| < 5 \times 10^{-2}$ ); the  $du$ -based key is equivalent in practice and consistent with the flux-based convergence decision (§2.4), while the residual remains the *target* of the JFNK finish. Convergence is linear (frozen-weight refresh and the chemistry noise floor). Planets with a breathing base (HD189733b) used to stop at a practical floor  $\|R\| \sim 3 \times 10^{-4}$ ; the 2026-08-11 line-search and base-energy fixes removed it, and a warm re-convergence of HD189733b now reaches  $\|R\| = 1.2 \times 10^{-5}$  in 11 outer iterations (re-measured 2026-08-15). The Newton finish is also *safe to fail*: the solver tracks the best iterate seen, aborts once the line search has failed to find any acceptable step in 12 consecutive outer iterations, and on failure the run prints JFNK failed, keeps the best iterate, and falls back to plain time-marching with the  $du$ -based stops re-armed: an unconverged Newton state is never accepted as the final answer. The associated diagnostic hooks (EXHALE\_RESIDUAL, EXHALE\_JAC\_TEST, EXHALE\_NEWTON\_TEST, ...) are environment variables, collected in §2.8 (Table 1). Full development record and validation: steady\_solver\_memo.pdf. Figure 1 shows the whole route end-to-end on WASP-121b (Roche): the monitored residual during the marching warm-up (left; note that the  $du$ -based stop would have fired long before the residual is actually small) and the JFNK finish (right).

**What certifies a converged state, and what the exit status says.** Reaching Resid tol is not the same as being certified. At the marching stop, at the JFNK final and at a pseudo-transient declaration, one evaluator builds an inventory of every balance the configuration activates and returns, for each, a status (not\_applicable / evaluated / unavailable), the row measure (the maximum over physical cells of  $|\text{res}| / \max(\text{scale}, \text{floor})$ ), the volume average, the worst cell and the scale it was divided by. The tolerances are  $3 \times 10^{-12}$  for the mass row,  $10^{-8}$  for momentum and  $10^{-6}$  for energy, the level balances and the eliminated-species closures.

The species rows are gated by regime. An element row and a carrier row are judged against  $10^{-5}$  for cells at  $r \geq 1.20R_p$  only; below that radius the row is measured and printed and does *not* gate. The gate is anchored on the element operator's own stated floor and sits a decade above the discretization error a smooth manufactured column shows at the production spacing ( $1.5 \times 10^{-6}$ ); the layer is excluded because its element fluxes are not conserved there at all (their radial spread over  $r < 1.10$  is 9.8 times the median), so no threshold means anything against them yet. The band  $1.10 \leq r < 1.20$  is likewise reported and does not gate. A state that passes with species rows present reports that it is certified *in the wind*, and its state file records `cert_reason=certified_in_wind`. The two radii are properties of the flow and are constants of certification.f90, not input keys; the five measurements behind the number are in `docs/certification_tolerance_anchoring_20260910.md`.

A run that DECLARED a stationary state and then failed certification exits with status 2; a run that ended on a cap makes no such claim, writes `certified=F cert_reason=no_stationary_claim` in its state files and exits 0. The regression harness never tests the status.

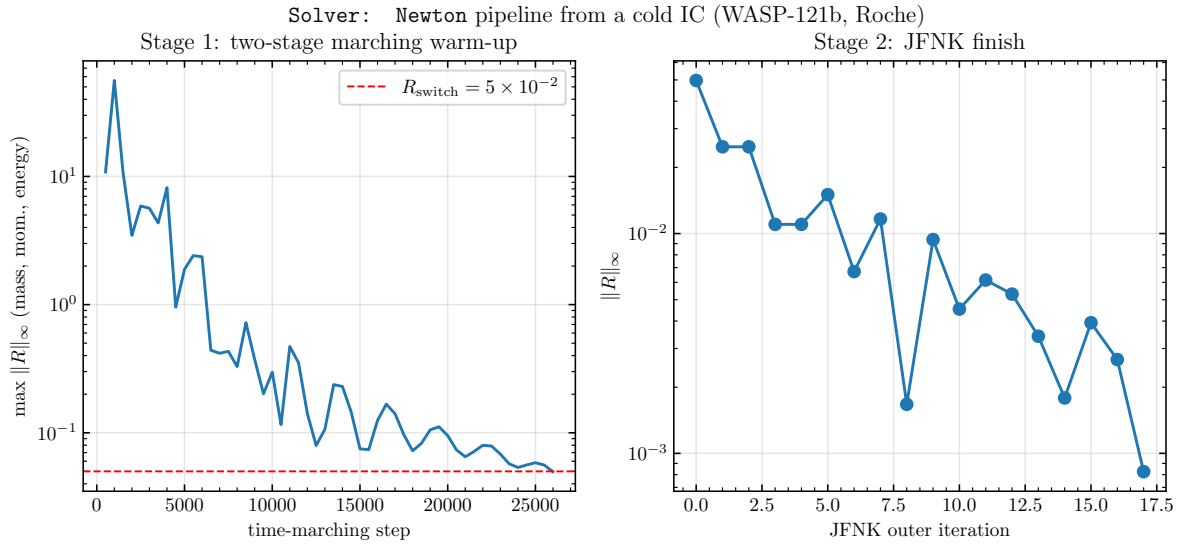


Figure 1: The Solver: Newton route from a cold IC (WASP-121b, Roche). Left: maximum steady residual  $\max \|R\|_\infty$  over the mass, momentum, and energy equations during the two-stage (Reconstruction scheme: PLM+WENO3) marching warm-up; the dashed line is the legacy residual hand-off threshold  $R_{\text{switch}} = 5 \times 10^{-2}$  used when this figure was produced (the current default keys the hand-off on  $du < 10^{-2}$  instead, at a comparable point on the warm-up). Right: residual vs. JFNK outer iteration during the Newton finish, down to `Resid tol` ( $10^{-5}$ ).

## 2.8 Environment-variable hooks (diagnostics and expert runs)

A small set of *environment variables* expose diagnostic and expert entry points without touching `input.inp` (Table 1). They are read once at startup; when unset they have strictly zero effect, so normal runs are unaffected. All of them are diagnostics or validation aids, apart from `EXHALE_ROOT`, which the lower-atmosphere pre-step needs: the only other one used in a production workflow is the `EXHALE_PTC` family, which implements the “Newton from an existing state” recipe of §2.7 (and `examples/04_newton_from_state/`). Typical usage:

```
EXHALE_RESIDUAL=1 ./EXHALE.x # rank a "converged" state
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x # re-converge it
```

Table 1: Environment-variable hooks. All are read once at startup and have zero effect when unset. The four diagnostic tests (top block), EXHALE\_PARSE\_DUMP and EXHALE\_PTC runs stop after printing/writing; they do not march.

| Variable                              | Effect                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EXHALE_DUMP_IC=1                      | After Load IC? True, write the restored state to output/Hydro_ioniz.txt and output/Ion_species.txt and stop, <i>before</i> the first ionization-equilibrium solve re-equilibrates the species fractions (verifies a snapshot/restart round-trip).                                                                     |
| EXHALE_RESIDUAL=1                     | Evaluate the steady residual $R$ of the loaded state (WENO3, boundary conditions applied first), print the norms for each equation (mass, momentum, energy), and stop. The criterion-independent way to rank candidate “converged” states (a premature-dip state shows $\ R\ _E \sim O(10)$ despite $du < 10^{-3}$ ). |
| EXHALE_NEWTON_TEST=1                  | Self-test of the Newton infrastructure (pack_U/unpack_U round-trip and the frozen residual against the EXHALE_RESIDUAL values); prints and stops.                                                                                                                                                                     |
| EXHALE_JAC_TEST=1                     | Verify the colored-FD banded Jacobian against a directional finite difference $Jr \simeq [F(Y+\epsilon r) - F(Y)]/\epsilon$ ; a correct band layout matches to $\sim 10^{-6}$ , a wrong one is $O(1)$ off. Prints and stops.                                                                                          |
| EXHALE_PTC=1                          | Solve the <i>loaded</i> state directly with the steady solver (no marching), write output/, and stop. Requires Load IC? True; set Resid tol: $1.0\text{e-}5$ (§2.7). Default engine: banded PTC.                                                                                                                      |
| EXHALE_PTC_JFNK=1                     | With EXHALE_PTC=1: use the full JFNK solver (GMRES on the matrix-free $J$ ) instead of the banded PTC, the recommended engine.                                                                                                                                                                                        |
| EXHALE_PTC_DTAU0= $\langle v \rangle$ | With EXHALE_PTC=1: override the initial pseudo-time step $\Delta\tau_0$ (default: the CFL $\Delta t$ of the loaded state). A large value (e.g. $1.0$ ) probes the pure-Newton regime directly.                                                                                                                        |
| EXHALE_PTC_NFIX= $\langle n \rangle$  | With EXHALE_PTC=1: anchor the first $n$ physical cells (frozen-base experiment; solves the wind on top of a fixed base).                                                                                                                                                                                              |
| EXHALE_DIAG_BASE=1                    | Write output/base_diag.txt ( $r, n, v, T$ , heating, cooling of the first few cells over the first steps) to trace IC-startup transients at the base.                                                                                                                                                                 |
| EXHALE_FORCE_HYBRD1=1                 | Force every ionization-equilibrium solve to use MINPACK hybrd1, skipping the analytic-Jacobian Newton: lets the same binary produce both solvers’ outputs for an A/B validation (Update_EXHALE_solver.pdf §6).                                                                                                        |
| EXHALE_DIFFUSION_CHECK=1              | Write the face-by-face elemental fluxes of every element-diffusion step to output/element_flux_profile.txt (§4.), and the composition profile of each direct-steady outer pass to output/diffusion_pass_profiles.txt. With a lower-atmosphere profile in use the flux file is written anyway.                         |

*continued on the next page*

| Variable                                                                                                                                                                                            | Effect                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EXHALE_DIFF_OMEGA= $\langle v \rangle$                                                                                                                                                              | Pin the starting under-relaxation factor of the direct-steady composition outer loop (default 0.5, halved down to 0.125 on a pass whose drift failed to fall), so the undamped loop can be compared with the damped one without a rebuild.                                                                                                                                                                                    |
| EXHALE_MAXSTEPS= $\langle n \rangle$                                                                                                                                                                | Deterministic step cap: leave the marching loop after $n$ steps and write the outputs. Separate from the Max steps: key (§2.4) and only ever earlier; it is what the regression harness uses to pin a relaxation-snapshot case.                                                                                                                                                                                               |
| EXHALE_PROFILE=1                                                                                                                                                                                    | Print a timing breakdown: the <code>ioniz_eq</code> share of the step every 500 steps, and the totals at the end of the run.                                                                                                                                                                                                                                                                                                  |
| EXHALE_PARSE_DUMP=1                                                                                                                                                                                 | Write every value <code>input_read</code> derived from <code>input.inp</code> (plus any <code>base.inp</code> override) to <code>parse_dump.txt</code> and stop, before any <code>grid/IC/output</code> work. Feeds the parser regression corpus.                                                                                                                                                                             |
| EXHALE_ROOT= $\langle \text{path} \rangle$                                                                                                                                                          | Install directory for the lower-atmosphere pre-step generators (§3.7); required when <code>Lower atmosphere:</code> is set on a relocated install. Not a diagnostic.                                                                                                                                                                                                                                                          |
| EXHALE_JFNK_MAXIT= $\langle n \rangle$                                                                                                                                                              | Cap a stationary solve at $n$ outer iterations in place of the number its caller passed (the default is 3000 at all three call sites). It also caps the <code>run:</code> under this variable the run takes its FIRST stationary solve and refuses later entries by name, so that a bounded measurement is one solve of one state rather than the sum of several from different states. Nothing about the step depends on it. |
| EXHALE_INVENTORY_REPORT=1                                                                                                                                                                           | Print the elemental inventory of every cell (what the carriers hold, what is locked, what the reservoir specifies, what the closure supplies, and the budget each side spends) at the carrier operator's entry.                                                                                                                                                                                                               |
| EXHALE_ELEMENT_ASSERT= $\langle 1 2 \rangle$                                                                                                                                                        | Verify the element census after each write-back and report (1) or stop on (2) a breach; off by default, and the only check that sees a write-back that creates nuclei.                                                                                                                                                                                                                                                        |
| <i>Measurement options of the stationary solve. Each was implemented, measured against both species-row fixtures and left in the default the measurement supported; none changes a default run.</i> |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| EXHALE_ELEMENT_CONSTRAINT_ROWS=0                                                                                                                                                                    | Limit a species trial by the five coordinate-box corners instead of the shared element constraint rows (the state before 2026-09-09; the corners let two carriers of one element each spend the whole budget).                                                                                                                                                                                                                |
| EXHALE_ELEMENT_WRITE_BACK                                                                                                                                                                           | The element-conserving write-back of each Newton iterate.                                                                                                                                                                                                                                                                                                                                                                     |
| EXHALE_TR_RESTART_STALL= $\langle n \rangle$                                                                                                                                                        | Restart the trust-region state after $n$ outer iterations without improvement. Delivered disarmed: no reset measured helped either fixture.                                                                                                                                                                                                                                                                                   |
| EXHALE_KRYLOV_ON_THE BALL=1                                                                                                                                                                         | Steihaug–Toint truncation of the Krylov cycle at the trust-region radius. Much cheaper in products, worse on the carrier fixture; default off.                                                                                                                                                                                                                                                                                |
| EXHALE_GM_REORTHO=1                                                                                                                                                                                 | Reorthogonalize the Arnoldi basis. The measured orthogonality loss is $10^{-13}$ to $3 \times 10^{-12}$ , so it is not the limiter, and the option is worse on both fixtures; default off.                                                                                                                                                                                                                                    |

*continued on the next page*

| Variable                     | Effect                                                                                                                                                                                                                              |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EXHALE_COLUMN_EQUIL=1        | Equilibrate the columns of the model as well as of the factored band. It moves the trust-region coordinates and trials ascend; measured and not adopted.                                                                            |
| EXHALE_CAUCHY_LEG_BY_IMAGE=1 | Let the image-share test alone decide whether the approximate-gradient leg is dropped. Worse on both fixtures; default off.                                                                                                         |
| EXHALE_CERT_ANCHOR=1         | Run and print the five anchors behind the species-row certification tolerances (reproducibility, closure seed, state perturbation, derivative, manufactured column). The default certification report is byte-identical without it. |
| EXHALE_ELEM_DIAG             | The diagnostic line the species-row step prints at every trial.                                                                                                                                                                     |

## 2.9 Metal cooling on/off comparison

Metal-line cooling is active *iff* a `metals.inp` is present (§3.); remove or rename it for a metals-off run. Because the metal abundances are read at run time, the executable is built once and the two cases differ only by whether `metals.inp` exists: no source edits or rebuild. To run the *same* planet both ways:

```
make                                # build once
# metals ON (solar C/N/O from metals.inp):
printf 'CI 2.69e-4\nNI 6.76e-5\nOI 4.90e-4\n' > metals.inp
./EXHALE.x ; mv output output_metals_on
# metals OFF (no metals.inp):
rm -f metals.inp
./EXHALE.x ; mv output output_metals_off
```

The notebooks for each planet `metals_comparison*.ipynb` read the two folders and plot the temperature, velocity / mass-flux, H and metal ionization, and heating/cooling differences (cf. §7.).

## 2.10 Wind-AE warm-start initial conditions

A 1-D steady-state Parker-wind solver (a Fortran port of *Wind-AE*; Murray-Clay et al. 2009 / Broome et al. 2025) is included under `src/modules/wind_ae/` and can supply the initial condition for a hard-to-launch EXHALE run. There are two ways to use it; the underlying algorithm, ramp, and BC/IC are documented in `docs/wind_ae_solver.tex`.

**In-process (recommended): IC mode: windae.** Add the single line

```
IC mode: windae
```

to `input.inp` (with `Load IC? False`). On startup EXHALE maps the planet's parameters to Wind-AE, ramps a shipped seed solution to the planet, solves the steady wind, and writes the IC onto the EXHALE grid (`output/*_IC.txt`), which is then loaded automatically: the whole input → Wind-AE solve → IC → run sequence is one `./run_EXHALE.sh` (one `EXHALE.x`) invocation. This is a *warm start*: EXHALE re-solves the ionization (and any metals) from the first

step, so the Wind-AE IC need not be exactly self-consistent. Always finish with Solver: Newton for a quantitative  $\dot{M}$  (§2.7). The shipped seed and spectrum live in `inputdata/windae_seed.csv` and `inputdata/windae_spectrum.inp`. The ramp runs in two stages, and the second one is what makes far-from-seed planets reachable. It first tries the *static-BC* ramp, which moves the system parameters (flux, gravity, star) while holding the seed's base boundary conditions fixed: fast, and enough for a seed-adjacent hot Jupiter such as HD 209458 b. If that stalls, the seed is reloaded and the ramp is retried with the *self-consistent-BC* continuation, which re-converges the base  $R_{\min}/T/\rho$ , the sonic-point column density, and the bolometric/molecular layer turn-off as the parameters move (`wae_continuation.f90`). That is the default path, and with it the ramp converges the far-from-seed HD 189733 b as well as the seed-adjacent HD 209458 b (`examples/11_windae_ic` and `examples/12_windae_ic_hd209`). Only if *both* stages fail does the run stop with a message advising IC mode: auto. What the Wind-AE IC does not fix is EXHALE's own HD 189733 b base-breathing instability, which limits the warm start there whatever the IC source.

Because a generated IC is an analytic profile that the marching loop has not touched, its  $du$  can start far below every threshold and would once have tripped a false convergence stop within a few steps. Both the  $du$  stop and the JFNK hand-off now require  $du$  to have been seen at or above their own threshold first, so only a descending crossing counts (§2.4). Measured on `examples/12_windae_ic_hd209` (2026-08-15):  $du$  at step 2 is  $1.43 \times 10^{-4}$ , neither trigger fires, and the two arm at steps 23 and 222 on the first ascending crossing. The false stop is structurally gone for this IC mode; an  $\dot{M}$  read off a  $du$  stop still carries its usual path-dependent spread, so the Newton finish remains the prescription.

**Standalone generator: `wind_ae_ic.x`.** The same solver builds an IC out of process:

```
make wind_ae_ic      # builds ./wind_ae_ic.x (separate from EXHALE.x)
./wind_ae_ic.x <input.inp> <seed.csv> <spectrum.inp> \
                <IC_dump_grid> <outdir> [rhoscale]
```

It reads an EXHALE `input.inp`, ramps from the seed, and writes `<outdir>/{Hydro_ioniz,Ion_species}_IC.txt` on the grid given by an `IC_dump.txt` from a prior EXHALE run with the same domain settings. Run EXHALE on the result with Load IC? True + Solver: Newton. (The standalone tool is independent of the in-process path above; either can be used.)

## 3. Input files

### 3.1 `input.inp`: the main parameter file

`input.inp` is read *line by line in a fixed order* for the core block (Table 2), followed by an *optional keyword block* (Table 3) whose lines are matched by keyword and may be omitted (older files without them keep the default behavior). A few lines are *conditional*: the spectrum-property line depends on Spectrum type, and the energy-band / X-ray-luminosity lines depend on Use only EUV?.

Table 2: input.inp core block (fixed order). Conditional lines are marked (\*).

| Label                                                    | Value                     | Meaning                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planet name:                                             | string                    | Label for the run (cosmetic).                                                                                                                                                                                                                                                    |
| Log10 lower boundary number density [cm <sup>-3</sup> ]: | real (log <sub>10</sub> ) | log <sub>10</sub> $n_0$ , the H number density at the base. <i>Optional since 2026-09-03</i> : a base.inp carrying p_base states the base level itself and $n_0$ follows from it. Give one or the other; if both are given they must agree to 1% or startup refuses the pair.    |
| Planet radius [R_J]:                                     | real ( $R_J$ )            | Base radius $R_p$ (the 1 $\mu$ bar level); the grid radius $r$ is in units of this. The unit is the IAU 2015 nominal Jupiter <i>equatorial</i> radius, $R_J = 7.1492 \times 10^9$ cm (parameters.f90); the Python tools take the same value from RJ_CM of examples/exhale_io.py. |
| Planet mass [M_J]:                                       | real ( $M_J$ )            | Planet mass $M_p$ (sets gravity / the Jeans parameter $b_0$ ). The unit is the IAU 2015 nominal $M_J = 1.8982 \times 10^{30}$ g; the stellar mass below is in $M_\odot = 1.98842 \times 10^{33}$ g.                                                                              |
| Equilibrium temperature [K]:                             | real (K)                  | Base temperature $T_0$ .                                                                                                                                                                                                                                                         |
| Orbital distance [AU]:                                   | real (AU)                 | Star–planet separation $a$ (sets the XUV flux $\propto a^{-2}$ and the tidal/Roche potential).                                                                                                                                                                                   |
| Escape radius [R_p]:                                     | real ( $R_p$ )            | Inner radius of the constant-momentum convergence window; must lie <i>inside</i> the domain ( $< r_{\max}$ ). For L1-truncated Roche runs use a value below the lobe.                                                                                                            |
| He/H number ratio:                                       | real                      | Helium abundance $n_{\text{He}}/n_{\text{H}}$ ; $> 0$ enables He chemistry.                                                                                                                                                                                                      |
| 2D approximate method:                                   | see text                  | Day–night averaging of the XUV flux and $\dot{M}$ : Rate/2 + Mdot/2 (dayside, $\xi = \frac{1}{2}$ ), Mdot/4 (full-sphere), Mdot, or alpha (then a trailing number sets the rate-correction $a_\tau$ ).                                                                           |

Table 2, continued

| Label                                                          | Value                                       | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parent star mass [M_sun]:                                      | real ( $M_{\odot}$ )                        | Stellar mass $M_{\star}$ (tidal potential, Roche-lobe size).                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Spectrum type:                                                 | Power-law<br>Planck   Load<br>Monochromatic | The incident spectrum. One type builds every band of the photon grid, the sub-13.6eV band and the Balmer continuum included (§3.5); no band is filled from a type the input did not select.                                                                                                                                                                                                                                                                                         |
| (*) Power-law index: / Spectrum file: /<br>Photon energy [eV]: | real / path / real                          | Property line of the chosen type: the power-law slope $E^{\text{PLind}}$ , the numerical SED file, or the monochromatic energy. Planck has no property line: it is built from Stellar Teff [K] and Stellar radius [R_sun] of the optional block, and startup refuses Planck without both of them. With metals or the He 2 <sup>3</sup> S triplet, mind the required wavelength coverage: a Loaded table that does not reach the band an active absorber needs stops the run (§3.5). |
| Use only EUV? True/False:                                      | logical                                     | True: EUV band only. False: include X-rays (adds the band's third energy and an X-ray luminosity line).                                                                                                                                                                                                                                                                                                                                                                             |
| (*) [E_low,E_mid,E_high] = [ . . . ]                           | reals (eV)                                  | Band edges: two values (EUV only) or three (with X-rays).                                                                                                                                                                                                                                                                                                                                                                                                                           |
| (*) Log10 of X-ray luminosity [erg/s]:                         | real (log <sub>10</sub> )                   | Only if X-rays are included; log <sub>10</sub> $L_X$ .                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Log10 of EUV luminosity [erg/s]:                               | real (log <sub>10</sub> )                   | log <sub>10</sub> $L_{\text{EUV}}$ (with $a$ , sets the incident XUV flux).                                                                                                                                                                                                                                                                                                                                                                                                         |
| Grid type:                                                     | Uniform<br>Stretched   Mixed                | Radial grid spacing. Mixed (fine near the base, stretched outward) is the usual choice.                                                                                                                                                                                                                                                                                                                                                                                             |

Table 2, continued

| Label                   | Value                | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Numerical flux:         | HLLC   ROE   LLF     | Riemann solver for the hydrodynamic fluxes (Num_Fluxes.f90). HLLC is the usual choice; LLF (local Lax–Friedrichs) is the most diffusive and is kept for solver-isolation tests. MEASURED 2026-09-17 (item L25): on a wind whose base is subsonic to $M \sim 10^{-7}$ the HLLC flux can fail to reach the stationary root on a 500-cell grid where it reaches it on 1000 cells, and ROE reaches it on 500; where both fluxes solve the same case they differ by $-0.3$ to $-0.6$ percent in $\dot{M}$ and by 6 to 8 percent of opposite sign in the first cell's $T$ and $\rho$ . Record: docs/lhs1140b_stationary_L25_20260916.md. |
| Low Mach velocity jump: | logical (def. False) | Scale the normal velocity jump of the ROE branch by $\min(\text{Ma}, 1)$ after Rieper (2011, J. Comput. Phys., 230, 5263); no effect under HLLC or LLF. Delivered off and NOT recommended on a near-hydrostatic column: the factor vanishes exactly where two neighbouring velocities cancel in the Roe average, so it removes the damping of the odd-even velocity mode at the faces that mode lives on, and it was measured to prevent a solve that unmodified ROE completes. Rieper's premise, that the local and the global Mach number are comparable, is violated by eight decades on such a column.                         |

Table 2, continued

| Label                  | Value                      | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reconstruction scheme: | PLM   WENO3  <br>PLM+WENO3 | Spatial reconstruction order and staging. PLM or WENO3 runs a single stage (only the first <code>du_th</code> value is used as the convergence threshold); PLM+WENO3 selects the two-stage mode (march in PLM to the first <code>du_th</code> value, then switch to WENO3 and converge at the second).                                                                                                                                      |
| Include He23S?:        | logical                    | Include the metastable He $2^3S$ triplet (needed for the He 10830 line). <i>On by default</i> , and the only line of the core block that may be omitted: a file without it gets the triplet. False is the deliberate opt-out, used to compare against the atomic helium limit. Forced off when the gas carries no helium.                                                                                                                   |
| Load IC?:              | logical                    | Restart from the previous run's output/ <code>*_IC.txt</code> (warm start). What the restart <i>continues</i> is stated by Restart intent: below; the state files also carry the configuration they were produced under, and a restart under another composition, grid, constant set or option set is refused naming the field (docs/input_schema.md appendix D.2), unless the differing options are named by Restart option change: below. |

Table 2, continued

| Label        | Value   | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Do only PP:  | logical | Skip the time loop and only run the post-processing on existing output. It still takes one time step before it stops, so it is not an identity on the state it reads, it reconstructs with whatever Reconstruction scheme: states rather than with the scheme the solution was reached under, and it writes the stepped state back over Hydro_ioniz.txt. For the advection-corrected products of a <i>solved</i> state use Restart intent: stationary evaluate instead, which takes no step at all (item L9, 2026-09-17; docs/lhs1140b_stationary_L9_20260916.md). |
| Force start: | logical | Force the first 1000 iterations regardless of the convergence test (useful when the IC is close to a fixed point).                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**The constants these units mean.** Every physical constant has one definition, in `src/modules/init/parameters.f90`, and the Python tools take the same numbers from `examples/exhale_io.py`. The ones a user input is quoted in, or that set the scale of a reported number, are

$$\begin{aligned}
 R_J &= 7.1492 \times 10^9 \text{ cm} & M_J &= 1.8982 \times 10^{30} \text{ g} & M_\odot &= 1.98842 \times 10^{33} \text{ g} \\
 R_\odot &= 6.957 \times 10^{10} \text{ cm} & \text{AU} &= 1.495978707 \times 10^{13} \text{ cm} & G &= 6.67430 \times 10^{-8} \text{ cgs}
 \end{aligned}$$

$R_J$ ,  $M_J$  and  $M_\odot$  are the IAU 2015 nominal values (Prša et al. 2016, AJ 152, 41, Table 1),  $R_J$  being the nominal *equatorial* radius, which is what a transit depth measures. The mass unit of the code is the hydrogen *atom*,  $m_H = 1.67353284 \times 10^{-24}$  g, and one helium nucleus weighs  $m_{\text{He}}/m_H = 3.9715$ : the helium-4 atom mass  $6.6464790722 \times 10^{-24}$  g divided by  $m_H$ , not the integer 4, so the mass density, the mean molecular weight and the base composition are the helium-4 gas and not a 0.7% heavier one. The ionization thresholds are the NIST ASD values, one definition each and every cross section turning on exactly there:  $e_{\text{th}}(\text{H I}) = 13.598434599$ ,  $e_{\text{th}}(\text{H}_2) = 15.425927$ ,  $e_{\text{th}}(\text{He I}) = 24.587389$ ,  $e_{\text{th}}(\text{He II}) = 54.417765$  and  $e_{\text{th}}(\text{He } 2^3S) = 4.767775$  eV.

Table 3: `input.inp` optional keyword block (any line may be omitted; defaults restore the standard behavior).

| Keyword               | Value                                         | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Domain mode:          | Spherical                                     | If present and Spherical: pure planetary potential $-b_0/r$ , domain extended to Outer radius. Absent (default): full Roche (tidal) potential; outer boundary at the Hill/L1 radius unless Outer radius is given (§1.3).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Outer radius [R_p]:   | real $> 1$ ( $R_p$ )                          | Outer boundary (e.g. 10–15 $R_p$ ). In Spherical mode it sets the pure $-b_0/r$ domain; in Roche mode (no Domain mode line) it extends the domain past $L_1$ while <i>keeping</i> the tidal potential (Yan/Huang-style; pair with IC mode: auto).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Base grid [dr,cells]: | real ( $R_p$ ), int (def. $2.0\text{e-}4$ 50) | Resolution of the uniform region of the Mixed grid: cells uniform cells of size dr stacked on the base, then $N - \text{cells}$ stretched cells out to $r_{\text{max}}$ . The product $\text{dr} \times \text{cells}$ is the extent of the uniform region (0.01 $R_p$ by default), so refining the base at fixed extent means dividing dr and multiplying cells by the same factor (5.0e-5 200 is the 4× refinement). Requirement: dr must resolve the base density scale height $H = kT/(\mu g)$ ; below $\sim 5$ cells per $H$ the scheme carries an undamped stationary $2dr$ entropy mode at the base (§2.5). Ignored by Grid type: Uniform and Stretched. At fixed Grid cells: a finer base leaves fewer cells for the stretched region; raise the two together to hold the upper stretch (§2.5). |

Table 3, continued

| Keyword     | Value                    | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grid cells: | int $\geq 10$ (def. 500) | Total number of computational cells $N$ (ghost cells not counted). The default reproduces the value that used to be compiled in, so an <code>input.inp</code> without the key is unchanged. A restart ( <code>Load IC? True</code> ) must load an IC written on <i>this run's grid</i> : the row count is checked against $N + 2N_g$ , and then the file's cell centers are compared with the ones <code>define_grid</code> built over the physical cells $1 \dots N$ . A relative difference above $10^{-10}$ anywhere stops the run, naming the cell and both radii. The reader does not interpolate, so a state written under a different Grid type, <code>Base grid</code> or <code>Outer radius</code> has to be mapped onto this grid before it is handed back ( <code>load_IC.f90 verify_restart_radii</code> ). |
| Max steps:  | int (def. $10^6$ )       | Hard cap on marching iterations ( <code>count_max</code> ); the run reports that it stopped without converging when the cap is reached. Lower it to bound an exploratory run, raise it for a high-gravity case whose base time step makes $10^6$ steps too short (§2.4).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Table 3, continued

| Keyword               | Value                   | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coronal cutoff width: | real (def. 0.1)         | Roll-off width $w$ of the coronal-excitation guard below the $10^3$ K CHIANTI fit floor: the coronal part of every metal-line coefficient is multiplied by $\exp(-x^2)$ with $x = (T_{\text{floor}}/T - 1)/w$ , so smaller $w$ cuts the extrapolated coronal cooling off more sharply. Since the ground-term fine-structure floors were solved in statistical equilibrium (§8.) the results no longer depend on $w$ : the base-cell balance temperature of all four paper planets is identical to every digit printed over $w = 0.02$ – $1.2$ , against a 500–650 K spread before. The default is kept at 0.1, and the guard is kept because extrapolating a fit below its fitted range is the wrong thing to do, not as a tuning knob (docs/coronal_cutoff_width.md, section 7).                                                                                                                              |
| Base IR field:        | True/False (def. False) | Let the infrared coolants of the molecular layer see the thermal radiation of the atmosphere below the base instead of emitting into vacuum. The lower atmosphere is taken to be black at those wavelengths and to radiate $B_\nu(T_0)$ over the sky fraction $1 - \sqrt{1 - (R_p/r)^2}$ , and the cooling becomes the net rate, emission minus absorption of that field, so each band stops cooling at its own radiative-equilibrium temperature (576–642 K for the CI/OI fine-structure lines and 941 K for the $\text{H}_3^+$ bands, at $T_0 = 1140$ K and half-sky coverage). Applies to the eight ground-term fine-structure lines of CI, CII, NII, OI and to the $\text{H}_3^+$ infrared cooling; every other channel keeps the optically thin, no-incident-field limit. Off by default, and with no effect on an atomic run, where those channels carry no cooling (docs/lower_atmosphere_coupling.md). |

Table 3, continued

| Keyword             | Value                | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Molecular IR bands: | True/False<br>False) | (def. Add the infrared coolants an H <sub>2</sub> atmosphere carries below the H <sub>2</sub> → H front and the atomic channels do not: the H <sub>2</sub> quadrupole and magnetic dipole line spectrum (Roueff et al. 2019, A&A 630, A58) and the H <sub>2</sub> O and CO vibration–rotation bands (HITEMP, through the correlated- <i>k</i> coefficients distributed with Photochem). Each is computed in LTE, in the optically thin limit, as the <i>net</i> exchange $4\pi n_X \int \sigma_\nu(T) [B_\nu(T) - W B_\nu(T_0)] d\nu$ with the same diluted field Base IR field supplies, so each stops cooling at its own radiative-equilibrium temperature: 889, 918 and 984 K for H <sub>2</sub> O, CO and H <sub>2</sub> at $T_0 = 1140$ K and half-sky coverage. The closure keeps stimulated emission, so the net rate is identically zero for gas at $T_0$ with the whole sky black. H <sub>2</sub> needs only Molecular chemistry; H <sub>2</sub> O and CO exist only with Oxygen chemistry, and the key is inert on species the run does not carry. Used together with Base IR field: on its own the new bands emit into vacuum, which deepens the collapse of the molecular layer instead of holding it, and input_read warns. Writes three net columns (H2_IR, H2O_IR, CO_IR, negative where the band heats) and the Planck-mean optical depths of the H <sub>2</sub> O and CO columns to output/Cooling_breakdown.txt, and a column-integrated infrared block to output/FUV_bands.txt. Cross sections tabulated 50–2000 K and clamped outside it (src/modules/lower_atmosphere/molecular_infrared_cooling.f90, cooling_data/molecular_infrared_bands.py). |

Table 3, continued

| Keyword                  | Value                        | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Molecular reaction heat: | True/False<br>(def.<br>True) | Deposit the energy the <i>collisional</i> reactions of the H <sub>2</sub> /He network release into the gas. <b>On by default; the key exists to turn it off.</b> In an atomic gas the ionization energy a photon spends comes back as the Lyman photon of a radiative recombination and leaves, so it is correctly absent from the heating. In a molecular gas it does not: the cycle $\text{H}_2 + h\nu \rightarrow \text{H}_2^+ + e$ (the photon pays $I(\text{H}_2) = 15.43 \text{ eV}$ and the photoelectron keeps the rest, which the photoheating already deposits), $\text{H}_2^+ + \text{H}_2 \rightarrow \text{H}_3^+ + \text{H}$ (+1.70 eV), $\text{H}_3^+ + e \rightarrow \text{H}_2 + \text{H}$ (+9.25 eV, <i>dissociative</i> ) returns to H <sub>2</sub> having turned one H <sub>2</sub> into H + H and left $I(\text{H}_2) - D_0(\text{H}_2) = 10.95 \text{ eV}$ in the gas with no photon to carry it away. Measured at 81 percent of the total heating rate at $1.02 r_{\text{base}}$ on the converged He/H=0.0793 hot Uranus. The heat is built from one table of species enthalpies, so a closed chemical cycle releases exactly zero; the photon-driven reactions, the radiative recombinations, the collisional ionizations, the Penning channels and the H/He charge exchange are excluded because their energy is already in the ledger. Needs Molecular chemistry, so an atomic run is unaffected to the bit. Writes the column heat_mol_chem to output/Heating_breakdown.txt (src/modules/lower_atmosphere/molecular_reaction_heat.f90, Update_EXHALE.md Sec. 141). |

Table 3, continued

| Keyword                                   | Value         | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stellar LW flux [erg/cm <sup>2</sup> /s]: | real (def. 0) | <p>Band-integrated stellar flux in the H<sub>2</sub> Lyman–Werner bands (912–1201 Å, i.e. 10.3–13.6 eV) <i>at the planet’s orbit</i>. Switches on H<sub>2</sub> photodissociation, <math>\text{H}_2 + h\nu \rightarrow \text{H} + \text{H}</math>, in the molecular network. The rate is the incident band photon fluence <math>F_{\text{LW}}/\overline{h\nu}</math> contracted with a dissociation cross section <math>\sigma_{\text{diss}}(N_{\text{H}_2}, T, n_{\text{H}})</math> read from a level-resolved CLOUDY table (line self-shielding and the trapping of the fluorescent decay photons are both inside it) times <math>e^{-\tau}</math> of the continuum of the same interval, and it is averaged over each cell between its two faces rather than taken at one of them (the cross section falls by four decades across the self-shielding transition). Plus 0.4 eV of heating per dissociation (Black &amp; Dalgarno 1977). The band is a separate input because the photon grid need not carry it: the grid floor is the lowest ionization threshold of an <i>active</i> absorber (§3.5), so a molecular run with no metastable, no low-IP metal and no armed excited-H coupling has no grid point in 11.2–13.6 eV at all, and where the grid does reach that far, a power-law type gives only the XUV fit extrapolated downward rather than a stellar FUV spectrum. With Spectrum type: Load the key may be omitted: the band is then integrated from the spectrum file over 912–1201 Å (the exact integral of the piecewise-linear table, the flux interpolated at each clipped edge; one read of the file serves the LW, B3 and B4 bands; the file is at the planet already) and the setup report says so. A stated value always wins, and a stated 0 switches the band off even when the file could supply it (the comparison with a model that has no H<sub>2</sub> photodissociation). A file that does not reach both edges of the band stops the run and asks for the key: a positive field is never turned into zero silently. Without a spectrum file, 0 (default) = no photodissociation. The key is inert without Molecular chemistry: True. Writes output/Lyman_Werner.txt</p> |

Table 3, continued

| Keyword                                       | Value          | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stellar FUV B1 flux [erg/cm <sup>2</sup> /s]: | <i>retired</i> | Band B1 was 1110–1201 Å, between the Lyman–Werner interval and Ly $\alpha$ . The H <sub>2</sub> Lyman and Werner lines pump on both sides of 1110 Å at the 700–3200 K of a planetary base, so a band edge there normalized the H <sub>2</sub> pumping per photon of a band narrower than the one the lines absorb from. On 2026-09-06 the band was merged into the Lyman–Werner interval, which is now 912–1201 Å, and this key was retired: a file that still carries it stops the run with a message asking for the two numbers added into Stellar LW flux. |
| Stellar FUV B3 flux [erg/cm <sup>2</sup> /s]: | real (def. 0)  | The same over 1231–1450 Å (band B3, branching 0.89/0.11/0.00). Band B2 is the Ly $\alpha$ line and comes from Stellar Lya flux. With Spectrum type: Load and Oxygen chemistry: True, a run that does not state this key gets the band integrated from the spectrum file over 1231–1450 Å (the exact piecewise-linear integral, as for Stellar LW flux; a file that does not reach the band stops the run), and the setup report says so; a stated value, zero included, always wins.                                                                          |
| Stellar FUV B4 flux [erg/cm <sup>2</sup> /s]: | real (def. 0)  | The same over 1451–2304 Å (band B4, branching 1.00/0.00/0.00), integrated from the spectrum file under the same rule when not stated. <i>Read the caveat in §3.7:</i> the H <sub>2</sub> O cross section falls three decades across this band, so a single band average misrepresents it by a factor of several for a red spectrum.                                                                                                                                                                                                                           |

Table 3, continued

| Keyword                       | Value                       | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stellar Teff [K]:             | real (K)                    | Stellar $T_{\text{eff}}$ . Two consumers: it arms the excited-H model when both this <i>and</i> Stellar radius are $> 0$ , and it is the temperature of the photosphere under Spectrum type: Planck, which is refused without both. Arming the excited-H model makes $H(n=2)$ an absorber of the grid-floor rule, so the photon grid then reaches down to the Balmer head $e_{\text{th,H}}/4 = 3.400 \text{ eV} = 3647 \text{ \AA}$ (§3.5). |
| Stellar radius [R_sun]:       | real ( $M_{\odot}$ -radius) | Stellar radius: sets the dilution $(R_{\star}/a)^2$ of the Planck field and of the Balmer continuum, and in EXHALE_transit.py it normalizes every transit depth and caps the impact-parameter integration at $R_{\star}/R_p$ . Supply it for any run whose spectrum will be synthesized: absent, the transit script falls back to a built-in radius that need not match the system, and the depths scale accordingly.                       |
| Jlya RT file:                 | path                        | Read an external $\bar{J}_{\text{Ly}\alpha}(r)$ profile to pump $H(n=2)$ (sets jlya_mode=1).                                                                                                                                                                                                                                                                                                                                                |
| Jlya escape-prob:             | True                        | Compute $\bar{J}_{\text{Ly}\alpha}$ in-line each step by escape-probability RT (jlya_mode=2).                                                                                                                                                                                                                                                                                                                                               |
| Stellar Lya flux [erg/cm2/s]: | real                        | Incident stellar $\text{Ly}\alpha$ flux at the planet (the stellar source for jlya_mode=2).                                                                                                                                                                                                                                                                                                                                                 |
| Lya stellar halfwidth [km/s]: | real (km/s)                 | Stellar $\text{Ly}\alpha$ line half-width; controls how deep the trapped stellar beam penetrates (default 70).                                                                                                                                                                                                                                                                                                                              |
| Lya stellar boost [-]:        | real                        | Bounded buildup cap for the trapped stellar $\text{Ly}\alpha$ beam (default 5).                                                                                                                                                                                                                                                                                                                                                             |
| Lya absorbing bottom:         | logical (def. False)        | Treat the lower boundary of the jlya_mode=2 closure as a pure absorber (Huang et al. 2017: the $\text{H}_2$ layer beneath the base destroys $\text{Ly}\alpha$ through accidental resonances): a downward wing-escape channel $\beta_{\text{bot}}$ on the depth to the bottom joins the upward and Sobolev channels, lowering $\bar{J}_{\text{Ly}\alpha}$ (and $n_2$ ) in the bottom few scale heights only. No effect on jlya_mode 0/1.     |

Table 3, continued

| Keyword            | Value               | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deexc heat:        | logical (def. True) | Collisional de-excitation of $H(n=2)$ returns 10.2eV to the electrons. It is not a double count of the H I collisional-excitation cooling, which is the one-way coronal rate of Cen (1992) with no de-excitation term in it; most of the $n=2$ population is maintained by $Ly\alpha$ pumping, so the term is absorbed $Ly\alpha$ thermalized by a collision. Active only with the excited-H model; False reverts to the one-way ledger.                                                                |
| Transonic IC:      | True                | Use a transonic isothermal-wind initial condition (solved from the potential via the Bernoulli integral) instead of the hydrostatic IC; needed for deep-RLOF bases that do not launch from rest. Auto-enables Force start.                                                                                                                                                                                                                                                                              |
| Newton solver:     | logical (def. True) | Solve the ionization equilibrium with the analytic-Jacobian Newton method (with a MINPACK hybrd1 fallback); False reverts to the legacy hybrd1. Default is faster ( $\sim 25\%$ ) and reproduces hybrd1 to $\sim 10^{-7}$ .                                                                                                                                                                                                                                                                             |
| Brent solver:      | logical (def. True) | Solve the post-process energy equation with a bracketing (Brent) root-finder that selects the physical root; False reverts to legacy hybrd1 + a $2\times$ -band reject.                                                                                                                                                                                                                                                                                                                                 |
| du_th [PLM,WENO3]: | two reals           | Convergence thresholds (e.g. $0.5 \quad 1.0e-3$ ). Which are used is set by Reconstruction scheme: above. With PLM+WENO3 (two-stage) EXHALE marches in PLM until $du < \text{first value}$ (or a PLM stall), then switches to WENO3 and converges at $du < \text{second value}$ ; with single-stage PLM or WENO3 only the first value is used and the second is ignored. The PLM+WENO3 mode replaces the manual PLM $\rightarrow$ stop $\rightarrow$ Load IC+WENO3 restart of the official ATES recipe. |

Table 3, continued

| Keyword                      | Value                                                                                                                       | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reconstruction continuation: | real $\Delta\lambda$ [, real $\tau$ ]<br>[, fixed   adaptive]<br>(def. $\Delta\lambda = 0$ , i.e. the<br>one-step hand-off) | Walk the PLM $\rightarrow$ WENO3 hand-off of a two-stage run along the homotopy $R_\lambda = (1 - \lambda)R_{\text{PLM}} + \lambda R_{\text{WENO3}}$ instead of changing the discrete operator in one step. $\Delta\lambda$ is the advance in $\lambda$ per marching step ( $\leq 0$ off; 1.0 reproduces the one-step switch, byte-identically). $\tau$ (def. 1.2) is the step control: $\lambda$ advances while $du$ stays within $\tau$ times its value at the start of the ramp, and a step that exceeds it halves $\Delta\lambda$ instead of advancing. fixed disables that control, adaptive is the default; the two optional fields are told apart by content, not position. Requires Reconstruction scheme: PLM+WENO3 with a non-empty PLM stage, otherwise a warning is printed and the key is ignored. Once $\lambda$ reaches 1 the continuation disarms and the run continues in pure WENO3, so the JFNK finish and every acceptance gate see the production operator (docs/f_plm_weno_continuation.md). |
| Stall [tol,N]:               | real, int                                                                                                                   | Stall-detector override (relative $du$ change and window; defaults $10^{-6}$ , 2000).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| CFL:                         | real (def. 0.6)                                                                                                             | CFL-number override.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Hot Parker IC:               | real (K)                                                                                                                    | Warm-seed initial condition: stable hydrostatic density with a warm $T$ /ionization/velocity overlay.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Table 3, continued

| Keyword          | Value       | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IC mode:         | word        | Initial-condition family: cold (default), transonic, hot_parker (synonyms for the two keys above), auto, which picks automatically from the cold sonic-point topology of the actual potential: an interior sonic point (deep-RLOF via the Roche L1 topology, or low-gravity boil-off, $r_c \simeq b_0/2c_0^2$ inside the domain) selects the transonic IC, otherwise the cold hydrostatic IC; or windae, which builds the IC in-process with the ported Wind-AE steady-wind solver (§2.10). No tunable threshold for auto; the choice and the escape parameter $b_0$ are logged. Explicit Transonic IC/Hot Parker IC keys override auto. See docs/auto_ic_design.md and, for windae, docs/wind_ae_solver.tex. |
| Level tol:       | real        | Mass-flux <i>level</i> -stability gate: a converged/stalled stop additionally requires the mean $ \rho v r^2 $ over the wind window to change by less than this (relative) across the last <code>N_stall</code> steps. Guards against stopping on a transient $du$ dip while the flux level still drifts ( $\leq 0$ = off, default; §2.7).                                                                                                                                                                                                                                                                                                                                                                    |
| Flux spread tol: | real [real] | FLUX gate (§133): a steady solve is accepted only when the radial spread of $\rho v r^2$ over $r \geq r_{\text{flux}}$ is below this tolerance AND the residual is below <code>Resid tol</code> . Defaults $2 \times 10^{-5}$ ( <code>flux_spread_th_default</code> ) and $r_{\text{flux}} = 1.2 R_p$ ; $\leq 0$ disables it.                                                                                                                                                                                                                                                                                                                                                                                 |

Table 3, continued

| Keyword        | Value         | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resid tol:     | real          | Residual-based convergence: stop when the steady residual falls below the value (monitored every 500 steps), <i>replacing</i> the <i>du</i> test ( $\leq 0$ = off, default; the JFNK finish then uses the code's own $10^{-5}$ ). The number compared against it is $\max_k \max_j  R_{j,k} /s_{j,k}$ , each row divided by a bound on the largest term that row itself contains: mass $\max( F r^2 \text{ at the two faces})/\Delta V$ , momentum $\max( \Delta F_2 ,  S_2 )$ , energy $\max( \Delta F_3 ,  S_3 , \text{heat, cool})$ , plus the operator-split viscous and conduction sources where those are on. The maximum is over cells, so one cell out of tolerance keeps the state out of tolerance; the volume-weighted norm is reported beside it and no longer decides acceptance. <i>du</i> alone is blind to the energy imbalance (§2.7); the scales and the reason for each are in docs/input_schema.md K35. |
| Time stepping: | Local         | Pseudo-time steps in each cell (steady-state acceleration; not time-accurate). Reaches the deep state fast but leaves residuals in each cell; pair with a Newton finish.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Energy solver: | Explicit      | Revert to the original forward-Euler source update (solver-isolation tests).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Solver:        | Newton [real] | Production steady-state solve: marching warm-up until the flux metric <i>du</i> (the radial spread of $\rho v r^2$ ) drops below the optional value <i>newton_du_switch</i> (default $10^{-2}$ ), then the JFNK Newton finish to <i>Resid tol</i> (default $10^{-5}$ ), then the standard final outputs and post-processing. See §2.7.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

Table 3, continued

| Keyword   | Value                      | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Run mode: | init   phys (def.<br>init) | What this run is doing, stated rather than inferred. init: initialization or continuation. It claims no elapsed time and permits local pseudo-time, pseudo-transient continuation and a chemistry that is not at its root, all as numerical devices; the physical clock <code>t_phys</code> stays where it started and no output reports it. phys: physical integration, one global $dt$ per step, the clock advanced only by a step accepted in full, the temporal error of the step sampled every 20 accepted steps, and the certification refusing a state whose chemistry has no root. Any other word is a fatal error stop 1. Refused: phys with Time stepping: Local, since cells advanced by different intervals do not form one trajectory. The mode sets the default of Restart intent: below, is written into the state file's metadata block as mode and read back on a restart (a phys run loading a file that says mode=phys without a finite non-negative <code>t_phys</code> is refused), and is echoed in EXHALE_setup.out together with whether it was given or defaulted. See docs/input_schema.md K45. |

Table 3, continued

| Keyword         | Value                                                                                                                                 | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Restart intent: | trajectory<br>relaxation<br>  stationary<br>[evaluate<br>equilibrate] (def.<br>trajectory with<br>Run mode: phys,<br>else relaxation) | What a loaded state is continued as. trajectory:<br>the physical clock continues from the t_phys<br>the state file records. relaxation: march toward<br>stationarity and then hand off to the solver, which<br>is what every restart did before this key existed.<br>stationary: rebuild the derived quantities of<br>the loaded state without taking a step, measure<br>its stationary residual and its certification <i>as<br/>loaded</i> , and enter the stationary solve at once;<br>the second word evaluate writes the measured<br>state back with its certification and stops, and<br>equilibrate puts the loaded composition on<br>its own fixed point before the measurement (a<br>change of the state, reported when taken). This<br>is the route that certifies a written state without<br>the two CFL steps a marching hand-off takes<br>first. Since 2026-09-17 (item L9) evaluate<br>also names the three states it handles apart (the<br>state as loaded, the work state whose pressure<br>and temperature are derived from the loaded<br>conserved state, and the advection-derived one)<br>and writes every product a solved case is read for<br>from the work state with no time update: the two<br>state files, the heating and cooling breakdowns,<br>the *_adv profiles and the mass-loss line. The<br>*_adv files carry a # derived_from: provenance<br>line rather than a # coupling: header, because<br>they describe a composition the certification was<br>not made on. Refused: the key with Load IC?<br>False, trajectory with Run mode: init,<br>stationary without Solver: Newton. Design:<br>docs/restart_contract_design_20260909.md. |

Table 3, continued

| Keyword                | Value                                                                  | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Restart option change: | token list, separated by commas or blanks (def.: no option may differ) | Which switches of the state file's # options line are <i>allowed</i> to differ between the loaded state and this run. <code>carrier_newton</code> is not among the tokens that have to be named: it states the ROUTE and not the equation set, the rows, columns and tolerances being the same either way, so a difference in it is admissible with nothing named, the load says so and the change is recorded as a # <code>route_change</code> line (item L22 increment I3, 2026-09-17). The restart contract otherwise refuses any option difference, which forbids the ladder of settings this code is converged with: converge without an option, restart with it on, converge again. Naming a token ( <code>sec_ion</code> , <code>He23S</code> , <code>mol_heat</code> , <code>base_ir</code> , <code>cond</code> , ...) permits exactly that token to differ; every other difference still refuses the load and names the token. The loaded state is then a solution of the <i>former</i> option set and a starting point for this one, so the change is written into the state this run produces as a # <code>option_change</code> line and inherited by the rungs that follow, and EXHALE_setup.out states it. Refused: an unknown token; a token that decides how many unknowns the state has ( <code>metals</code> , <code>mol</code> , <code>oxychem</code> , <code>carrier</code> , <code>carrier_newton</code> , <code>iontrans</code> ), whose change makes the rows of the file not the rows of this run and is a cold start and not a restart; a word naming the grid, the reservoir or the constant set; the key with <code>Load IC? False</code> . See docs/input_schema.md appendix D.2. |

Table 3, continued

| Keyword           | Value                                                                                                | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-Mach damping: | real $\varepsilon_4$ [, real $M_{\text{th}}$ ]<br>(def. -1, i.e. off;<br>$M_{\text{th}} = 10^{-3}$ ) | Artificial dissipation of the $2\Delta r$ contact/entropy mode. The HLLC flux carries the middle wave at $S^* \simeq v$ , so the dissipation it applies to the entropy family vanishes as the flow stagnates and that mode becomes marginally damped; where metal cooling stalls a shell (HD 209458 b with solar C/N/O, $r \simeq 1.02$ , $M \sim 10^{-5}$ ) it grows to several times the local mean $ v $ . This key adds a gated fourth-difference (Jameson, Schmidt & Turkel 1981) stress $\varepsilon_4 g(M) \rho_f \lambda_f (v_{j+2} - 3v_{j+1} + 3v_j - v_{j-1})$ to the numerical <i>momentum</i> flux, and its work $v_f D_p$ to the energy flux, with the gate $g = [\max(0, 1 - M^2/M_{\text{th}}^2)]^2$ exactly zero for $M \geq M_{\text{th}}$ . It enters the numerical flux, so the marching loop and the JFNK steady residual see the <i>same</i> equation. Explicit stability requires $\varepsilon_4 < 1/(16\text{CFL})$ (0.10 at the default CFL); the classical range 1/64–1/32 works. $\varepsilon_4 \leq 0 =$ off, bit-identical. See <code>src/modules/flux/low_mach_dissipation.f90</code> . |

Table 3, continued

| Keyword        | Value                | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Well balanced: | logical (def. False) | <p>Carry the <i>departure</i> from the cell's own local hydrostatic equilibrium through the reconstruction, the Riemann jumps and the pressure force, instead of the state itself. Inside cell <math>j</math> the equilibrium is the constant-density one through that cell's own density and pressure, <math>p_{\text{eq},j}(r) = p_j - \rho_j[\phi(r) - \phi(r_j)]</math>, so no thermal, entropy or compositional stratification is assumed; the pressure stencil acts on the departure of a neighboring cell's pressure from that equilibrium continued <i>through</i> the face between them (this cell's density up to the face, the neighbor's beyond it), which vanishes at a discrete equilibrium, and the momentum row is assembled from the face pressure measured against that equilibrium while the gravitational source term is zero. The equilibrium's flux difference and its source then cancel in the algebra rather than in floating point, and a discrete hydrostatic equilibrium is preserved to rounding (Käppeli &amp; Mishra 2016, A&amp;A 587, A94; the face-pressure-difference form of the source is Käppeli &amp; Mishra 2014, J. Comput. Phys. 259, 199). Exact preservation also needs a numerical flux that resolves a stationary contact, so it holds with Numerical flux: ROE and HLLC but <i>not</i> with LLF, and not at a face the positivity repair has replaced with a first-order flux. It applies to both reconstructions and to the PLM <math>\rightarrow</math> WENO3 continuation. Turning it on <i>moves every result</i>: the pressure/gravity pair is in every run, and the two discretizations are both second-order consistent with neither a subset of the other. It is stated in EXHALE_setup.out and EXHALE_resolved.out, and written as the option token wellbal in a state file's # options line, where Restart option change: may name it, since it does not decide how many unknowns a state has. See src/modules/states/Reconstruction.f90.</p> |

Table 3, continued

| Keyword                    | Value                | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ATES_photoionization_rate: | True                 | Revert the He I ( $1^1\text{S}$ ) photoionization cross section to the legacy ATES two-term fit (default: Verner+1996). Does <i>not</i> affect the He $2^3\text{S}$ Penning rate, which is always the temperature-dependent form (§8.).                                                                                                                                                                                                                                 |
| Legacy_HHe_rates:          | logical (def. False) | Revert the H/He case-B recombination and collisional ionization to the legacy ATES fits (Hui & Gnedin 1997 recombination; Abel+1997 / Hui & Gnedin 1997 collisional ionization). Default uses Badnell (2023) RR (+ He II DR) minus the Mao & Kaastra (2016) $\alpha_1$ for case B, and Voronov (1997) collisional ionization (§8.). The free-free Gaunt factor and its ion net-charge weighting are always the van Hoof et al. (2014) form, independent of this switch. |

Table 3, continued

| Keyword          | Value                                     | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Atomic rate set: | default<br>Koskinen2022<br>(def. default) | Swap the four atomic H/He rate coefficients for the fits Koskinen et al. (2022) list as R1–R4 of their Table 1, for a like-for-like comparison with their Model A: recombination $4.0 \times 10^{-12}(300/T)^{0.64}$ for $\text{H}^+$ and $4.6 \times 10^{-12}(300/T)^{0.64}$ for $\text{He}^+$ (Storey & Hummer 1995), and collisional ionization $2.91 \times 10^{-8}U^{0.39}e^{-U}/(0.232+U)$ with $U = 13.6\text{eV}/kT$ for H and $1.75 \times 10^{-8}U^{0.35}e^{-U}/(0.180+U)$ with $U = 24.6\text{eV}/kT$ for He (Voronov 1997). The two collisional fits are the ones EXHALE already uses, so only the two recombination coefficients change value; the branch does pin the Voronov form against Legacy_HHe_rates. The default set is the physically preferred one here, case B recombination, the Lyman continuum being optically thick in this gas, so the key exists to reproduce their choice and not to replace ours. It swaps <i>rate</i> coefficients only, with one deliberate exception: $\lambda_{\text{rec}}(\text{He II})$ is $k_B T$ times the He II recombination coefficient and therefore follows the switch, since removing electrons at one rate and charging the gas at another would not conserve the ledger. Any other word stops the run (§8.). |

Table 3, continued

| Keyword               | Value                                        | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Secondary_ionization: | False   True   Immediate (def. True, staged) | Secondary ionization by fast photoelectrons in the main loop: a photoelectron ejected with $E_0 > 30\text{eV}$ deposits only $f_{\text{heat}}(x, E_0)$ of its excess as heat and drives further HI / He I / H <sub>2</sub> ionizations, where $x$ is the free electron density over the H+He nuclei. The ionization budget is Shull & van Steenberg's (1985) with the energy dependence of Dalgarno, Yan & Liu (1999); the heating fraction, the H <sub>2</sub> target and the molecular heat returns are Dalgarno et al.'s. The default is <i>staged</i> (§2.4): the coupling switches on only after the wind first converges without it, then the run re-converges. False restores full photoelectron thermalization, bit-identical to the legacy path; Immediate applies it from step 0 (A/B tests only). |
| He_rec_coupling:      | logical (def. True)                          | Couple He II $\rightarrow$ He I recombination radiation to HI ionization (Draine 2011 on-the-spot $y/z$ ; §8.). Adds an HI photoionization rate $n_{\text{HeII}}n_e[z\alpha_B + y\alpha_1]$ with its photoelectron heating and shifts the He II recombination to $\alpha_B + y\alpha_1$ ; $y$ is the local $\geq 24.6\text{eV}$ ground-capture share ionizing H, $z$ the density-dependent cascade share. False = pure case B (photons lost locally, the legacy path; in the He 2 <sup>3</sup> S mode the singlet recombination is then neither case A nor case B).                                                                                                                                                                                                                                          |

Table 3, continued

| Keyword               | Value                                       | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| He_H_charge_exchange: | logical (def. True)                         | The $\text{He} \leftrightarrow \text{H}$ charge-exchange pair (Koskinen 2013 rates via Huang 2023 Table 4): $\text{He}^0 + \text{H}^+ \rightarrow \text{He}^+ + \text{H}^0$ (endothermic) and the $\text{He}^+ + \text{H}^0 \rightarrow \text{He}^0 + \text{H}^+$ loss channel where neutral H dominates ( $\sim 40\%$ of He II at $r \simeq 1.05$ on WASP-121 b, $< 1\%$ by $r \geq 1.2$ ). Applied by dedicated routines in <i>every</i> helium-bearing ionization system, independent of <code>cx_full</code> (which still gates the metal groups). False = legacy no-He-CX path, bit-identical. |
| He_diffusion:         | logical (def. False)                        | Enable diffusive separation of helium: transport the helium <i>mass fraction</i> of the two-component H/He gas with advection + molecular-diffusion settling, so He/H falls with altitude. Off = He/H frozen at the input He/H number ratio (§3.6).                                                                                                                                                                                                                                                                                                                                                 |
| He_Kzz:               | real ( $\text{cm}^2\text{s}^{-1}$ , def. 0) | Eddy-diffusion coefficient used with He_diffusion: mixes the composition toward uniform below the homopause. Default 0 = pure molecular diffusion; an eddy term belongs to the atmosphere being modeled, so it is stated rather than inherited (Taylor+2025 use $10^9$ ).                                                                                                                                                                                                                                                                                                                           |
| He_ambipolar:         | logical (def. True)                         | Ambipolar-corrected settling: in the ionized wind the polarization field lifts He ions, reducing settling. The field is computed from the local electron-pressure gradient, $eE = -kT \partial \ln(n_e T) / \partial r$ ; False = $eE = 0$ , i.e. neutral settling $\Delta m = 3$ .                                                                                                                                                                                                                                                                                                                 |
| He_alphaT:            | real (def. 0)                               | Thermal-diffusion factor $\alpha_T$ added to the settling drift ( $\alpha_T \partial \ln T / \partial r$ ); 0 = off.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| He_metal_diffusion:   | logical (def. False)                        | Also diffuse each trace metal against the hydrogen background with its own mass, mean charge and stage-resolved diffusion coefficient; heavier elements settle more. Off = metals frozen to H (§3.6).                                                                                                                                                                                                                                                                                                                                                                                               |

Table 3, continued

| Keyword              | Value                | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Molecular chemistry: | True/False<br>False) | (def. Add $\text{H}_2$ , $\text{H}_2^+$ , $\text{H}_3^+$ and $\text{HeH}^+$ to the coupled ionization equilibrium, with the $\text{H}_2$ photoionization opacity and heating and the $\text{H}_3^+$ infrared cooling. Requires $\text{He}/\text{H}$ number ratio $> 0$ . $\text{He\_diffusion}$ is allowed (since 2026-08-26): the element transport closes over the molecular carriers. Turning it on also turns <code>Molecular base</code> on (see there). A <code>metals.inp</code> may be present: the metal stages are then solved in the same system (§3.7).      |
| Molecular base:      | True/False<br>False) | (def. Remove the $\text{H}_2$ -bound particles from the base particle count <code>ntot_bc</code> using the chemical-equilibrium $\text{H}_2/\text{H}/\text{He}$ partition, giving a lower base pressure and a heavier base $\mu$ with the species arrays still atomic (Tier 2a). Forced on by <code>Molecular chemistry: True</code> , because an atomic <code>ntot_bc</code> under a molecular base inflates the base temperature by the particle-count ratio and dissociates the layer the chemistry just built, so with the chemistry on the key is redundant (§3.7). |

Table 3, continued

| Keyword           | Value                | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oxygen chemistry: | True/False<br>False) | (def. Add OH, H <sub>2</sub> O and CO to the same coupled system, with the FUV photolysis of H <sub>2</sub> O and OH, so that the code computes its own base H <sub>2</sub> /H partition instead of importing one. Requires Molecular chemistry: True, He/H > 0 and a non-zero oxygen abundance; refuses q_H2_base (it computes that quantity) and He_metal_diffusion (that option counts an element over its ion stages alone, and CO carries an oxygen <i>and</i> a carbon nucleus, so it cannot follow two element factors at once). He_diffusion alone is accepted. Supply the band fluxes with Stellar LW flux, Stellar FUV B3/B4 flux and Stellar Ly $\alpha$ flux; with Spectrum type: Load the three continuum bands (LW, B3, B4) are integrated from the spectrum file when their keys are absent, and only Ly $\alpha$ has to be stated. Without band fluxes the run returns the chemical equilibrium of the O/OH/H <sub>2</sub> O family rather than a photochemical partition, and says so at startup. Writes output/Oxygen_chemistry.txt and output/FUV_bands.txt and appends OH H2O CO to Ion_species.txt (§3.7). |

Table 3, continued

| Keyword                      | Value                                                  | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Molecular carrier transport: | True/False (def. True whenever Oxygen chemistry is on) | Transport the molecular carriers $H_2$ , OH, $H_2O$ and CO vertically instead of closing them by a local steady state: one implicit backward-Euler diffusion-advection step per pass, solved together with the same chemistry rows the local equilibrium uses, with molecular diffusion from Blanc's law over the background carriers, the eddy coefficient $K_{zz}$ of He_Kzz or of a lower-atmosphere profile, and settling. False restores the local steady state, which isolates the chemistry for testing and is not a model of a base: at a cool base the $H_2$ chemical time and the flow time are comparable. With $K_{zz} = 0$ everywhere the transport is pure molecular diffusion, the wrong limit for a lower atmosphere, and the startup report says so. It acts only with Molecular chemistry: True, not with Oxygen chemistry: $H_2$ is the first carrier and is transported with or without the oxygen cycle, which only adds OH, $H_2O$ and CO to the set, and Ionization transport adds $H^+$ . With the molecular network off there are no carriers and the key changes nothing; input_read reports that at startup rather than refusing the run, the state produced being the correct atomic one (§3.7). |

Table 3, continued

| Keyword                | Value                                                                                                                      | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coupled carrier solve: | False   True   On stall (def. False; a word the key has no meaning for stops the run rather than falling through to False) | Solve the transported balances as unknowns of the stationary system itself, $(\rho, \rho v, E, n(\text{H}_2)/n_0, \dots)$ per cell, instead of alternating a wind solve with a fixed-wind carrier relaxation. The alternation was measured not to converge: the two halves are coupled through the particle count, which sets the temperature, so a splitting that evaluates each at the other's old state gets the sign of the $\text{H}_2$ front's motion wrong. <b>On a molecular configuration this key requires Molecular carrier transport: True, and the combination without it is refused at startup.</b> With the carriers eliminated, $n(\text{H}_2)$ is not an unknown but whatever the local-equilibrium sweep returns for the composition it was seeded with, and in the shielded layer the sweep cannot return a content at all: the fast chemistry cycles $\text{H}_2 \rightarrow \text{H}_2^+ \rightarrow \text{H}_3^+ \rightarrow \text{H}_2$ without changing the number of $\text{H}_2$ nuclei, so the local balance rows fix the partition among the molecular species and leave their sum where the seed put it. The stationary residual is then not a function of its unknowns (measured seed dependence $3.2 \times 10^5$ times <code>Resid tol</code> per unit relative seed change, against $9.7 \times 10^{-10}$ with the $\text{H}_2$ row carried), so it has no root. A run whose only transported balance is an element ( <code>He_diffusion</code> in an atomic gas) is not affected, and neither is marching. The carrier densities of the coupled system are carried as $\ln n$ , so that the positivity of a carrier is a property of the unknown space rather than a bound the step control has to enforce; the floor under $\ln n$ is $10^{-20}$ of the cell's own element budget, the fraction of its element below which the carrier row scale already calls a row converged ( <code>EXHALE_CARRIER_LOG_UNKNOWN=0</code> restores the density unknown for measurement). |

Table 3, continued

| Keyword               | Value                | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ionization transport: | True/False<br>False) | (def. Carry the hydrogen ionization state with the flow: $H^+$ becomes a fifth transported carrier of the same operator that carries $H_2$ , and the ionization sweep is handed the transported fraction instead of solving the $H/H^+$ partition as a local root. The local partition is the right answer only where a parcel is ionized faster than it leaves its shell, $Pr/ v  \gg 1$ . On the Koskinen et al. (2022) Model A comparison that ratio is 0.15–0.35 above $1.5 r_{\text{base}}$ , and the local root then gives $x(H^+) = 0.129/0.327/0.585$ at $2.0/2.4/3.0 r_{\text{base}}$ where integrating along the same flow gives 0.057/0.078/0.106 and Model A, which advects every species, has 0.020/0.055/0.080 (docs/k22_electron_density_excess.md §7). The proton's molecular diffusion is set to zero, an ion in a neutral gas being ambipolar- or Coulomb-limited rather than hard-sphere and the term this option exists for being the bulk advection, while the eddy coefficient $K_{zz}$ still applies, it being a bulk mixing coefficient. There is no base Dirichlet value, no handoff stating an ionization fraction, so the lower ghost cells stay with the sweep.<br><b>It requires Molecular carrier transport: True and Molecular chemistry: True;</b> each is a fatal refusal at startup. With Solver: Newton both stationary routes carry it: with Coupled carrier solve: True the stationary system carries the proton's row and the sweep is handed the Newton unknown; without it the route is the partitioned alternation, whose hydrodynamic solves hold the composition and whose sweeps are handed the transported proton fraction of that composition, the carrier relaxation between the solves transporting it (until 2026-09-12 this combination was refused, on the premise, true of the plain Newton finish and not of the alternation. <b>Kwang-II Seon / Last updated: 2026-09-17</b> that the last sweep would put the proton back on its local root). Marching requires neither. The direct steady route (EXHALE PTC=1) is refused |

Table 3, continued

| Keyword                   | Value                                                         | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lower atmosphere:         | none   analytic  <br>vulcan [, real ( $R_f$ )]<br>(def. none) | Lower-atmosphere pre-step run before the wind: generate the base composition over the 1-bar to 1 $\mu$ bar column, analytically (src/utls/run_lower.py) or by calling VULCAN (src/utls/vulcan_driver.py, hours on the first run, cached after), and hand it over through base.inp (Tier 3). The second token is the 1-bar (transit) radius and is required. An existing base.inp is used as is and nothing is regenerated; the generator needs EXHALE_ROOT set to the install directory (§3.7).                                                                                                                                                                                                                                                                                                                               |
| Lower column:             | real ( $R_f$ )                                                | Tier-1 consistency <i>report</i> only: integrate the isothermal- $T_{eq}$ hypsometric column with the chemical-equilibrium $H_2/H/He$ partition from the given 1-bar radius up to 1 $\mu$ bar and print the derived base radius, base fractions and $\mu$ next to the input Planet radius. Nothing is overridden (§3.7).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Lower atmosphere profile: | path (def. none)                                              | Take the lower atmosphere from a <i>profile</i> (the lower model's own solution tabulated over an interval of pressure) instead of the single-level scalars of base.inp. The named file supplies the base state at its matching pressure ( $T_0$ , $R_0$ , $p_{base}$ , the $H_2$ fraction), the elemental reservoirs (He/H and every element column it carries, overriding metals.inp), and the eddy coefficient $K_{zz}(r)$ as a profile, which makes He_Kzz inert. The profile also fixes the base <i>level</i> : $n_0$ follows from p_match_bar exactly as it does from p_base of base.inp, and a Log10 lower boundary number density line given beside a profile is refused unless the two agree to 1%. A missing file, a malformed table, or a base.inp beside it that still states a physics key stops the run (§3.8). |

Table 3, continued

| Keyword         | Value                                                            | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Base BC:        | density   pressure<br>[, real ( $\mu$ bar)] (def.<br>density)    | What the lower boundary anchors. density (legacy) fixes the base number density at the $n_0$ of the core block. pressure (CETIMB-style) fixes the base pressure at the optional second token (default 1 $\mu$ bar) and <i>derives</i> $n_0$ from $n_0 k_B T_0 n_{\text{tot},bc} = p_{\text{base}}$ ; the same isothermal ghost then pins $p$ and $T_0$ and derives $\rho$ . It is a re-parameterization of the same ghost, whose physical effect is a much less dense base.                                                                                                                                                                                                                          |
| Shapiro filter: | real $\varepsilon$ [, int $n$ ] (def.<br>-1, i.e. off; $n = 4$ ) | Apply a weak Shapiro (1970) 1–2–1 low-pass filter $u_j += (\varepsilon/4)(u_{j-1} - 2u_j + u_{j+1})$ to the conservative variables every $n$ steps, damping the gravity-unbalanced sound-wave (base-breathing) mode the same way CETIMB does. $\varepsilon \leq 0 = \text{off}$ , and off is the default on purpose: with the filter on, an HD209458b cold-IC sweep left the transonic-wind saddle for an infall state ( $v < 0$ everywhere) regardless of the velocity BC, while filter-off relaxed to a clean outflow. Enable it only for a case that actually breathes. Unlike Low-Mach damping it touches only the marching state, not the numerical flux, so the JFNK residual does not see it. |
| Viscosity:      | True/False, or real<br>$\mu_0$ [, real $s$ ] (def.<br>False)     | True adds the radial viscous force $(\nabla \cdot \tau)_r$ and its dissipation $q_\mu = \frac{4}{3}\mu(\partial_r v - v/r)^2$ with the calibrated $\mu(T) = \frac{4}{15}(m_H/k_B)\kappa(T)$ (monatomic Chapman–Enskog at Prandtl 2/3, tied to the Watson et al. 1981 conductivity): the molecular transport that damps the near-base momentum imbalance the inviscid HLLC scheme lacks. A numeric value is instead a diagnostic power law $\mu = \mu_0 T^s$ in code units ( $s = 0.7$ by default), kept for sensitivity scans. Off = byte-identical to the inviscid code.                                                                                                                            |

Table 3, continued

| Keyword                | Value                                                   | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conduction:            | logical (def. False)                                    | Add heat conduction $r^{-2}\partial_r(r^2\kappa\partial_r T)$ with the Watson et al. (1981) atomic-hydrogen $\kappa(T) = 4.45 \times 10^4 (T/10^3 \text{ K})^{0.7} \text{ erg cm}^{-1} \text{ s}^{-1} \text{ K}^{-1}$ . Off = byte-identical.                                                                                                                                                                                                                                                                                                                |
| Caloric EOS:           | ladder   monatomic<br>(def. ladder)                     | The caloric equation of state. ladder: every atom, ion and electron stores $\frac{3}{2}kT$ and $\text{H}_2$ its rovibrational energy (the Roueff et al. 2019 ladder), so $\gamma_{\text{eff}}$ depends on composition and $T$ ; in an 80% $\text{H}_2$ layer that is not 5/3. monatomic: $\text{H}_2$ too stores $\frac{3}{2}kT$ and $\gamma = 5/3$ everywhere, kept as a <i>comparison</i> option for a published model whose energy equation is $u = c_v T$ with a monatomic $c_v$ . In an atomic gas the two agree exactly. Any other word stops the run. |
| Photoelectron heating: | excess   full  <br>real $f \in (0, 1]$ (def.<br>excess) | What a photoionization deposits as heat. excess (physical): the photoelectron's $h\nu - I$ . full: the whole photon energy $h\nu$ ; a number: the fraction $f$ of $h\nu$ . The last two are <i>comparison</i> options. Koskinen et al. (2022) Figure 9 shows a stellar heating rate 2.5–3 times what $h\nu - I$ gives at their ionization rate and spectrum, i.e. between the two accountings, and a fraction reproduces their profile without claiming a physics; a run that uses one must say so. Anything else stops the run.                             |
| Resid norm:            | <b>retired</b>                                          | Selected between a volume-weighted and a max-over-cells combination of the scaled residual rows. The acceptance norm is now the cellwise maximum and the volume-weighted number is reported beside it every time, so there is nothing left to select. The label is still matched and a file carrying it draws a warning saying so, rather than failing the unknown-key scan.                                                                                                                                                                                 |

Table 3, continued

| Keyword           | Value                | Meaning                                                                                                                                                                                                                                                                                                                                 |
|-------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wind-AE seed:     | path, or grid   auto | Which converged Wind-AE solution IC mode: windae ramps from; the default is the shipped inputdata/windae_seed.csv. grid/auto instead picks the nearest solution in inputdata/windae_grid/ to this planet's ( $M_p, R_p, F_{XUV}$ ), which is a shorter and more robust ramp than always starting from the single fiducial seed (§2.10). |
| Wind-AE seed out: | path (def. none)     | Write the converged Wind-AE relaxation solution to this file as a reusable seed, e.g. to grow inputdata/windae_grid/. Omitted = nothing is written.                                                                                                                                                                                     |

### 3.2 metals.inp: trace-metal abundances

`metals.inp` is *optional*: if absent, the run is metals-off (H/He only). If present, it lists the total elemental abundance  $n_X/n_H$  (summed over ionization stages; the equilibrium solver distributes each element over its stages) for any of C, N, O, Mg, Si, Ca, Na, K, S, Fe. Blank lines and # comments are ignored; any non-zero abundance turns metals on. Five control keys are also accepted: `pp_metals` (metal treatment in the advection post-process: 0=metal-free, 1=frozen at the converged equilibrium (default), 2=re-solve the metal stages inside the post-process iteration, at fixed element totals), `cx_full` (1=full Huang+2023 charge-exchange network; default metal-H only), `cno_cool` (C/N/O line-cooling source: 1=closed-form fits to CHIANTI v11, including N I/N II cooling (default); 0=legacy AIOLOS analytic fits, which have no N cooling and deviate by 40–70% from CHIANTI for O I/O II in the wind-launch region, see `cooling_data/cno_cooling_comparison.ipynb`), and `eos_metals` (bulk-gas metal policy: 1=metals contribute their mass to the gas density and base density, their electrons to the free-electron density and the base pressure correction, and their nuclei to the total particle density, a  $\sim 1\%$  mass effect at solar abundances, consistent with the charge balance inside the ionization solver (default); 0=legacy trace approximation, H/He-only gas budget), and `cx_02p_H` (default 1 = the published rate). Huang+2023 Table 4 has no doubly ionized oxygen row, so re-sourcing the charge-exchange network to that paper dropped  $O^{2+} + H^0 \rightarrow O^+ + H^+$ . EXHALE carries it separately at the published Barragán et al. (2006, ApJ 636, 544) rate (the calculation CHIANTI v11 and Cloudy both adopt), and `cx_02p_H <scale>` rescales it, with 0 reproducing the Table-4-only reaction set exactly and other values serving as bounding experiments. It is deliberately *not* controlled by `cx_full`, which means “all of Table 4”. Measured on HD 209458 b, carrying the reaction suppresses the O III fraction by 4.7 decades at  $1.05 R_p$ , 3.7 at 1.1 and 2.7 at 1.2, and raises O I by 3% below  $1.2 R_p$  and 52% near  $2\text{--}3 R_p$ , while  $\dot{M}$  moves by 0.01 dex (§103 and §107 of Update\_EXHALE). The reverse direction is not carried: it is endothermic by 21.5 eV, hence Boltzmann suppressed by  $10^{-11}$  at  $10^4$  K. Solar reference values (Asplund+2009):

```
# metals.inp -- solar n_X/n_H (Asplund+2009)
C 2.69e-4 N 6.76e-5 O 4.90e-4
Mg 3.98e-5 Si 3.24e-5 Ca 2.19e-6
Na 1.74e-6 K 1.07e-7 S 1.32e-5 Fe 3.16e-5
pp_metals 1
```

### 3.3 opacity.inp: opacity model (optional)

opacity.inp selects the bound-free opacity model via `OPACITY_MODEL = A|C|P|T`: **A** analytic (default, the standard ATES cross sections); **C** constant (threshold  $\sigma$  times the factors for each species `OPA_CONST_HI/HEI/HEII/HEIIR`); **P** pressure-broadened (Robinson & Catling  $f(p) = 1 + a(p/p_0)^n$  via `OPA_PB_FACTOR=a`, `OPA_PB_EXPONENT=n`, `OPA_PB_PIVOT=p_0`); **T** tabulated (two-column .opa files per species, energy [eV] vs.  $\sigma$  [ $10^{-18} \text{ cm}^2$ ], given by `OPA_FILE_HI` etc.). Missing keys keep their defaults, so an absent opacity.inp is equivalent to model A.

### 3.4 Excited hydrogen and the Ly $\alpha$ pumping field

The non-LTE  $H(n=2)$  model (the metastable  $2s$  + resonance  $2p$  levels, Christie+2013) feeds two things: in EXHALE.x the  $H(n=2)$  photoionization adds a Balmer proton source and photoelectric heating to the energy/ionization balance; in EXHALE\_transit.py the  $n=2$  population sets the  $H\alpha/H\beta$  transmission. The model is *off by default*; it switches on only when both `Stellar_Teff` and `Stellar_radius` are given, so the standard run is unchanged. Those two keys *arm* the coupling; the Balmer continuum that ionizes and heats  $n=2$  is then the run's own spectrum over  $[e_{\text{th,HI}}/4, 13.598] \text{ eV}$  (the power law for a Power-law run, the photospheric blackbody for Planck, the loaded table for Load), integrated against the hydrogenic  $\sigma_2(E) = \sigma_2(e_{\text{th,HI}}/4)(e_{\text{th,HI}}/4E)^3$ , and not a blackbody of its own (§3.5). Arming the coupling also lowers the photon-grid floor to 3.400 eV, so a Loaded table that stops above 3647 Å stops the run.

The  $2s/2p$  populations are driven by the **Ly $\alpha$  mean intensity**  $\bar{J}_{\text{Ly}\alpha}(r)$ , which pumps  $1s \leftrightarrow 2p$ . How  $\bar{J}_{\text{Ly}\alpha}$  is obtained is selected by three mutually-exclusive options:

(mode 0) Parameterized: the default with no external input. If neither `Jlya_RT` file nor `Jlya_escape-prob` is present,  $\bar{J}_{\text{Ly}\alpha}$  is the Huang et al. (2017, Eq. 6) estimate computed internally from the run's own state,

$$\bar{J}_{\text{Ly}\alpha}(r) = \frac{0.1 F_{\text{LyC}}}{\Delta\nu_D(r)} \frac{1}{1 + \tau_{\text{Ly}\alpha}(r)},$$

where  $\Delta\nu_D = \nu_0 \sqrt{2k_B T / \mu} / c$  is the local Doppler width,  $\tau_{\text{Ly}\alpha}(r)$  is the *top-down* line-center Ly $\alpha$  optical depth integrated from the outside in through the HI column, and

$$F_{\text{LyC}} = \xi \frac{10^{L_{\text{EUV}}}}{4\pi a^2} \left(1 - e^{-\sigma_{\text{LyC}} N_{\text{HI}}}\right)$$

is the deposited Lyman-continuum flux (the incident EUV flux times the dayside factor  $\xi$  from 2D approximate method and the absorbed fraction set by the neutral-H column  $N_{\text{HI}}$ ). This needs *no* external file: it is a self-contained, approximate “staging” field, adequate to get the coupling and signs right but not a true radiative-transfer solution.

(mode 1) External RT file: `Jlya RT file: <path>`. Read a tabulated  $\bar{J}_{\text{Ly}\alpha}(r)$  profile (two columns,  $r/R_p$  and  $\bar{J}$  in cgs) produced by a real  $\text{Ly}\alpha$  radiative-transfer calculation (e.g. a Monte Carlo run); it is interpolated onto the grid and used directly.

(mode 2) In-line escape probability: `Jlya escape-prob: True`. Compute  $\bar{J}_{\text{Ly}\alpha}$  every step from the current state by escape-probability radiative transfer: a trapped *internal* field (recombination + collisional sources closed by the escape probability  $\beta$ ) plus a trapped *stellar beam* set by `Stellar Lya flux` (with the penetration controlled by `Lya stellar halfwidth` and the buildup capped by `Lya stellar boost`).  $\beta = 1/(1 + \langle N \rangle)$  with  $\langle N \rangle$  the mean number of scatterings before escape of the static plane-parallel damping-wing slab (Neufeld 1990, ApJ 350, 216, eq. 3.27 at zero continuum destruction; for a mid-plane source, Harrington 1973, MNRAS 162, 43, eq. 40,  $\langle N \rangle = 0.909316\tau$ ), so  $\beta$  goes as  $1/\tau$  in the wings and tends to one when the column thins. By default the lower boundary reflects, which makes the emitting plane a symmetry plane; `Lya absorbing bottom: True` instead puts the planet-ward face of that slab at the bottom of the domain, where the  $\text{H}_2$  layer beneath the base absorbs  $\text{Ly}\alpha$  (a pure absorber, as in the Huang et al. 2017 Monte Carlo), which shortens the random walk and lowers  $\bar{J}_{\text{Ly}\alpha}$  in the bottom few scale heights only. This is the most physical of the three but still an approximation; see `Update_EXHALE.tex` §12 for the full formulae.

`EXHALE_transit.py` uses the *same* logic for its  $\text{H}\alpha/\text{H}\beta$   $n=2$  population: if a `Jlya_file` is set it is read (mode 1), otherwise the mode-0 parameterized estimate above is used (with  $F_{\text{LyC}}$  rebuilt from the run's  $\text{H I}$  column and the stellar EUV). So a converged run with no  $\text{Ly}\alpha$  input still yields  $\text{H}\alpha/\text{H}\beta$  predictions, via the default field.

### 3.5 The photon grid and the spectrum that fills it

Neutral metals have ionization thresholds *far below* the hydrogen Lyman limit ( $13.6 \text{ eV} \equiv 912 \text{ \AA}$ ): the lowest is K I at  $4.34 \text{ eV}$  ( $2856 \text{ \AA}$ ), followed by Na I, Ca I, Mg I, Fe I, Si I, S I, and C I (Table 4). So does the  $\text{He } 2^3\text{S}$  metastable ( $e_{\text{th,HeTR}} = 4.768 \text{ eV} = 2600 \text{ \AA}$ ) and, whenever the excited-hydrogen coupling is armed, hydrogen in  $n = 2$  ( $e_{\text{th,H I}}/4 = 3.400 \text{ eV} = 3647 \text{ \AA}$ , the head of the Balmer continuum). To photoionize any of them the photon grid has to reach their thresholds, and the spectrum has to state a flux there: the EUV/X-ray band alone ( $\lambda < 912 \text{ \AA}$ ) leaves every neutral metal un-ionized and artificially abundant.

**One spectrum type builds every band.** `Spectrum type`: states the spectrum of the *whole* photon grid, the part below  $13.6 \text{ eV}$  and the Balmer continuum included. No band is filled from a type the input did not select.

**Power-law.** The power law  $E^{\text{PLind}}$ , normalized on  $[E_{\text{low}}, E_{\text{mid}}]$  and  $[E_{\text{mid}}, E_{\text{high}}]$  from  $L_{\text{EUV}}$  and  $L_X$ , evaluated wherever the grid reaches. Below  $E_{\text{low}}$  that is an *extrapolation* of an EUV fit, which is what the type means: real stars do not follow it into the near-ultraviolet (their flux rises steeply there). Treat power-law metal and  $\text{He } 2^3\text{S}$  photoionization as indicative.

**Planck.** The photospheric blackbody  $\pi B_\nu(T_{\text{eff}})(R_\star/a)^2$ , from `Stellar Teff` [K] and `Stellar radius` [ $R_{\text{sun}}$ ], on the whole grid. Startup refuses the type without both. It is an upper bound in the near-ultraviolet, where line blanketing and the Balmer jump depress a real F or G photosphere below the curve, and it is orders of magnitude

*below* the chromospheric and coronal EUV of an active star: a run that needs both bands right needs a measured spectrum.

Load (Spectrum file:). The tabulated SED, on the whole grid. The file's own rows are the grid points, log-log interpolated where a band asks for a flux between them; no extrapolation.

Monochromatic. One energy, Photon energy [eV], and no grid at all ( $N_\lambda = 1$ ).

**The grid floor is the lowest active threshold, and a loaded table that does not reach it stops the run.** The floor (photon\_grid\_floor\_eV) is the lowest ionization threshold over every *active* absorber: the He 2<sup>3</sup>S metastable when the triplet is included, the neutral stage of every element metals.inp gives a nonzero abundance, and H(*n*=2) when Stellar Teff and Stellar radius arm the excited-hydrogen coupling. With none of them active the floor is the H I edge. The two analytic types are evaluated down to it, and a loaded table is read down to it.

A loaded table that *ends above* the band an active absorber needs stops the run at startup. There is no key to continue: a truncated table gives that absorber the photoionization rate of a truncation of the stated spectrum, not of the spectrum. The message names the file, the lowest photon in it, the uncovered band in eV and in Å, every absorber whose threshold lies below that lowest photon (H(*n*=2), the metastable, each active neutral metal), and the remedies: turn the absorber off (Include He23S? False; remove the element from metals.inp; remove Stellar Teff and Stellar radius, which is what arms the excited-hydrogen coupling), or supply an SED that reaches the stated wavelength.

**How the grid is partitioned (analytic types).** For Power-law and Planck the code builds the grid itself, from *bin edges*: e\_v holds the *geometric center* of each bin and de\_v its exact width, so the bins are contiguous and  $\sum de_v$  is exactly the width of the covered band. Six bands, each with its own budget of bins, put the resolution where the absorbers are rather than spreading it over the X-ray decades:

| Band                          | Bins | Present when                           |
|-------------------------------|------|----------------------------------------|
| $[e_{th,H I}/4, e_{abs,low}]$ | 20   | the excited-H coupling is armed        |
| $[e_{abs,low}, 13.598]$       | 89   | a metastable or low-IP metal is active |
| $[13.598, 24.587]$            | 50   | always                                 |
| $[24.587, 54.418]$            | 50   | always                                 |
| $[54.418, E_{mid}]$           | 50   | always                                 |
| $[E_{mid}, E_{high}]$         | 50   | always                                 |

$e_{abs,low}$  is the lowest threshold of an absorber of the *bulk gas* (the metastable or a low-IP metal), so the first band is the one no active absorber has a cross section in; its 20 bins only have to state it, because the H(*n*=2) photoionization rate is an integral over its own band and not a sum over these bins. The 89 bins of the second give it the bin ratio of the pure-H I band ( $\Delta \ln E = 0.0118$ ), which is what the neutral-metal cross sections need: every metal photoionization rate the grid forms is then within  $5 \times 10^{-4}$  of the exact integral.

**Every ionization threshold of an active absorber is a bin edge.** A photoionization rate is the rectangle-rule sum  $\sum F\sigma/E de_v$ , so a bin *containing* a threshold would charge that absorber's cross section, read at the bin center, over the part of the bin lying below the threshold, where it cannot absorb at all. Each band is therefore cut at every threshold

strictly inside it: H I, He I, He II, the metastable, H<sub>2</sub> and its dissociative- and double-ionization channel thresholds when the molecular chemistry is on, and every photo-ionizable stage of an element with a nonzero abundance, and the band's budget is shared over the resulting sub-intervals in proportion to their logarithmic widths, at least one bin each, so the budget is met exactly. A band with no threshold inside it keeps its budget in one piece.

The total is therefore  $N_\lambda = 200 + N_{\lambda, \text{TR}}$ , where  $N_{\lambda, \text{TR}}$  counts the bins below the H I edge: 109 for a run with both sub-13.6 eV bands ( $N_\lambda = 309$ ; He 2<sup>3</sup>S or metals plus an armed excited-H coupling), 89 without the  $n = 2$  band ( $N_\lambda = 289$ ), and 0 for a pure H/He run with no absorber below the edge ( $N_\lambda = 200$ ). The startup report states the type, what the band below 13.6 eV was built from, the grid floor, and the integrated grid flux  $\sum F \Delta E$  against the nominal  $(10^{L_X} + 10^{L_{\text{EUV}}})/(4\pi a^2)$ .

The energy bands set by [E\_low, E\_mid, E\_high] and the luminosities  $L_{\text{EUV}}/L_X$  still describe the *ionizing* (H/He-driving) part of the spectrum above 13.6 eV; the bands below it are in addition to them, which is why the integrated grid flux and the nominal flux need not agree.

Table 4: Neutral-metal photoionization thresholds (first ionization stage of each modeled element), as  $E_{\text{th}}$  and the corresponding wavelength  $\lambda_{\text{th}} = hc/E_{\text{th}}$ . A loaded SED must extend to at least the longest  $\lambda_{\text{th}}$  among the *active* metals, or the run stops (§3.5). The He 2<sup>3</sup>S metastable (4.768 eV, 2600 Å) and H( $n=2$ ) (3.400 eV, 3647 Å) floor the grid the same way. Values are the VFKY96/NIST thresholds used in the code (species\_table.f90).

| Species | $E_{\text{th}}$ [eV] | $\lambda_{\text{th}}$ [Å] | Note                                          |
|---------|----------------------|---------------------------|-----------------------------------------------|
| K I     | 4.34                 | 2856                      | lowest, sets the grid floor when K is present |
| Na I    | 5.14                 | 2413                      |                                               |
| Ca I    | 6.11                 | 2028                      |                                               |
| Mg I    | 7.65                 | 1622                      |                                               |
| Fe I    | 7.90                 | 1569                      |                                               |
| Si I    | 8.15                 | 1521                      |                                               |
| S I     | 10.36                | 1197                      |                                               |
| Cl I    | 11.26                | 1101                      |                                               |
| (H I)   | 13.60                | 912                       | Lyman limit (reference)                       |
| O I     | 13.62                | 910                       | $\simeq$ Lyman limit                          |
| NI      | 14.53                | 853                       | above the limit (already in the EUV band)     |

### 3.6 Diffusive separation of helium and metals

By default EXHALE is single-fluid: the composition is re-solved each step by *local* ionization equilibrium, which conserves the element ratios, so  $n_{\text{He}}/n_{\text{H}}$  (and each metal/H) stays frozen at the input value at every radius. Setting He\_diffusion: True adds a genuine transport equation for the helium *mass fraction*  $X = \rho_{\text{He}}/\rho$  of the two-component H/He gas: advection at the bulk velocity plus a molecular-diffusion *settling* drift of He relative to H,

$$\Phi_{\text{He}} = -\rho(D + K_{zz}) \frac{\partial X}{\partial r} - \rho X(1-X)DG, \quad G = \frac{\Delta m_{\text{eff}} m_{\text{H}} g}{kT},$$

with eddy diffusion  $K_{zz}$  (He\_Kzz, default 0) acting on the gradient term only. The binary coefficient  $D$  is **stage-resolved**: each element is split into its ionization stages and the friction is built pair by pair, hard-sphere for neutral–neutral pairs (Banks & Kockarts 1973), a combined polarization + hard-sphere form for non-resonant ion–neutral pairs, and Coulomb for ion–ion pairs, then mixed over the stage fractions (Blanc’s law,  $1/D_{\text{eff}} = \sum_{s,t} y_s y_t / D_{st}$ ). This matters above the ionization front, where Coulomb friction suppresses the separation by orders of magnitude relative to the neutral hard sphere and keeps the metal mixing ratios flat in the wind. Helium settles out relative to the escaping hydrogen and *He/H declines with altitude*, as in Taylor et al. (2025) and Xing et al. (2023). The equation is solved conservatively (implicit, tridiagonal) and the new split is written back into the composition conserving mass. Physics refinements: He\_ambipolar (default on) reduces the settling in the ionized wind via the polarization field, computed from the electron-pressure gradient; He\_alphaT adds thermal diffusion; He\_metal\_diffusion extends the same operator to each trace metal (heavier elements settle more, so they deplete faster than He).

The strength of the separation scales as (settling speed)/(wind speed): it is suppressed by fast escape and is stronger for heavier species. On a gentle escaper (HD 209458b) helium itself separates and the He 10830 line weakens; on a furious escaper (WASP-121b) helium is dragged out almost unfractionated while the heavy metals still settle strongly. Full physics, derivation, and a two-planet before/after comparison: docs/design\_hehe\_diffusion.md and docs/version\_compare.{md,tex,pdf}; the binary formulation and the stage-resolved coefficient are derived and measured in docs/binary\_diffusion\_design.md and docs/Update\_EXHALE.{md,pdf} §68 and §71. The flags are **default off**, so standard runs are byte-identical to before.

### 3.7 Lower-atmosphere connection (Tier 1–3)

Three opt-in layers connect EXHALE’s 1- $\mu$ bar base to the atmosphere below (survey, implementation, examples and figures: docs/lower\_atmosphere\_coupling.pdf). All are **default off**.

**Tier 1: analytic lower column (consistency report).** Add

Lower column: 1.36 # radius of the 1-bar level [R\_J], e.g. transit radius

and EXHALE integrates the isothermal- $T_{\text{eq}}$  hypsometric column with the chemical-equilibrium  $\text{H}_2/\text{H}/\text{He}$  partition (Koskinen et al. 2022) from 1 bar to 1  $\mu$ bar and *reports* the derived base radius (equilibrium value + a fully-atomic bracket), the base  $\text{H}_2/\text{H}/\text{He}$  fractions and  $\mu$ , next to the input “Planet radius”. Nothing is overridden. A large base  $q_{\text{H}_2}$  on a genuinely cool planet signals that the molecular physics below is required.

**Tier 2a: molecular base (EOS only).** Molecular base: True removes the  $\text{H}_2$ -bound particles from the base particle budget (ntot\_bc) using the same equilibrium fit: a lower base pressure and heavier base  $\mu$ , with the species arrays still atomic. Crude on its own, and it is what Tier 2 turns on for itself, so the key is worth setting by hand only for a run that wants the base EOS without the chemistry.

**Tier 2: molecular chemistry.**

Molecular chemistry: True #  $\text{H}_2$ ,  $\text{H}_2^+$ ,  $\text{H}_3^+$ ,  $\text{HeH}^+$  in the coupled equilibrium

adds the four molecular species to the coupled ionization equilibrium (the Koskinen et al. 2022 reaction network), the  $\text{H}_2$  photoionization opacity and heating (Yan et al. 1998 cross section), and the optically-thin  $\text{H}_3^+$  infrared cooling (Miller et al. 2013 with their non-LTE factor). The output `Ion_species*.txt` gains four columns ( $\text{H}_2$   $\text{H}_2\text{p}$   $\text{H}_3\text{p}$   $\text{HeHp}$ ). It also turns `Molecular` base on automatically, so that key need not be given: with an atomic `ntot_bc` under a molecular base the base temperature is inflated by the particle-count ratio ( $\sim 1.8\times$  for a fully molecular base) and thermally dissociates the layer the chemistry just built, so the parser couples the two rather than leaving the inconsistent combination reachable. Requires `He`. `He_diffusion` is no longer refused beside it (milestone M4, 2026-08-26): the element transport of `binary_element_diffusion` closes over the molecular carriers, and the `mol_diffusion` regression case runs the combination. A `metals.inp` may be present: the trace metals are then solved in the same system as the molecular network (`System_HeH_mol_metals`), because the two blocks share the free electron density and the metals are the dominant electron donors in the shielded molecular base.  $\text{H}_2$  photodissociation in the Lyman–Werner bands is available and opt-in: add `Stellar LW flux [erg/cm2/s]`: with the band flux at the planet. Caveats: local *equilibrium* chemistry only (no molecular advection),  $\gamma = 5/3$  retained. On hot Jupiters the result is a thin molecular base under a sharp  $\text{H}_2 \rightarrow \text{H}$  front with an atomic wind above: the atomic assumption becomes a result rather than an input.

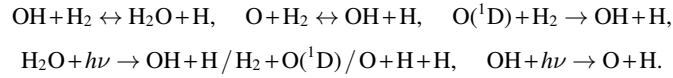
**Converging a molecular run takes three keys an atomic run does not need:** Solver: Newton  $5.0\text{e-}2$ , Resid tol:  $1.0\text{e-}7$  and Max steps: 150000. The *du* descent of a molecular run is not monotonic, so the default  $10^{-2}$  hand-off threshold can consume the whole step budget; and  $\|R\|$  is set by the two or three cells just above the base, so a run stopped at the default Resid tol:  $1.0\text{e-}5$  leaves the molecular layer still cooling (its energy residual there is still  $0.7\times$  the local  $\text{H}_3^+$  cooling rate). Marching alone never reaches that layer (its  $\text{H}_3^+$  cooling time is  $\sim 10^7$  CFL steps), so the Newton finish is not optional. The converged layer is much colder than the relaxation snapshot ( $\sim 400$  K on HD 209458 b, 190–300 K on a hot Uranus) and, without `Base IR field`, that collapse is the model’s own steady state rather than a prediction; see docs/lower\_atmosphere\_coupling.pdf §7–§9 and item (G) of `T0_BE_DONE.md`.

### Tier 2b: oxygen chemistry.

```
Oxygen chemistry: True           # OH, H2O, CO in the same coupled system
Molecular carrier transport: True # the default: carriers transported
Stellar LW flux [erg/cm2/s]:      1248.5 # 912–1201 Å at the planet (LW)
Stellar Lyα flux [erg/cm2/s]:     14957.2 # 1215.67 Å, the Lyα line (B2)
Stellar FUV B3 flux [erg/cm2/s]:  1374.1 # 1231–1450 Å (B3)
Stellar FUV B4 flux [erg/cm2/s]:  46294.8 # 1451–2304 Å (B4)
```

With `Spectrum type`: Load the three continuum lines (LW, B3, B4) may be omitted: each band is then integrated from the spectrum file over its own interval, a trapezoid over the file’s rows (the file is already at the planet), and the setup report names the source of every band. Only the  $\text{Ly}\alpha$  line is always a key, because a line flux reconstructed from observations is a better number than a trapezoid over the file’s rows across 1202–1230 Å. A stated value, zero included, always wins over the file. Tier 2 can build a molecular base but not *decide* how molecular it is: the  $\text{H}_2/\text{H}$  partition is either imposed through the `q_H2_base` handoff or taken from a chemical-equilibrium fit, neither of which is photochemistry. Tier 2b is the layer that computes it, by adding the oxygen carriers that the catalytic cycle runs

on. Measured on a photochemical model of the HD 189733 b base, the oxygen family carries 96–99.6% of the net  $\text{H}_2$  destruction there, and the whole of it sits in three reactions and four photolysis channels:



Every coefficient is transcribed from a named evaluation with its validity range at the code site, and the reverse of each two-body pair is generated by detailed balance against a NIST-JANAF Shomate table rather than transcribed independently: the pairs run close to cancellation on a hot base, where an inconsistent pair gives an arbitrary net rather than a small error. The audit behind the numbers is `docs/a2_reaction_audit.md`; the design is `docs/a2_oxygen_option_design.md`.

**$\text{O}(^1\text{D})$  is closed by a local steady state**, not carried as a transported species: its only sink in this set is  $\text{O}(^1\text{D}) + \text{H}_2 \rightarrow \text{OH} + \text{H}$ , and that gives it a chemical lifetime of  $1.6 \times 10^{-3}$  s at the HD 189733 b base (shorter than every other time scale in the problem by many decades), so the steady state is the accurate treatment. Its density is reported as a diagnostic column.

**CO is an oxygen reservoir, not a reactant.** It locks 45–46% of the oxygen at every measured level, so a network that gave the whole oxygen abundance to the water family would over-supply the OH cycle by about a factor two. It has no chemical source in this network: with `Molecular carrier transport: False` it sits at its own  $\text{CO} \leftrightarrow \text{C} + \text{O}$  chemical equilibrium, which keeps it associated below  $\sim 3000$  K and dissociates it by  $\sim 4000$  K, above the  $\text{H}_2 \rightarrow \text{H}$  front and below the wind, so the carbon and the oxygen of the wind stay atomic. With the transport on it is a transported reservoir, and “inert” would then mean *indestructible*: measured before the ceiling below was added, the wind carried CO out to  $1.67 R_p$  and  $2 \times 10^4$  K, where it held 55% of the oxygen and switched off the O I and C II cooling of the whole wind. The transported CO is therefore capped at the chemical equilibrium of its own  $(n, T)$ : a one-sided constraint, inactive through the molecular layer and exact far above the turnover, which over-suppresses *quenched* CO in the narrow interval around 3000–4000 K where the dissociation is neither fast nor negligible. **Neither limit is kinetics** (the truth is quenching, and the audited set has no CO rate to give it), so a run whose *outer* wind is cool can hand the transit tool an O I column short by whatever fraction is still molecular. Every run therefore prints the molecular oxygen fraction at the top of its domain, and the number of cells the ceiling touched.

**The FUV arrives as four bands with named fluxes**, not as part of the photon energy grid: that grid is floored by the ionization thresholds of the run’s active absorbers (§3.5) and carries their opacity, not the  $\text{H}_2$  line system or the  $\text{H}_2\text{O}$  and OH continua, and a power-law type below 13.6 eV gives only an XUV fit extrapolated downward rather than a stellar FUV spectrum. The edges are fixed by the  $\text{H}_2\text{O}$  branching ratios, not chosen: the three-body channel  $\text{H}_2\text{O} + h\nu \rightarrow \text{O} + \text{H} + \text{H}$  opens only in the interval containing  $\text{Ly}\alpha$ , where it is 12%.  $\text{H}_2\text{O}$  and OH absorb these bands as *continua*, so the attenuation is  $\exp(-\tau)$  on their star-ward columns and no self-shielding function is needed (unlike the  $\text{H}_2$  Lyman–Werner band).

**One interval, one flux, one beam.** Over 912–1201 Å the same photons are absorbed by  $\text{H}_2$  in the Lyman and Werner lines and by  $\text{H}_2\text{O}$  and OH in a continuum. That interval is therefore band LW, entered once through `Stellar LW flux`: supplying it twice would count its energy twice on the way in, and letting each absorber attenuate a private copy of the beam would count it twice on the way down. Both are closed. The  $\text{H}_2$  rate is multiplied by  $\exp(-\tau_{\text{cont}})$ , the  $\text{H}_2\text{O} + \text{OH}$  continuum term of Draine & Bertoldi (1996) eq. (40) (identically 1 for a run without this option), and the  $\text{H}_2\text{O}$  and OH rates in that band are multiplied by  $1 - A$ , the fraction of the band the  $\text{H}_2$  lines have *not* removed.  $A$  is the

column integral of the *pump* cross section  $\sigma_{\text{diss}}/p_{\text{eff}}$  of the same level-resolved table the  $\text{H}_2$  rate comes from, so the photons the beam loses to the lines and the photons the dissociation rate spends are one number:  $p_{\text{eff}}$  is by definition the dissociations per photon removed from the beam.  $A$  cannot exceed 1, and measured over the whole table it reaches 0.69–1.00 at the top of its column axis. Until 2026-09-06  $A$  was DB96 eq. (39), an equivalent width of the narrower 912–1110 Å band, and the two normalizations disagreed by 45 per cent. The split is exact cell by cell, so what the three absorbers take is what the beam loses, and output/FUV\_bands.txt reports that ledger. A run with the oxygen chemistry on, no spectrum file and Stellar LW flux left at 0 is warned that it gets no  $\text{H}_2\text{O}$  or OH photolysis over 912–1201 Å, where those cross sections peak; with a spectrum file that band, and B3 and B4, are integrated from it unless stated.

**Three limitations to read before quoting a result.** (i) The band average assumes a flat  $F_\lambda$  inside each band. Measured against a real stellar spectrum (the HD 189733 b flux at the planet), that is good to 3% for  $\text{H}_2\text{O}$  and to 18–22% for OH on both halves of LW, and to about 35% on B3, but **off by a factor 4.6–6.1 on B4**, whose cross section falls three decades across the band; a run in which B4 matters is outside the treatment. Inside LW the sensitivity is not uniform: nearly all of it sits shortward of 1110 Å, where  $\text{H}_2\text{O}$  and OH move in *opposite* directions (–60% and +89% for a 3000 K blackbody), while the 1110–1201 Å half is the best-behaved interval of the set for  $\text{H}_2\text{O}$  ( $\leq 2\%$ ). One band average cannot show that; the two half-band numbers of water\_photolysis.f90 can. (ii) The Ly $\alpha$  band uses the *incident* stellar flux, attenuated only by the  $\text{H}_2\text{O}$  and OH continuum: the H I resonance scattering that actually decides how much Ly $\alpha$  reaches a molecular base is not applied, so that band’s rate is an upper bound. Note also that setting Stellar Ly $\alpha$  flux for the H( $n=2$ ) pumping therefore drives the photolysis as well; the startup report prints all four band fluxes so this is visible. (iii) The molecular ions  $\text{H}_2^+$ ,  $\text{H}_3^+$  and  $\text{HeH}^+$  and  $\text{O}(^1D)$  are closed by their local balance even with the transport on, and the band average applies one transmission to a whole band, so the rates are lower bounds deep in the water layer. Every run writes its own chemical, advective and diffusive time scales (output/Oxygen\_chemistry.txt), which say where the answer is chemistry, where it is flow, and where the two are comparable.

**The carriers are transported by default** (Molecular carrier transport: True).  $\text{H}_2$ , OH,  $\text{H}_2\text{O}$  and CO are advanced by an implicit backward-Euler diffusion-advection step solved together with the same chemistry rows the local equilibrium uses, so a rate that changes in one changes in the other. The molecular coefficients come from Blanc’s law over the background carriers with the rigid-sphere pair coefficients of Banks & Kockarts (1973) that the element-diffusion operator already carries; the eddy coefficient is read from He\_Kzz or from a lower-atmosphere profile and from nowhere else; settling uses the same  $g$  as the hydrodynamics, with no thermal diffusion. Each cell’s *element* totals are unchanged by the step (only the partition among an element’s carriers moves), so element conservation holds by construction and  $\rho$  is untouched; every positivity or element limiter that fires is counted and reported in the trailer of output/Oxygen\_chemistry.txt. This operator therefore transports *speciation* and not elements, which is why it cannot be run together with He\_metal\_diffusion: that option transports elements, and CO carries an oxygen and a carbon nucleus and so cannot follow two element factors at once. Molecular carrier transport: False restores the earlier local steady state; it isolates the chemistry for testing and is not a model of a base, because at the HD 189733 b 1  $\mu\text{bar}$  level the  $\text{H}_2$  chemical and flow times are comparable. With  $K_{zz} = 0$  and no profile the transport is pure molecular diffusion (the wrong limit for a lower atmosphere, where eddy mixing is what holds the composition well mixed below the homopause), and the run says so at startup rather than refusing.

**The option removes coolants it does not replace.** Where the oxygen is in  $\text{H}_2\text{O}$  and CO the [O I] 63/145/44  $\mu\text{m}$  fine-structure lines and the C I/C II line cooling correctly stop (those atoms are not there), but EXHALE carries no  $\text{H}_2\text{O}$  or CO infrared bands to put in their place, and  $\text{H}_3^+$  is its only molecular coolant. Measured on the hot-Uranus molecular gate, the layer just above the base cools  $15\times$  more slowly with the option on than with it off, at comparable heating. That is item (G) of TO\_BE\_DONE.md arriving through the composition instead of through the temperature, and it is why the resulting molecular-layer temperature is not a prediction. Every oxygen-chemistry run reports how much of its oxygen and carbon is molecular, where, and that those nuclei no longer cool.

The option is **default off** and refuses combinations that would give one quantity two owners: `q_H2_base` (it computes that partition), and `He_metal_diffusion` (the trace-metal transport counts each metal element over its ion stages alone, so it would transport an oxygen reservoir missing everything bound into OH,  $\text{H}_2\text{O}$  and CO (about half the element), and moving the carriers with it needs a rule for CO, which carries an oxygen *and* a carbon nucleus). `He_diffusion`: True on its own is accepted: the H/He element operator scales every H-bearing species by the same factor, so the molecular carriers already move with hydrogen there. A lower-atmosphere *profile* is accepted beside it: the profile owns the region below the matching level and the chemistry the region above it.

**Tier 3: base.inp handoff (and VULCAN).** An optional `base.inp` in the run directory overrides the base state before the derived constants are formed (each override is echoed; absent file = no-op):

|                        |         |           |                                                           |
|------------------------|---------|-----------|-----------------------------------------------------------|
| <code>T_base</code>    | 1140.0  | # [K]     | -> overrides <code>T0</code>                              |
| <code>r_base</code>    | 1.402   | # [R_J]   | -> overrides "Planet radius" (1-ubar radius)              |
| <code>p_base</code>    | 1.0e-6  | # [bar]   | -> the level every other value refers to                  |
| <code>q_H2_base</code> | 0.42    | #         | -> H2 volume mixing ratio at the base                     |
| <code>HeH_base</code>  | 0.083   | #         | -> overrides the He/H number ratio                        |
| <code>O_H_base</code>  | 4.90e-4 | #         | -> O/H nuclei ratio, overrides <code>metals.inp</code>    |
| <code>Kzz_base</code>  | 1.0e9   | # [cm2/s] | -> sets <code>He_Kzz</code> (element + carrier transport) |

Keys are matched as labels, by the same anchored rule the `input.inp` reader uses, and every key carries a **category** that states what the value is allowed to do to the wind. There are five, and the same table is carried by `read_base.inp`'s header comment and by `docs/input_schema.md §2c`:

| Key                               | Category            | What it sets, and where the value is consumed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (comments only)                   | provenance          | Nothing. Which code, network and profile produced the file. The one comment that is parsed is <code># solution_id</code> , and only beside a profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| T_base, r_base, p_base, q_H2_base | EOS boundary        | T0, R0, p_base_bar, q_h2_base: the state of the gas at the matching level. p_base also <i>fixes</i> that level:<br>$n_0 = p_{\text{base}} / (k_B T_0 n_{\text{tot,bc}})$ , so Log10 lower boundary number density is redundant beside it and is refused if it disagrees by more than 1%. q_H2_base replaces the chemical-equilibrium fit in <code>comp_ntot_bc</code> , i.e. it sets the base <i>particle count</i> , and it is also imposed on the H <sub>2</sub> partition of the inflowing ghost species, the particle count then coming from that same species state. It is refused above the attainable ceiling $0.5 / (0.5 + \text{He}/\text{H})$ rather than capped. |
| HeH_base, <El>_H_base             | elemental reservoir | HeH, and X_<El> hence <code>melem_ab</code> : the nuclei ratio of an element to hydrogen, which the wind then transports.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| (none today)                      | initial guess       | A key here would seed a profile the solver is free to move away from. None exists yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Kzz_base                          | boundary constraint | <code>he_kzz</code> : the eddy coefficient the element-diffusion operator imposes at the base; inert with <code>He_diffusion</code> off.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| q_H2O, q_CO, ...                  | diagnostic          | Nothing; metadata written as comments by the generators.<br>Unknown keys are ignored.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

The distinction the categories draw is a real one, and q\_H2\_base is where it bites: it is an **EOS anchor**, not a composition pin. Nothing holds H<sub>2</sub> at that value, no H<sub>2</sub> profile is seeded from it, and the molecular network moves away from it in the first cell.

<El>\_H\_base is the elemental form: C\_H\_base, N\_H\_base, O\_H\_base, S\_H\_base and the same spelling for Mg Si Ca Na K Fe (any of the ten element symbols of the species table), each the El/H *nuclei* ratio at the handoff level. It overrides `metals.inp` for that element, it turns the metal system on by itself when it is the only nonzero abundance, and on a restart (`Load IC? True`) it renormalizes that element's whole loaded column onto the stated ratio, preserving the ionization split and the radial shape. An element that reached the run from `metals.inp` alone is loaded untouched. With a lower-atmosphere profile in use (§3.8) the EOS boundary, elemental reservoir and boundary constraint keys of this file are **refused** with an error `stop` naming the key and its category: the profile states all of them at the matching level, and a scalar beside it would be the same quantity said twice. Provenance comments and diagnostic keys remain allowed.

The pre-step can be invoked *by EXHALE itself*: with

```
Lower atmosphere: vulcan 1.36      # or: analytic 1.36  (arg = 1-bar radius R_J)
```

in `input.inp`, the code runs the VULCAN photochemistry (or the analytic column) at startup when no `base.inp` exists, then proceeds with the wind solve on the generated base, subroutine-style, with the first VULCAN run taking hours

and later runs reusing the cached result in `<run_dir>/vulcan_work/`. Omitting the key disables the pre-step entirely. The VULCAN/ tree (obtained by `setup_vulcan.sh`; it ships FastChem inside it) carries its own citation requirements, see §3.9. Two generators are provided in `src/utis/`:

- `run_lower.py <run_dir> --r1bar <R_J> [--guillot]`: the analytic column (isothermal  $T_{\text{eq}}$ , or the Guillot 2010 semi-grey  $T(p)$ ); pure Python, no dependencies beyond numpy.
- `vulcan_to_base.py <out.vul> <run_dir> --mp <M_J> --r1bar <R_J>`: converts a **VULCAN** photochemical-kinetics output into the same `base.inp`, with the *photochemical*  $\text{H}_2/\text{H}$  dissociation state (on HD 189733 b photochemistry dissociates  $\sim 11\times$  more H at 1  $\mu\text{bar}$  than chemical equilibrium) and a hypsometric base radius using VULCAN's own  $\mu(p)$  and  $T(p)$ .

*Running VULCAN* is handled automatically by the driver `src/utis/vulcan_driver.py`, so no manual VULCAN setup is required. When the pre-step is requested it (i) obtains VULCAN if needed (via `src/utis/setup_vulcan.sh`; see §3.9) and copies the EXHALE/VULCAN/ tree into `<run_dir>/vulcan_work/` (once, then reused), (ii) builds a Guillot (2010) semi-grey  $T(p)/K_{\text{zz}}$  atmosphere for the planet and selects a stellar UV spectrum from the shipped spectra by the host  $T_{\text{eff}}$  ( $> 6800\text{ K} \rightarrow 51\text{ Eri}$ ;  $5300\text{--}6800\text{ K} \rightarrow \text{solar}$ ;  $4300\text{--}5300\text{ K} \rightarrow \epsilon\text{ Eri}$ ; cooler  $\rightarrow \text{GJ 436}$ ), (iii) compiles FastChem if its binary is missing, (iv) runs VULCAN to steady state, and (v) converts the result to `base.inp`. The first run takes hours; the converged `<name>-photo.vul` is cached and reused on subsequent runs (regenerate with `--force`). The code root is taken from EXHALE\_ROOT when set (for relocated installs). VULCAN's networks are H/C/N/O/S: it supplies the  $\text{H}_2/\text{H}$  state and molecular mixing ratios, *not* the atomic-metal release (Na/Mg/Ca/Fe): `metals.inp` stays user-supplied.

### 3.8 The lower atmosphere as a profile (Lower atmosphere profile:)

The Tier-3 handoff of §3.7 states the lower atmosphere at one level, as a handful of scalars in `base.inp`. A *profile* states it over an interval of pressure instead: the lower model (photochemistry, with or without a climate solution) is solved on a column that reaches *above* the level where EXHALE places its base, and the whole overlap is handed over as one table. Point EXHALE at it with

Lower atmosphere profile: `lower_atmosphere_profile.dat`

in `input.inp`; the file lives in the run directory and the name is free. Two things ask for the interval rather than the level. The first is eddy mixing: the homopause is where  $K_{\text{zz}}$  crosses the molecular diffusion coefficient, which rises by nearly five decades between the base and  $1.2R_p$ , so a single `He_Kzz` cannot state where it lies (§3.6). The second is the elemental-flux closure below, in which the composition at the base stops being an input and becomes an output.

**Schema.** A block of # header lines of the form `# key value`, then a fixed-column table in full double precision running **deep to shallow**: the pressure column must be strictly decreasing, and at least two levels must be present. Three header keys and one schema line are required:

- `solution_id`: a sha256 fingerprint of the lower model's configuration (code and version, mechanism and thermodynamic data, stellar flux and dilution, planet mass and reference radius, elemental abundances, matching pressure, trial fluxes). It is what makes “the same lower-atmosphere solution” a checkable statement.
- `p_match_bar`: the matching pressure, i.e. where EXHALE places its base. It must lie inside the pressures the table covers.
- `p_top_bar`: the shallowest level carried. It must be smaller than `p_match_bar` (otherwise the two models never overlap) and must not lie above the shallowest tabulated level.
- `# columns::` the names of the columns, in file order. Every column is found *by name*, because the producers order their element list differently from run to run.

The remaining header keys are optional and are provenance or metadata: `source_code`, `source_version`, `mechanism`, `stellar_flux`, `p_deep_bar`, `trial_flux_H`, `trial_flux_He`, `iteration`, `reached_steady_state` and `notes`. A profile with no iteration index runs perfectly well; it is simply not drivable by the closure iteration, which the run reports as `lower_profile_iterable`. Header keys this version has no consumer for are ignored rather than refused. Nine columns are required: `p` (bar), `r` ( $R_J$ ), `T` (K), `n_tot` ( $\text{cm}^{-3}$ ), `rho` ( $\text{g cm}^{-3}$ ), `Kzz` ( $\text{cm}^2\text{s}^{-1}$ ), `q_H2`, `q_H` and `X_He`. Optional columns are the other elemental ratios `X_C`, `X_O`, `X_Mg`, ... (E/H *nuclei* ratios, summed over every carrier, for any element species\_table knows), the elemental fluxes `F_H` and `F_He` ( $\text{g s}^{-1}$ , outward positive), and any number of diagnostic mixing ratios such as `q_H2O`. Columns the reader has no consumer for are **kept, not dropped**, so a file survives a round trip through the tools. A data row that does not carry as many numbers as the schema line names stops the run.

Every quantity is interpolated **linearly in**  $\log p$  and is never extrapolated; a target that falls exactly on a tabulated level returns that level's value bit for bit, which is what lets a profile with a node at the matching pressure reproduce the scalar handoff exactly.  $K_{zz}$  is the one quantity taken as a profile rather than at a point: it is mapped onto the radial grid through the file's own radius column, and above the shallowest level carried each cell takes that level's value, the eddy coefficient being a lower-atmosphere property about which the file makes no statement above its top.

**What it supplies, and what `base.inp` may no longer say.** At the matching level the profile sets the base temperature  $T_0$  (column `T`), the base radius  $R_0$  (column `r`), the base pressure  $p_{\text{base}}$  (`p_match_bar`), the base  $\text{H}_2$  fraction (column `q_H2`), the He/H ratio (column `X_He`) and each elemental reservoir it carries, which overrides `metals.inp`. It also fixes the base *level*:  $n_0$  follows from  $p_{\text{match_bar}}$  through  $n_0 k_B T_0 n_{\text{tot,bc}} = p_{\text{base}}$ , the same relation a `base.inp` carrying `p_base` uses, because the composition,  $q_{\text{H}_2}$  and temperature the profile hands over all refer to that one level. A `Log10 lower boundary number density` line given beside a profile is then saying the same thing twice: the two must agree to 1%, and if they do not, startup stops and prints both numbers, the  $n_0$  each implies,  $n_{\text{tot,bc}}$ , and the two ways to fix it. The columns `n_tot` and `rho` are carried but not imposed: the base particle count and mass follow from  $T_0$ , He/H,  $q_{\text{H}_2}$  and the elemental ratios exactly as they do on the scalar path.

Because the profile is now the single source of those quantities, a `base.inp` sitting beside it may no longer state them. Three categories of key are **refused**, and each stops the run naming the offending key: the EOS boundary (`T_base`, `r_base`, `p_base`, `q_H2_base`), the elemental reservoirs (`HeH_base` and any `El_H_base`) and the boundary constraint

(Kzz\_base). Provenance comments and diagnostic keys are still allowed, and indeed a `base.inp` whose whole content is provenance is what the adapters write. Such a file must carry a `# solution_id` comment, and it must match the profile's: a `base.inp` beside a profile with no `solution_id` line, or with a different one, stops the run, because two files describing the same gas at the same level from two different lower-atmosphere solutions are not a model of anything. `He_Kzz`: is treated differently, because it belongs to `input.inp` and not to the handoff: it is not refused, but it is *inert* while a profile is in use (the eddy coefficient comes from the Kzz column), and the run prints a warning saying so. The resolved-configuration report records what was used: `lower_profile_present`, and with a profile in use the file name, the `solution_id`, the source, `p_match_bar`, `p_top_bar`, the trial fluxes, the iteration index and whether the profile is iterable.

**Producing a profile.** Three tools in `src/utils/` do this, sharing one command line and one schema module (`lower_profile_schema.py`, which also computes the `solution_id` and writes the paired provenance-only `base.inp`). `photochem_to_lower_profile.py` is the production path: it solves a gas-giant photochemistry model to steady state on the column and writes the handoff, with the temperature structure either prescribed (`--tp-file`) or solved for by the radiative-convective climate model (`--climate`), in which case the tropopause, and with it the water cold trap, is part of the solution rather than an input. `vulcan_to_lower_profile.py` converts a finished VULCAN `.vul` output into the same format; it is the cross-check, not a second production path, because VULCAN carries no climate model and so can never be the route away from a prescribed  $T(p)$ .

Both take the same statement of the eddy coefficient. By default the Kzz column is whatever  $K_{zz}(p)$  the underlying solution carried; `--kzz-const`  $K$  replaces it with one value at every level, and `--kzz-power`  $\alpha$ , with its amplitude `--kzz-ref`  $K_{\text{ref}}$  and the pressure that amplitude is stated at `--kzz-ref-bar`  $P_{\text{ref}}$  (default 1 bar), replaces it with the saturated gravity-wave form  $K_{zz}(p) = K_{\text{ref}}(p/P_{\text{ref}})^{-\alpha}$ :  $\alpha = 1/2$  being the Lindzen (1981) slope fitted by Parmentier et al. (2013), 0.4 that of Charnay et al. (2015). The two are mutually exclusive, `--kzz-power` without `--kzz-ref` is refused, and the power-law settings enter the `solution_id` only when they are in use, so stating neither leaves a run and its fingerprint exactly as they were. On the Photochem path a stated coefficient is the profile the chemistry is *solved* on and not only the one reported. There is no ceiling on the power law: the file's own top bounds it, and EXHALE gives every cell above the shallowest tabulated level that level's value, so the level the amplitude is anchored at decides what the wind inherits above the profile. Finally `element_flux_closure.py` drives the pair of codes to their fixed point: the trial elemental fluxes of the header are imposed on the chemistry's upper boundary, the wind's own elemental flux is measured face by face and reduced over a radial window, and the two are iterated (under-relaxed) until they agree, at which point the base composition is an output of the coupled problem instead of an input to it.

**See also.** `examples/17_lower_profile/` is a ready-to-run HD209458b case whose `make_example_profile.py` re-generates the synthetic column it ships (§6.3). The authoritative column-by-column schema is `docs/input_schema.md` section 2d, and the design of the handoff and of the closure is `docs/phase_e_flux_closure_design.md`.

### 3.9 Obtaining VULCAN and FastChem

The lower-atmosphere pre-step uses the **VULCAN** photochemical-kinetics code, which ships the **FastChem** equilibrium-chemistry code inside it. These are third-party open-source codes and are *not* committed to the EXHALE repository (VULCAN/ is git-ignored). Fetch and prepare them with

```
src/utils/setup_vulcan.sh
```

which clones VULCAN into EXHALE/VULCAN/, applies the two modifications EXHALE needs, builds FastChem, and removes stale run products. EXHALE also runs this automatically the first time the vulcan pre-step is used and VULCAN/ is missing. Obtain updates from, and *cite*, the original sources:

- **VULCAN**: [github.com/exoclimate/VULCAN](https://github.com/exoclimate/VULCAN) (mirror [github.com/shami-EEG/VULCAN](https://github.com/shami-EEG/VULCAN)). Cite Tsai et al. (2017, ApJS 228, 20) and Tsai et al. (2021, ApJ 923, 264).
- **FastChem**: [github.com/NewStrangeWorlds/FastChem](https://github.com/NewStrangeWorlds/FastChem) (shipped inside VULCAN). Cite Stock et al. (2018, MNRAS 479, 865), and Stock et al. (2022, MNRAS 517, 4070) for FastChem 2.

The modifications `setup_vulcan.sh` applies to the cloned VULCAN tree (i.e. the only changes needed if you set it up by hand):

| File (in the VULCAN clone)     | Modification                                                                                                                                                 |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>make_chem_funs.py</code> | add <code>encoding=None</code> to the <code>np.genfromtxt(vulcan_cfg.com_file, ...)</code> call so the element-conservation check does not crash on Python 3 |
| <code>vulcan_cfg.py</code>     | set <code>use_photo = True</code> and <code>use_live_plot = False</code> (this file is the driver's template)                                                |
| <code>output/*.vul</code>      | delete (run products; regenerated in <code>&lt;run_dir&gt;/vulcan_work/</code> )                                                                             |
| <code>fastchem_vulcan/</code>  | make to build the fastchem binary                                                                                                                            |

On the EXHALE side (in this repository) the pre-step adds `src/utils/setup_vulcan.sh`, `vulcan_driver.py` and `vulcan_to_base.py`, and the Lower atmosphere: key in `input_read.f90/parameters.f90`.

## 4. Output files

All profiles are written to `output/` as plain text, one row per radial cell, in *cgs* units. Two flavors exist: the equilibrium files (`*.txt`, written during the run) and the advection-corrected files (`*_adv.txt`, written once at the end by the post-processor, which re-solves the ionization balance including advection of the wind). The `*_adv.txt` files are the ones the analysis tools read.

**Caveat: the `_adv` profile at a “breathing” base.** The advection-corrected (`*_adv.txt`) *temperature and ionization* would otherwise develop a spurious spike / sawtooth in the dense lower atmosphere (typically  $r \lesssim 1.1\text{--}1.2R_p$  for strongly Roche-filling planets such as WASP-121b). Two compounding causes, both confined to the base, are responsible:

1. *A breathing, inflowing base (a 1-D limitation).* When the planet nearly fills its Roche lobe the 1-D wind is subsonic / only marginally launched at  $L_1$ , so the dense lower atmosphere does not settle into a clean monotonic outflow but into a slowly recirculating state with small *negative* (inflow) velocities and a stagnation point ( $v = 0$ ) near the wind base. This is intrinsic to the spherically-symmetric (1-D) geometry (removing it would require a 2-D/3-D model), and is not a numerical bug.
2. *A spurious hot root in the metal-cooled post-processor.* The post-processor re-solves the energy/ionization balance in each cell *including* advection, assuming an outflow (each cell takes the next-inner cell as its upwind state). In the dense base the non-monotonic metal line-cooling curve gives the energy equation a second, spurious *hot* root that the Newton (MINPACK) solver can land on; through the upwind coupling one bad cell then cascades outward, producing the sawtooth and an overshoot spike (post\_process\_adv.f90). A guard rejects solves that fall outside a band around the converged equilibrium temperature, but mild spikes *within* that band can survive.

The post-processor guards against cause 2 directly: wherever the flow is *not* a clean outflow ( $v \leq 0$ , the breathing inflow base) it skips the advection correction and keeps the converged equilibrium ionization and temperature (post\_process\_adv.f90). This removes the spurious spike and sawtooth at the source (it breaks the upwind cascade), while preserving the genuine few-percent `_adv` correction in the outflow above the stagnation point. Cause 1 (the breathing base itself) is intrinsic to the 1-D geometry and remains, but it is now represented there by the smooth, physical equilibrium solution. The shipped `_adv` profiles are therefore clean, and the temperature panel of Figure 6 plots the `_adv` profile directly. (Velocity and density are *identical* in the `eq` and `_adv` files: the post-processor corrects only the ionization and temperature.)

**Validity range of the advection correction.** The inflow condition above is one of four under which the post-processor keeps the converged equilibrium state instead of correcting it. A cell is left at equilibrium when

1.  $v \leq 0$  on either face (inflow): the upwind discretization has no upstream cell, and the residence time  $\Delta r/v$  is negative;
2.  $\text{Da} = (\Delta r/v)(\Gamma_{\text{HI}} + \alpha_{\text{HII}} n_e) > 100$ : the gas relaxes to local ionization equilibrium many times over while it crosses the cell, so the equilibrium solution already solves the advection ODE and the correction can only add integration error;
3.  $x_{\text{HII},\text{eq}} < 10^{-6}$ : the residuals carry the *neutral* fraction and report the ion density as  $(1 - x_{\text{HI}})n_h$ , so the ion fraction is resolved only to the solver's absolute tolerance  $\sqrt{\epsilon} = 1.5 \times 10^{-8}$ . In a shielded base with  $x_{\text{HII},\text{eq}} \sim 10^{-13}$  the returned value is numerical noise of arbitrary sign, which the upwind cascade would then carry outward as negative ion densities;
4. the mass flux through the cell changes by too large a fraction of itself. Every equation the post-processor solves is a steady equation integrated along the recorded flow, and the flow it integrates along carries a fractional change of the *face* mass flux across the cell,

$$m_j = |R_1(j)|/s_1(j),$$

the mass row of the state divided by the largest term that row itself holds. This is the same operator the stationary certification measures the mass row by (steady\_residual.f90, certification.f90), evaluated by calling those routines on the state handed in rather than by re-deriving it from centre-sampled  $\rho v r^2$ , which is the same quantity in the continuum and a different discrete test. The terms the steady equations drop are the ones the mass divergence puts into them, each of them  $m_j$  times a term they keep, so *the correction is first order in  $m_j$* : a cell with  $m_j$  at or below  $\text{adv\_conditional\_tol} = 10^{-2}$  is corrected, and its correction is a *conditional* one, accurate to that fraction of itself in the mass flux; a cell above it keeps the run's own temperature and the ionization equilibrium at that temperature. The value of  $m_j$  for every row is written out as the `adv_mass_row` column, so a reader has the condition of each row and not only its verdict.

The fraction above is *not* the tolerance of a certified stationary state. That is  $\text{cert\_tol\_mass} = 3 \times 10^{-12}$ , the number the stationary certification judges the mass row of the *whole* state by, and the verdict it reaches on the input state travels in the file's own `# adv_input_certified` line. The two are different statements about one measure and the file carries both: the rows of a certified state are corrected, while a corrected row is not thereby certified. Conditions 1 and 2 are properties of the flow, 3 of the representation of the unknown and 4 of the state the equations are integrated on, so none of them depends on whether metal cooling is active. Condition 4 refuses both the temperature and the composition of the cell, condition 1 both as well (the upwind difference has no upstream cell either way), and conditions 2 and 3 the composition alone. The energy solve additionally applies a *thermal* Damköhler condition of its own,  $\text{Da} = (\Delta r / v) |\Gamma - \Lambda| / u_{\text{th}} > 1$ : where the local radiative balance rather than the flow sets the temperature, the temperature to keep is the run's own, which is the root of that same balance with every channel the run solved. Conditions 1 to 3 trigger in the dense base; condition 4 follows the state, and it is the one that separates a converged wind from a snapshot. The counts are printed at the end of the run, and the verdict and the measure of every row are recorded in the `adv_T_status`, `adv_comp_status` and `adv_mass_row` columns of `Hydro_ioniz_adv.txt` (§4.1). Full account: `docs/postprocess_advection_validity.md`.

#### 4.1 Hydro\_ioniz.txt / Hydro\_ioniz\_adv.txt

Thermodynamic profiles (7 columns, and three more in the `_adv` file). The dimensionless code variables are restored to cgs with the normalizations  $n_0$  (base density),  $v_0 = \sqrt{k_B T_0 / \mu}$ ,  $p_0 = n_0 \mu v_0^2$ ,  $T_0$  (base temperature),  $q_0 = n_0 \mu v_0^3 / R_p$ .

| Col | Quantity              | Unit                                | Note                                                                |
|-----|-----------------------|-------------------------------------|---------------------------------------------------------------------|
| 1   | $r$                   | $R_p$                               | radius (base = 1)                                                   |
| 2   | $\rho$ (mass density) | $m_H \text{ cm}^{-3}$               | mass density ( $= \rho_{\text{code}} n_0$ )                         |
| 3   | $v$                   | $\text{cm s}^{-1}$                  | radial velocity ( $\times v_0$ )                                    |
| 4   | $p$                   | $\text{erg cm}^{-3}$                | pressure ( $\times p_0$ )                                           |
| 5   | $T$                   | K                                   | temperature ( $\times T_0$ )                                        |
| 6   | heating               | $\text{erg cm}^{-3} \text{ s}^{-1}$ | XUV photoionization heating ( $\times q_0$ )                        |
| 7   | cooling               | $\text{erg cm}^{-3} \text{ s}^{-1}$ | radiative cooling ( $\times q_0$ )                                  |
| 8   | adv_T_status          | none                                | _adv file only: integer, the validity of the temperature of the row |
| 9   | adv_comp_status       | none                                | _adv file only: integer, the validity of its composition            |
| 10  | adv_mass_row          | none                                | _adv file only: $m_j$ , the measure both fields were decided by     |

Column 2 is a *mass* density expressed in units of  $m_H$  per  $\text{cm}^3$  (metals included under the eos\_metals policy), not a number density; multiply by  $m_H$  to get  $\text{g cm}^{-3}$ .

**Columns 8 to 10: what each row of an \_adv file is.** Written only in Hydro\_ioniz\_adv.txt; the equilibrium file has seven columns, because its rows are the run's own solution by construction. Ion\_species\_adv.txt carries the two status fields as its last two columns, so a tool reading the species file alone sees the same verdicts.

The advection post-process solves the steady ionization and energy equations along the recorded flow. The temperature of a row and its composition are *two* statements: a row whose temperature was corrected can carry the run's own composition and the reverse, so each has its own field and neither is the first refusal of a single mixed integer. Both take the same five values.

| Value | Name          | Meaning                                                                                                                           |
|-------|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 0     | corrected     | the steady equation was solved and adopted for this quantity                                                                      |
| 1     | retained      | the run's own value was kept: $m_j$ above the fraction below, the local radiative balance rather than the flow sets $T$ (thermal) |
| 2     | failed        | the cell solve did not converge, or its root was out of band                                                                      |
| 3     | unsupported   | the closure does not cover this cell class: the omitted molecular and oxygen carriers hold more of the particle count than        |
| 4     | not_evaluated | the post-process did not reach this row (a ghost row below the first cell at which its upwind difference can be taken)            |

A retained quantity is the run's own value *exactly*, not to a tolerance: the composition of such a row is the ionization equilibrium at the run's temperature, which is what the uncorrected file carries. A corrected row is a *conditional* correction, accurate to adv\_conditional\_tol of itself in the mass flux, and column 10 carries the  $m_j$  it was judged by, so a reader can weigh a corrected row rather than only count it.

The file's own # header states all of it: the schema version, the legend of both fields, an # adv\_status\_counts line with the number of rows carrying each value in each field, the # adv\_stationarity\_operator the measure comes from, the # adv\_conditional\_tol line with the fraction a corrected row is accurate to, the # adv\_input\_certified verdict of the stationary certification on the whole input state (a separate statement from the fraction), and two declarations of what the product is not: # adv\_product, that this is a *one-way* correction on a fixed density and velocity field, so a converged temperature solve does not make the row a self-consistent solution and a profile assembled from corrected

and retained rows need not satisfy any interface balance, and # `adv_model_restrictions`, that the molecular and oxygen columns are the equilibrium solution, that those species are absent from the particle and electron counts every equation here is solved with, and that the  $H_3^+$  infrared cooling is absent from its energy balance. A file written before the schema carries no # `adv_schema` line, and the validity of its rows is UNKNOWN, not corrected.

`exhale_io.load_hydro` returns the three columns under their own names together with the header's statements, and `exhale_io.Run` exposes them as `adv_T_status`, `adv_comp_status`, `adv_mass_row` and `adv_conditional_tol` (None for an equilibrium file); `EXHALE_transit.py` prints the census of the rows its chords sample with the fraction and the largest measure among them, and copies both into the comment metadata of every saved curve. The two lowest rows are ghost rows below the first computational cell, whose temperature the energy loop never corrects whatever their status.

**Header lines.** The file opens with # comment lines that every loader skips: the schema tag, the # columns list, the # rows line saying which rows are ghost cells, and

```
# coupling: sec_ion=T sec_ion_step=2402 recon=WENO3 certified=T mode=init
```

which records the run state the profile was produced under: every switch that changes *while* a run converges rather than being fixed by `input.inp`. Whether the SvS85 photoelectron secondary ionization was being applied and at which step it was armed; the discrete reconstruction operator in force, which is not `rec_method` alone while the PLM  $\rightarrow$  WENO3 continuation is armed; whether the state passed the stationary certification, with `cert_reason=` beside it whenever the verdict is qualified (`no_stationary_claim` for a run that ended on a cap, `certified_in_wind` for a state whose species rows were gated at  $r \geq 1.20$  only); and the `Run mode` the run declared. Two fields that used to follow `sec_ion_step`, `valve` and `fluxconst`, were the switches the old base closure chose for itself and are gone with it (§2.6); a reader of an older file still finds them there, and the parser takes the fields it recognizes one at a time. This matters because the file is also the restart input: copied to `Hydro_ioniz_IC.txt` and read back under `Load IC?` True, its `sec_ion` field tells the new run to start the coupling where the old one left it, instead of re-running the staging on a state that no longer needs it (§144 of `docs/Update_EXHALE.md`). A file written before this line existed simply has no such line, and such a restart starts staged exactly as it did before. The other fields are reported, never imposed: `input.inp` states them for the new run.

**Provenance.** Three further # lines say what produced the file, so that a profile found on disk in a year can be tied to an executable and an input without asking anyone:

```
# provenance: git=6d07d48afd41 tree=dirty run=2026-09-03T20:19:09
# provenance: ck_input=1260237361560544917 ck_base=-1 ck_metals=4583919659595562347
# provenance: recon=WENO3 base_bc=isothermal carrier=none restart_schema=3 resid_def=145 N=500
```

`git` and `tree` are the revision the *executable* was built from and whether the working tree carried uncommitted changes at that moment; the Makefile stamps them in and the run never calls `git`. No build time is compiled in: a timestamp in a constant would make two builds of one source different files, and the md5 of the binary is what ties a bitwise regression verdict to a revision, so the time shown is the *run* time, from `date_and_time`. The `ck_` fields are checksums

of the input files as they were read (−1 means the file was absent), and the last line records the discretization, the lower boundary model, the molecular carrier model, the version of the restart layout, the version of the residual definition the acceptance gates used, and the grid size.

## 4.2 Ion\_species.txt / Ion\_species\_adv.txt

Ion number densities [ $\text{cm}^{-3}$ ] (34 columns). Column 1 is  $r$ ; columns 2–7 are the H/He ions; columns 8–34 are the 27 metal-ion stages, in this order:

```
r ; HI HII HeI HeII HeIII HeI* ; C I CII CIII OI OII OIII NI NII NIII MgI MgII MgIII SiI SiII SiIII CaI
CaII CaIII NaI NaII KI KII SI SII FeI FeII FeIII
```

(HeI\* is the metastable  $2^3\text{S}$  triplet.) The ionization fraction of any element is the stage density divided by the sum over that element's stages. With `Molecular chemistry: True` four more columns ( $\text{H}_2$   $\text{H}_2\text{p}$   $\text{H}_3\text{p}$   $\text{HeHp}$ ,  $\text{cm}^{-3}$ ) are appended after the metals, and with `Oxygen chemistry: True` three more after those ( $\text{OH}$   $\text{H}_2\text{O}$   $\text{CO}$ ,  $\text{cm}^{-3}$ ); the # columns header always names the layout. In `Ion_species_adv.txt` two further columns follow, `adv_T_status` and `adv_comp_status`, the validity of the row's temperature and of its composition (§4.1), with the same header block as the hydrodynamic file, so a tool reading the species file alone sees the same verdicts; the measure they were decided by is a column of that other file. Note that with the oxygen chemistry on the OI column means *free atomic* neutral oxygen: the element total is  $n(\text{OI}) + n(\text{OII}) + n(\text{OIII}) + n(\text{OH}) + n(\text{H}_2\text{O}) + n(\text{CO})$ , and the same holds for carbon with CO.

## 4.3 Cooling\_breakdown.txt

Radiative cooling by channel [ $\text{erg cm}^{-3} \text{s}^{-1}$ ] vs. radius. Columns: 1= $r$ , 2= $T$  [K], 3= $n_e$  [ $\text{cm}^{-3}$ ], 4=total cooling; 5--10= the six H/He channels (recombination, collisional ionization, then collisional excitation split by absorber into H I (the  $\text{Ly}\alpha$ -dominated H-line cooling), He I (which also carries the He  $2^3\text{S}$  metastable terms) and He II, and finally bremsstrahlung); 11= the  $\text{H}_3^+$  infrared cooling (zero unless the run tracks the molecular network); 12--14= the  $\text{H}_2$ ,  $\text{H}_2\text{O}$  and  $\text{CO}$  infrared bands, which are NET rates (emission less absorption of the field from below, so a negative value is that band heating the gas) and are identically zero unless `Molecular IR bands` is on; 15--41= the 27 metal-line cooling channels (same ion order as `Ion_species`). The channels sum to column 4. Useful for identifying the dominant coolant vs. altitude (cf. Huang+2023 Fig. 10), and the  $\text{H}_3^+$  column is the coolant of a molecular base. The # col header line always names the layout.

## 4.4 Heating\_breakdown.txt

Volumetric heating by channel [ $\text{erg cm}^{-3} \text{s}^{-1}$ ] vs. radius, the companion of `Cooling_breakdown.txt` and written unconditionally at the end of the run. 21 columns: 1= $r$  [ $R_p$ ], 2= $T$  [K], 3= $n_e$  [ $\text{cm}^{-3}$ ], 4=total heating; 5--10= the photoionization heating split by *absorber*: H I, He I, He II, He  $2^3\text{S}$ ,  $\text{H}_2$ , and the sum over the metal stages; 11= the photoelectric heating of the H( $n=2$ ) population (the excited-H model); 12= the collisional de-excitation of H( $n=2$ ), i.e. absorbed  $\text{Ly}\alpha$  thermalized by a collision; 13= the He-recombination coupling; 14= the Penning branch of He  $2^3\text{S} + \text{H}$ ; 15= its associative branch  $\text{He } 2^3\text{S} + \text{H} \rightarrow \text{HeH}^+ + \text{e}$ ; 16= He  $2^3\text{S} + \text{H}_2$  Penning ionization; 17= the fragment kinetic energy of the  $\text{H}_2$

Lyman–Werner photodissociation; 18= the collisionally de-excited Lyman–Werner fluorescence; 19= the collisional reactions of the H<sub>2</sub>/He network; 20= the FUV photolysis of H<sub>2</sub>O and OH; 21= the collisional reactions of the oxygen network including the O(<sup>1</sup>D) sink. A channel of physics the run does not carry is identically zero.

**One state per output file.** These columns are not recomputed for the dump: they are the channel array the ionization sweep filled for the state written beside them, through the one heating assembly `heating_of_composition` (`src/modules/radiation/util_ion_eq.f90`), whose total is the running sum of its channels and is the heat column of `Hydro_ioniz.txt`. Column 4 therefore reproduces that column to round-off: measured at  $1.7 \times 10^{-16}$  over the physical cells of the `mol_base_handoff` case (2026-09-06). The header names the layout and is built from the channel list, so a deposit added to the energy equation appears here by construction.

Rebuilding the rates on the written state instead, which earlier versions of this file did, is a different state: the rates a sweep contracts belong to the composition that *entered* it, and cannot be recovered from the composition it returned. That put a lag of one sweep between the two columns, measured at  $1.2 \times 10^{-5}$  in the median cell of the same case.

#### 4.5 Lyman\_Werner.txt

H<sub>2</sub> photodissociation diagnostic, written when a molecular run carries a `Stellar LW flux`. 8 columns: 1  $r$  [ $R_p$ ], 2  $T$  [K], 3  $x_{\text{H}_2} = 2n_{\text{H}_2}/n_{\text{H}}$ , 4  $n_{\text{H}_2}$  [cm<sup>-3</sup>], 5 the star-ward H<sub>2</sub> column  $N_{\text{H}_2}$  [cm<sup>-2</sup>], 6 the level-resolved self-shielding factor  $f_{\text{shield}}$ , a diagnostic,  $\sigma_{\text{diss}}(N_{\text{H}_2})/\sigma_{\text{diss}}$  at the bottom of the tabulated column axis, evaluated at that cell's own column, 7 the dissociation rate  $k_{\text{LW}}$  [s<sup>-1</sup>], which is the cell mean of the tabulated rate between the cell's two faces and is therefore *not*  $f_{\text{shield}}$  times a thin-limit rate, and 8 the photodissociation heating [erg cm<sup>-3</sup> s<sup>-1</sup>]. It shows directly how deep the band penetrates:  $f_{\text{shield}} \rightarrow 1$  in the thin wind above the H<sub>2</sub> → H front, collapsing by orders of magnitude inside the self-shielded molecular base.

#### 4.6 Oxygen\_chemistry.txt

The solved oxygen partition, written for the equilibrium state of a run with `Oxygen chemistry: True`. 18 columns: 1  $r$  [ $R_p$ ], 2  $T$  [K], 3  $n(\text{O})$  (*free atomic* neutral oxygen), 4  $n(\text{O II})$ , 5  $n(\text{O III})$ , 6  $n(\text{OH})$ , 7  $n(\text{H}_2\text{O})$ , 8  $n(\text{CO})$ , 9  $n(\text{O}(\text{<sup>1</sup>D}))$  from its local steady state, 10  $x_{\text{H}_2} = 2n_{\text{H}_2}/n_{\text{H}}$ , 11  $\tau_{\text{chem}}$  [s], 12  $\tau_{\text{adv}}$  [s], 13 the Damköhler number  $\tau_{\text{adv}}/\tau_{\text{chem}}$ , 14--15 the H<sub>2</sub>O and OH photodissociation rates summed over the bands [s<sup>-1</sup>], 16  $D(\text{H}_2)$  [cm<sup>2</sup> s<sup>-1</sup>], 17  $K_{\text{zz}}$  [cm<sup>2</sup> s<sup>-1</sup>], and 18  $\tau_{\text{diff}} = \Delta r^2/(D_{\text{H}_2} + K_{\text{zz}})$  [s]. Columns 3--8 sum to the oxygen element total.

$\tau_{\text{chem}} = 1/(k_{\text{O I}}n_{\text{OH}} + k_{\text{O II}}n_{\text{O}})$  is the H<sub>2</sub> lifetime against the oxygen cycle and  $\tau_{\text{adv}} = r/|v|$  the radial flow time. These are a *required* output rather than a diagnostic. With `Molecular carrier transport: True` a cell whose chemistry is slow is solved rather than assumed away, and the time scales say which part of the profile the reaction set is responsible for and which part the flow is; with `Molecular carrier transport: False` they say something stronger, namely where that limit is simply wrong.

With the transport on, a trailer at the end of the file records the last transport step: `newton_steps`, the relative residual `newton_resid` it stopped at, `limited_cells` (cells where a carrier would have taken more nuclei than its element has), `worst_overshoot`, and `co_ceiling_cells`, the cells in which the transported CO was cut back to the  $\text{CO} \leftrightarrow \text{C} + \text{O}$  equilibrium of their own ( $n, T$ ). A limiter count that is not small is a statement about the step and is reported rather than

absorbed.

Every run also closes the file with the **net H<sub>2</sub> loss budget at the base cell**, decomposed into eleven channels from the same coefficients the residual uses and net of its own reverse where a channel has one (so a negative entry is a net H<sub>2</sub> *source* and a share may exceed 1). The option exists to make the oxygen cycle set the base partition, and whether it does is a property of the run: on a 2331 K base the thermal channel  $\text{H}_2 + \text{M} \rightarrow \text{H} + \text{H} + \text{M}$  carries 70–97% of the net and the oxygen cycle a few percent, while on an 864 K base the oxygen family carries 96–99.6%. Read that block before reading the partition.

#### 4.7 FUV\_bands.txt

How deep each FUV band penetrates, and the band energy ledger. Written for an Oxygen chemistry run. 16 columns: 1  $r$  [ $R_p$ ], 2–3 the star-ward H<sub>2</sub>O and OH columns [ $\text{cm}^{-2}$ ], 4–7 the four band optical depths (LW, B2, B3, B4), 8–11 the H<sub>2</sub>O photodissociation rate in each band [ $\text{s}^{-1}$ ], 12–15 the same for OH, and 16 the total FUV photolysis heating [ $\text{erg cm}^{-3} \text{s}^{-1}$ ]. A header line above the data names the flux and the H<sub>2</sub>O and OH band cross sections of each band.

The trailer of the file carries the energy ledger. For a beam attenuated as  $\exp(-\tau)$  the photons absorbed over a column are exactly  $N_b(1 - e^{-\tau_b})$  whatever the density profile, and the file reports the grid sum against that closed form band by band. They agree to round-off at any grid spacing, because each cell's rate is the mean over the cell and not its face value; a residual there means the rates and the columns have come apart. It also reports the absorbed energy split into deposited heat and bond energy, against the incident band flux: the absorbed energy can never exceed it, which is the property that keeps the band average from over-heating.

The last block is the shared beam of the LW band. Over 912–1201 Å the same photons are absorbed by H<sub>2</sub> in the Lyman–Werner lines and by H<sub>2</sub>O and OH in a continuum, so the closed form the band's sum is checked against is  $N_b[1 - (1 - A)e^{-\tau}]$  with  $A$  the line-removed fraction, and the H<sub>2</sub> share belongs inside it: the H<sub>2</sub> photon count is the cell mean of the *pump* cross section  $\sigma_{\text{diss}}/p_{\text{eff}}$ , because the pumps that fluoresce back also take a photon out of the band. Since  $A$  is the column integral of that same cross section, the two sides are one normalization of one absorption and the LW row is a closure residual like B3 and B4. The block reports LW\_H2\_absorbed\_en, LW\_cont\_absorbed\_en, LW\_incident\_en and LW\_line\_removed\_frac. The run warns if the absorbers together take more energy out of the interval than it carries. A separate warning fires where the star-ward column passes the top of the self-shielding table's column axis; above it the table is clamped at its edge cross section, so  $A$  goes on growing past 1 and is capped there, which is the one place the LW closure residual does not go to round-off.

#### 4.8 OI\_levels.txt / OI\_levels\_adv.txt

Populations of the three fine-structure levels of the O I  $2p^4\ ^3P$  ground term, written whenever the run carries metals (no key: the presence of metals.inp, or of a lower-atmosphere profile with an oxygen reservoir, is enough). 11 columns: 1  $r$  [ $R_p$ ], 2  $T$  [K], 3  $n_e$ , 4  $n_{\text{HI}}$ , 5  $n(\text{OI})$ , 6–8 the fractional populations  $f(^3P_2)$ ,  $f(^3P_1)$ ,  $f(^3P_0)$ , and 9–11 the same three as densities [ $\text{cm}^{-3}$ ], which sum to column 5 by construction.

The term is inverted, so  $^3P_2$  is the lowest level. These are the lower levels of the O I  $\lambda\lambda 1302.168, 1304.858, 1306.029$  resonance triplet, which is why they are an output field: a transit forward model that applied the total O I density

to all three components would count the same atoms three times. They are not a second calculation – the file is filled by the same three-level statistical equilibrium, with the same electron and H-atom collisional rates and the same line escape probabilities and incident infrared field, that the [O I] 63.2/145.5/44.1  $\mu\text{m}$  cooling is assembled from (Cool\_coeff.f90). In the dense base the solution is the Boltzmann one to round-off; it departs from it in the thin outer wind, where the levels drain radiatively toward  $^3P_2$ .

#### 4.9 Excited\_H.txt

H( $n=2$ ) diagnostic (written when the excited-H model is on, i.e. stellar  $T_{\text{eff}}$  and radius are set). 16 columns: 1  $r$ , 2  $T$ , 3  $n_{\text{HI}}$ , 4  $n_e$ , 5  $\bar{J}_{\text{Ly}\alpha}$ , 6  $n_{2s}$ , 7  $n_{2p}$ , 8 Balmer proton source  $S_p$ , 9 photoelectric heating, 10 de-excitation heating, 11  $\tau_{\text{Ly}\alpha}$ , 12--14 ground-state HI photoionization / collisional-ionization / recombination rates (for the proton-budget comparison), 15--16 the internal and stellar contributions to  $\bar{J}_{\text{Ly}\alpha}$ .

#### 4.10 EXHALE\_setup.out and the mass-loss rate

EXHALE\_setup.out is a human-readable setup report (planet/star parameters, the Jeans parameter, composition, numerics) plus the final result. The **steady-state mass-loss rate** is  $\dot{M} = 4\pi\rho v r^2$  evaluated near the outer boundary, corrected by the 2-D-approximation factor from 2D approximate method ( $\text{Rate}/2 + \text{Mdot}/2 \Rightarrow \times \frac{1}{2}$ ,  $\text{Mdot}/4 \Rightarrow \times \frac{1}{4}$ ). It is reported as  $\log_{10} \dot{M}$  [ $\text{g s}^{-1}$ ]; divide by  $M_p$  and multiply by 1 Gyr to express it in  $M_p \text{ Gyr}^{-1}$ . The helper exhale\_io.mdot\_log10 / mdot\_Mp\_per\_Gyr reproduce this.

#### 4.11 EXHALE\_resolved.out: the configuration the wind actually used

A machine-readable record written in the run directory (next to input.inp, not inside output/) by write\_setup\_report.f90 once input.inp, metals.inp and any base.inp / lower-atmosphere-profile handoff have been resolved. Format: # comment lines, then one key value pair per line, reals at full double precision. It carries the base radius, planet mass, base temperature, He/H, orbital distance and stellar mass in effect; base\_inp\_present and he\_diffusion; the EOS factors mass\_per\_H\_amu and ntot\_bc\_per\_H; one abundance\_<El> line per element; and, with a profile in use (§3.8), the profile's provenance and the measured elemental fluxes. It exists so that consumers do not re-parse input.inp and inherit a value the handoff has since moved: EXHALE\_transit.py reads it in preference to input.inp (§5.2), and src/utls/element\_budget.py and src/utls/element\_flux\_closure.py read it as the authority on the resolved abundances.

#### 4.12 element\_flux\_profile.txt

Face-by-face elemental fluxes carried by the element-diffusion operator, written to output/ whenever a lower-atmosphere profile is in use (where they are the quantity the two models must agree on, not a diagnostic), and on demand with EXHALE\_DIFFUSION\_CHECK=1. Columns (named by the file's own # columns: line): the face index and radius, the face helium mass fraction, the diffusive and advective fluxes, then  $F_{\text{He}} = 4\pi r^2(\rho X v + J)$ ,  $F_{\text{H}} = 4\pi r^2(\rho(1-X)v - J)$  and  $\dot{M}_{\text{face}} = 4\pi r^2 \rho v$  in  $\text{g s}^{-1}$ , the face  $\Delta m_{\text{eff}}$ , the effective and neutral diffusion coefficients with their ratio, and the dominant friction pair. Replaced at every diffusion step, so it always describes the last one.

## 5. Analysis tools

### 5.1 EXHALE\_plots.py: live profile viewer

`python3 EXHALE_plots.py [--live N]` draws an 8-panel figure from output/: density, velocity, pressure, temperature, momentum  $\log_{10}(\rho v r^2)$ , the ion densities, and the H and He ionization fractions. With `--live N` it refreshes every  $N$  seconds (watch a run converge). If the `*_adv.txt` files exist they are overplotted (dashed) against the equilibrium profiles (solid). It also prints  $\log_{10} \dot{M}$ .

### 5.2 EXHALE\_transit.py: transmission spectra

`EXHALE_transit.py` post-processes a converged run (`*_adv.txt + input.inp`) into transit transmission spectra and reports the line-center absorption depths (theoretical, instrument-convolved, and instrument+rotation-convolved) for **He 10830**, **Ly $\alpha$** , **H $\alpha$ /H $\beta$**  (from the non-LTE  $n=2$  population, Christie+2013, pumped by the Ly $\alpha$  field of §3.4), and the metal resonance doublets **Mg II h&k**, **Ca II H&K**, **Na I D** and the **O I 1302/1304/1306** triplet: the metals get the same full treatment as the H/He lines (all components in one window, theoretical / instrument / +rotation spectra, written under the same `tpm_<line>` names), plus the validated single-component line-center and 4 Å-band depth table; a metals-off run skips them automatically.

**O I 1302: the one line whose components do not share a lower level.** The three components of the O I resonance triplet absorb out of the three *different* fine-structure levels of the  $2p^4\ ^3P$  ground term, so the script reads their populations from output/`OI_levels_adv.txt` (§4.8) rather than applying the total O I density three times over, which at the near-statistical populations of these winds would triple the column. A run whose output predates that file, like a metals-off run, simply skips the line. Because each absorption component owns its own lower-level density, the component descriptor is (`lambda0`, `f`, `A21`, `n_lower`) for every line, not only for this one.

The O I line is also the only one compared here against a published measurement, and the comparison is written into `tpm_OI_band_depths.txt` together with the conventions that make it legitimate: the depth is *band-integrated* over the observers' band (1300–1310 Å for Vidal-Madjar et al. 2004, 1299–1310 Å for Ben-Jaffel & Hosseini 2010), never at line center; the band weight is the stellar emission-line profile those authors measured (0.2 Å FWHM, peak ratios 1:1.5:1.17), because what was measured is a flux ratio and essentially all the flux in that band is in the three narrow chromospheric lines; the interstellar absorption is already inside those measured peak ratios and is therefore *not* applied a second time; the instrument LSF is not applied either, because a Gaussian LSF conserves the integral and cancels from a ratio over a band many LSF widths wide; and the published values, which are  $(R_{\text{abs}}/R_{\star})^2$  and include the opaque planetary disk, have each paper's own continuum band subtracted before comparison. Overrides: `EXHALE_TRANSIT_RES_OI` (default 1000, HST/STIS G140L), `_OI_WEIGHT (star | flat)`, `_OI_STAR_FWHM_A`, `_OI_STAR_PEAKS`, and `_OI_NSACLE`, which multiplies all three level densities at fixed ratios so the same code path can be driven up and down the curve of growth. It supports two geometries via `geometry = 'spherical'` (Phase 5a) or `'triaxial'` (Phase 5b, the 3-D Roche-equipotential reconstruction in `roche_recon.py`).

**Where the system parameters come from.** Every planet and star parameter is taken from the run directory, so the spectra always describe the same system as the simulation. The labels are matched exactly as `input_read.f90` does, so line order is irrelevant. Two sources are read, in this order:

EXHALE\_TRANSIT\_\* environment override > ./EXHALE\_resolved.out > ./input.inp > built-in default,

so an explicit override still wins for an experiment. The middle step matters whenever a `base.inp` or a lower-atmosphere profile moved the base: `EXHALE_resolved.out` (§4.11) is written by the wind solver *after* those overrides are applied, so it carries the  $R_p$ ,  $M_p$ ,  $T_0$ ,  $a$ ,  $M_*$  and He/H the wind actually used, while `input.inp` alone would be stale. When the file is present the script says so at startup and takes those six from it; when it is absent it says that too and falls back on `input.inp`. A present-but-unreadable file is a hard error, never a silent fallback. The stellar radius and  $T_{\text{eff}}$  are not in that file and come from `input.inp` (or their environment overrides) either way.

| Quantity          | input.inp label                          | Override                  |
|-------------------|------------------------------------------|---------------------------|
| $R_p$             | Planet radius [R_J]                      | -                         |
| $M_p$             | Planet mass [M_J]                        | -                         |
| $T_{\text{eq}}$   | Equilibrium temperature [K]              | -                         |
| $a$               | Orbital distance [AU]                    | -                         |
| $M_*$             | Parent star mass [M_sun]                 | -                         |
| $L_{\text{EUV}}$  | Log10 of EUV luminosity                  | -                         |
| $\xi$ (day–night) | 2D approximate method                    | xi_override               |
| $R_*$             | Stellar radius [R_sun]                   | EXHALE_TRANSIT_RSTAR_RSUN |
| $T_*$             | Stellar Teff [K]                         | EXHALE_TRANSIT_TSTAR      |
| $P_{\text{spin}}$ | from $a$ , $M_*$ + $M_p$ (tidal locking) | EXHALE_TRANSIT_ROTPT      |

The planet spin period has no `input.inp` key: a close-in giant is tidally locked, so the default is the Keplerian orbital period  $P_{\text{orb}} = 2\pi\sqrt{a^3/[G(M_*+M_p)]}$  built from the orbital distance and the masses already in `input.inp`. Only the instrument resolving powers, the wavelength windows, and the geometry switch remain script/environment settings: they describe the observation, not the system. At startup the script prints every resolved parameter together with its source (`input.inp`, `env ...`, or built-in default), and warns when Stellar radius/Stellar Teff are missing from `input.inp`; read that block first when a spectrum looks wrong. A missing Stellar radius is the one worth checking before anything else: transit depths are normalized by  $\pi R_*^2$  and the chord integration stops at  $R_*/R_p$ , so the built-in fallback shifts every depth, and not by a single factor. Measured on the tutorial planet (a  $1.13 M_\odot$  star, so  $R_* \simeq 1.155 R_\odot$ ), the fallback returned a He 10830 depth of 22.8% where the correct radius gives 4.0%.

**Outputs.** Every line carries one key (He10830, Lya, Halpha, Hbeta, MgII, CaII, NaI, OI), and both products of a line are named from that key and land in the run directory: the model curve `tpm_<line>.txt` (always written) and, when `EXHALE_TRANSIT_FIG_PREFIX` is set, the figure `<prefix><line>.png` together with its vector `.pdf`. A curve file has the columns `lambda[A]`,  $T_\lambda$  (theoretical),  $T_\lambda$  (instrument-convolved),  $T_\lambda$  (+rotation), so the excess absorption in percent is  $(1 - T_\lambda) \times 100$ ; a metals-off run simply writes no metal curves. `EXHALE_TRANSIT_SAVE_PREFIX` decorates

the curve name in the same way the figure prefix decorates the figure name; both are resolved against the run directory, so an absolute prefix writes elsewhere.

**Step-by-step usage.** EXHALE\_transit.py reads output/Hydro\_ioniz\_adv.txt and output/Ion\_species\_adv.txt from path, which is the current directory unless EXHALE\_TRANSIT\_PATH names the run directory:

1. *Ensure the \*\_adv.txt files exist.* A normal converged run writes them automatically. To post-process an existing state (e.g. a Newton solution produced by the EXHALE\_PTC hook, which writes only the eq files): copy Hydro\_ioniz.txt → Hydro\_ioniz\_IC.txt and likewise for Ion\_species, set Load\_IC? True and Do\_only\_PP: True in input.inp, and run ./EXHALE.x once (seconds).
2. *Set what describes the observation,* if the defaults do not: the instrument resolutions and wavelength windows (EXHALE\_TRANSIT\_RES\_\*, \_HE\_LMIN/LMAX/N), the azimuthal sampling of the rotation integral, and geometry for a Roche run. The star and the spin period need nothing: they come from input.inp.
3. *Run* MPLBACKEND=Agg EXHALE\_TRANSIT\_PATH=<run\_dir> python3 EXHALE\_transit.py (the Agg backend suppresses the interactive windows on headless machines). The script writes the tpm\_<line>.txt curves into the run directory and prints the line-center absorption depths (theoretical / instrument-convolved / +rotation) and the metal-line summary; add EXHALE\_TRANSIT\_FIG\_PREFIX=tpm\_ to save the figures as well.

Worked examples (the converged Newton solutions of WASP-121b and HD189733b, with all four H/He spectra and the metal lines): backup/regression/tpm\_wasp/ and backup/regression/tpm\_hd189/, collected in the notebook backup/regression/steady\_solver\_results.ipynb. Figure 2 shows the four H/He spectra EXHALE\_transit.py saves for the WASP-121b Newton solution of backup/regression/tpm\_wasp/ (run with Include\_He23S? True, so the He I 10830 line is populated).

**Transit-depth conventions (important).** Huang et al. (2023) quote transit depths as the *effective transit radius*  $R_{\text{eff}}/R_*$ , not a percent absorption; convert from the code's line absorption fraction  $h$  via  $R_{\text{eff}}/R_* = \sqrt{(R_p/R_*)^2 + h}$ . The metal NUV lines (Mg II) are quoted in a 4 Å bin, while the optical lines (Ca II, H $\alpha$ , Na) are line-center; line-center vs. 4 Å can differ by  $\sim 3\times$  for an optically-thick line, so always compare in the same definition.

### 5.3 exhale\_io.py: named, physical-unit loaders

examples/exhale\_io.py loads every EXHALE output file into named arrays in physical units, so analysis code need not remember column orders (Table 5). The central object is the Run container: load\_run(outdir, inputfile) fills it with the hydro profiles ( $r$  [ $R_p$ ],  $n$  [ $\text{cm}^{-3}$ ],  $v$  [ $\text{cm s}^{-1}$ ] with a  $v_{\text{kms}}$  shortcut,  $p$ ,  $T$ , heat, cool), the full ion-density dictionary run.ion (the 6 H/He plus 27 metal stages of §4.), and the parsed input.inp in run.inp. By default the \_adv (advection-corrected) files are read; pass adv=False for the equilibrium snapshot. Typical use:

```
import exhale_io as aio
run = aio.load_run('output', 'input.inp')      # _adv profiles by default
print(aio.mdot_log10(run))                     # log10 Mdot [g/s], 2D factor incl.
```

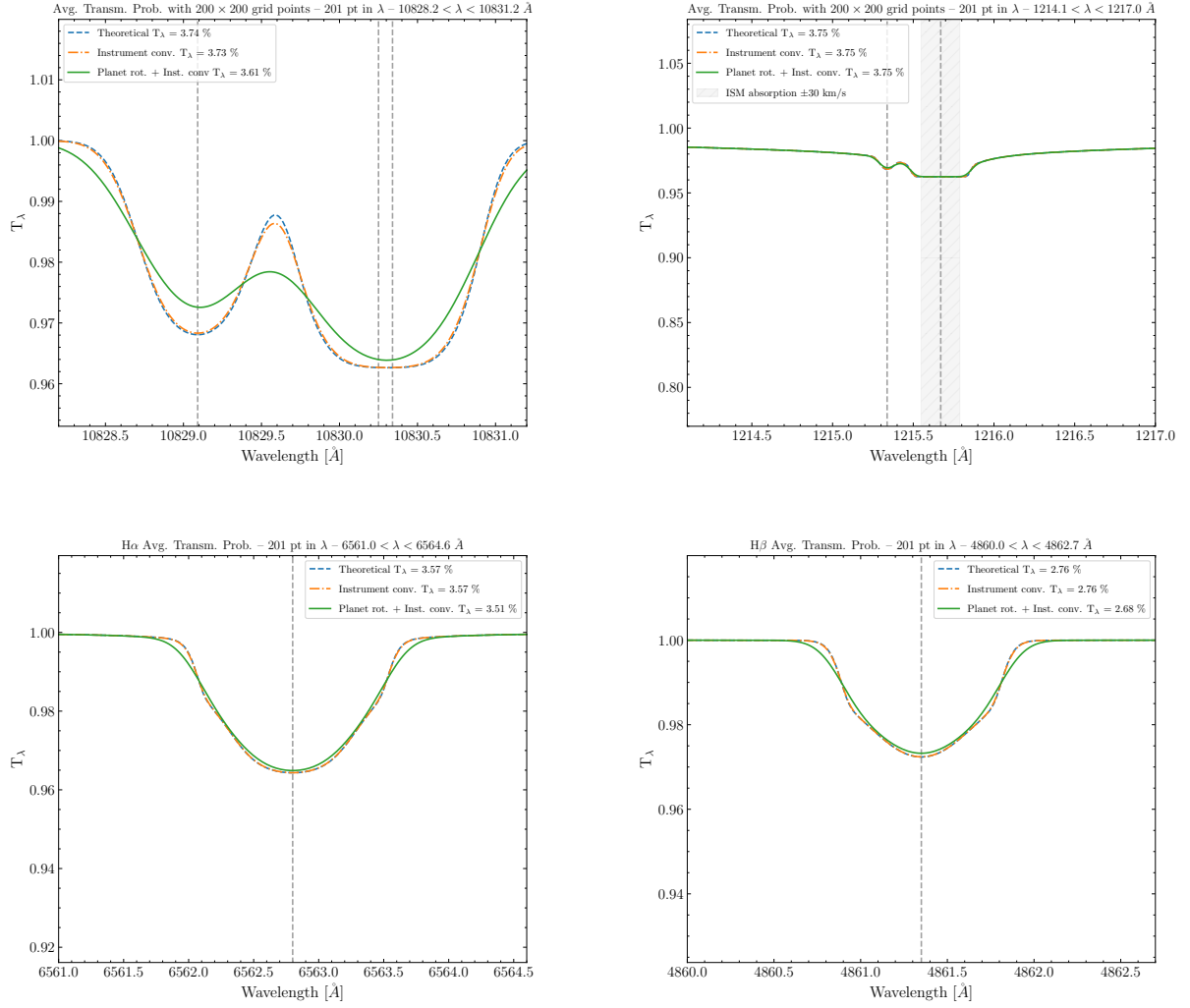


Figure 2: EXHALE\_transit.py output spectra for WASP-121b (Roche, Newton solution with He 2<sup>3</sup>S on; backup/regression/tpm\_wasp/): He I 10830 (top left), Ly $\alpha$  (top right), H $\alpha$  (bottom left), and H $\beta$  (bottom right). Each panel shows the theoretical line, the instrument-convolved line, and the instrument+rotation-convolved line as produced by the script.

```

print(aio.mdot_Mp_per_Gyr(run))          # same in M_p/Gyr
xh = run.x_ion(['HI', 'HII'])             # stage fractions of one element
plt.plot(run.r, run.T)
plt.plot(run.r, xh['HII'])
cool = aio.load_cooling('output/Cooling_breakdown.txt')
plt.semilogy(cool['r'], cool['chan']['FeII']) # one cooling channel

```

| Function                              | Returns                                                                                                                                                                                            |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| load_run(outdir, inputfile, adv=True) | a Run: hydro profiles + run.ion dict + parsed input.                                                                                                                                               |
| Run.x_ion([stages])                   | ionization fractions of one element, e.g.<br>$x_{\text{ion}}(['\text{HI}', '\text{HII}'])['\text{HII}'] = n_{\text{HII}}/(n_{\text{HI}} + n_{\text{HII}})$ .                                       |
| load_hydro(path)                      | one Hydro_ioniz(_adv).txt as a dict (r, n, v, p, T, heat, cool).                                                                                                                                   |
| load_ions(path)                       | (r, {ion: density}) from Ion_species(_adv).txt (all 33 stages).                                                                                                                                    |
| load_cooling(path)                    | r, T, ne, cool_total + chan dict of cooling in each channel (6 H/He channels rec/coll_ion/coex_HI/coex_HeI/coex_HeII/brems, the H3p infrared channel, + 27 metal-line channels).                   |
| load_excited_H(path)                  | Excited_H.txt as a dict (n2s, n2p, Jlya, tau_lya, Sproton, Hpe, . . .).                                                                                                                            |
| load_lyman_werner(path)               | Lyman_Werner.txt as a dict (x_H2, NH2, f_shield, k_LW, heat_LW).                                                                                                                                   |
| read_input(path)                      | input.inp as a dict (Rp_RJ, Mp_MJ, a_AU, HeH, LX, LEUV, . . . + raw lines).                                                                                                                        |
| mdot_log10(run, j_from_top=20)        | $\log_{10} \dot{M}$ [g/s]: $4\pi\rho v r^2$ near the outer boundary <i>including</i> the 2D approximate method factor (mirrors EXHALE_main.f90; see §4).                                           |
| mdot_Mp_per_Gyr(run)                  | the same in units of $M_p \text{ Gyr}^{-1}$ .                                                                                                                                                      |
| constants                             | RJ_CM (and its alias RJ), MJ, Msun, AU, mu (= $m_{\text{H}}$ ), GYR (cgs). These are the one Python definition of each; the tools import them from here rather than writing the number down again. |

Table 5: examples/exhale\_io.py public interface. Run analysis scripts either from examples/ or after adding it to the Python path (sys.path.insert(0, '<repo>/examples')).

The notebook examples/EXHALE\_analysis.ipynb uses these loaders to plot the profiles and the parameter-difference comparisons of §6., and examples/make\_figures.py (which generates every figure in this manual) is a further worked example.

## 5.4 exhale\_to\_lart.py: LaRT $\text{Ly}\alpha$ input from an EXHALE run

EXHALE\_transit.py estimates the  $\text{Ly}\alpha$  mean intensity  $\bar{J}_{\text{Ly}\alpha}$  with the simple Huang et al. (2017) prescription (§3.4). For a proper treatment of the resonant  $\text{Ly}\alpha$  radiative transfer that sets the  $n=2$  hydrogen population (and hence  $\text{H}\alpha$ ), the

converged atmosphere can be post-processed with the Monte Carlo code **LaRT** (Seon & Kim 2020; Seon et al. 2022). `examples/exhale_to_lart.py` builds a complete LaRT spherical-illumination input set directly from an EXHALE run.

Given a run directory it loads the `_adv` profiles (via `exhale_io.py`) and `input.inp`, and writes

- `dens_profile.txt`:  $r[R_p]$ ,  $n_{\text{HI}}[\text{cm}^{-3}]$  (the  $\text{Ly}\alpha$  scatterer is ground-state H);
- `temp_profile.txt`:  $r[R_p]$ ,  $T[\text{K}]$ ;
- `velo_profile.txt`:  $r[R_p]$ ,  $v_r[\text{kms}^{-1}]$ ;
- `line_profile.txt`: the incident stellar  $\text{Ly}\alpha$  profile, a double-Gaussian with peaks at  $\pm m$  and Gaussian width  $s$  (defaults  $m = 74$ ,  $s = 49 \text{ kms}^{-1}$ , the Huang et al. 2017 / Yan et al. 2022 values; pass `--fwhm` to read  $s$  as FWHM);
- `<name>_lya.in`: the LaRT namelist (geometry='spherical\_atmosphere', source\_geometry='stellar\_illumination');
- `run.sh`: a multi-host `mpirun` launcher.

The length unit is set to  $R_p$  (distance2cm =  $R_p$  in cm), so all LaRT radii are in  $R_p$  and match the EXHALE grid; the star-planet geometry (distance\_star\_to\_planet =  $a/R_p$ , stellar\_radius =  $R_*/R_p$ ) is read from `input.inp`. Because the stellar illumination breaks spherical symmetry, the scattering rate is requested as a cylindrical 2-D grid (geometry\_JPa = 2):  $P_\alpha(\rho, z)$  is axisymmetric about the star-planet axis, which is the form needed to build the 2-D  $n_{2p}(\rho, z)$  population for the  $\text{H}\alpha$  transit. To populate  $P_\alpha$ , build LaRT with `make CALCPnew=1` (producing `LaRT_calcP.x`).

```
# build the LaRT input from a converged EXHALE run, then run it
python examples/exhale_to_lart.py WASP-52b/fxuv0p25_he98 \
    -o WASP-52b/LaRT_lya --name WASP-52b \
    --m 74 --s 49 --rmax 10 --ngrid 201 --nphotons 1e7 \
    --hosts lart4,lart3,lart2
cd WASP-52b/LaRT_lya && ./run.sh
```

It is also importable: `import exhale_to_lart as e2l; e2l.build(run_dir, outdir, m_kms=74, s_kms=49, geometry_jpa=2, ...)`.

## 6. Worked examples

### 6.1 A first run: a generic hot Jupiter

The directory `examples/tutorial/` contains a minimal, self-contained model of a generic HD 209458b-like hot Jupiter: a spherical domain out to  $10R_p$ , solar metals, EUV+X-ray power-law SED. The full `input.inp` is:

```
Planet name: demoHJ
Log10 lower boundary number density [cm^-3]: 12.70
Planet radius [R_J]: 1.38
Planet mass [M_J]: 0.69
```

```

Equilibrium temperature [K]: 1320.0
Orbital distance [AU]: 0.047
Escape radius [R_p]: 2.00
He/H number ratio: 0.083
2D approximate method: Rate/2 + Mdot/2
Parent star mass [M_sun]: 1.13
Spectrum type: Power-law
Power-law index: -1.0
Use only EUV? False
[E_low,E_mid,E_high] = [ 13.60 , 123.98 , 1.24e3 ]
Log10 of X-ray luminosity [erg/s]: 27.0
Log10 of EUV luminosity [erg/s]: 29.30
Grid type: Mixed
Numerical flux: HLLC
Reconstruction scheme: PLM+WENO3
Include He23S? True
Load IC? False
Do only PP: False
Force start: False
Domain mode: Spherical
Outer radius [R_p]: 10.0
du_th [PLM,WENO3]: 0.5 1.0e-3
Solver: Newton
Stellar radius [R_sun]: 1.155

```

The `Reconstruction scheme: PLM+WENO3` line enables the two-stage reconstruction (with the two `du_th` values as the PLM and WENO3 thresholds), and the `Solver: Newton` line adds the steady-state Newton finish (§2.7), so the run converges to the true steady state without manual restarts. With `metals.inp` present (solar abundances, §3.), run `./EXHALE.x` in that directory. Figure 3 shows the converged profiles read back with `exhale_io`: the wind heats to  $\sim 10^4$  K, the velocity rises monotonically outward (transonic outflow), and hydrogen ionizes across a front near  $1.2\text{--}1.4R_p$ .

## 6.2 Turning physics on and off

The same model is re-run with single-line changes to illustrate each option: *metals off* by removing `metals.inp`; *Roche/RLOF* by removing the `Domain mode/Outer radius` lines (the default truncates at L1); the *atomic-helium limit* by setting `Include He23S? False` (the triplet is on by default, so this is the change that removes it); the *H $\alpha$ /Balmer* physics by adding `Stellar Teff/Stellar radius`; and the *transonic IC* by adding `Transonic IC: True` for a near-unbound (deep-RLOF) base; and *diffusive He/H separation* by adding `He_diffusion: True` (extend it to the metals with `He_metal_diffusion: True`; §3.6). The comparison studies below quantify the effect of the first two.

Tutorial run: generic hot Jupiter (spherical, H/He + metals)

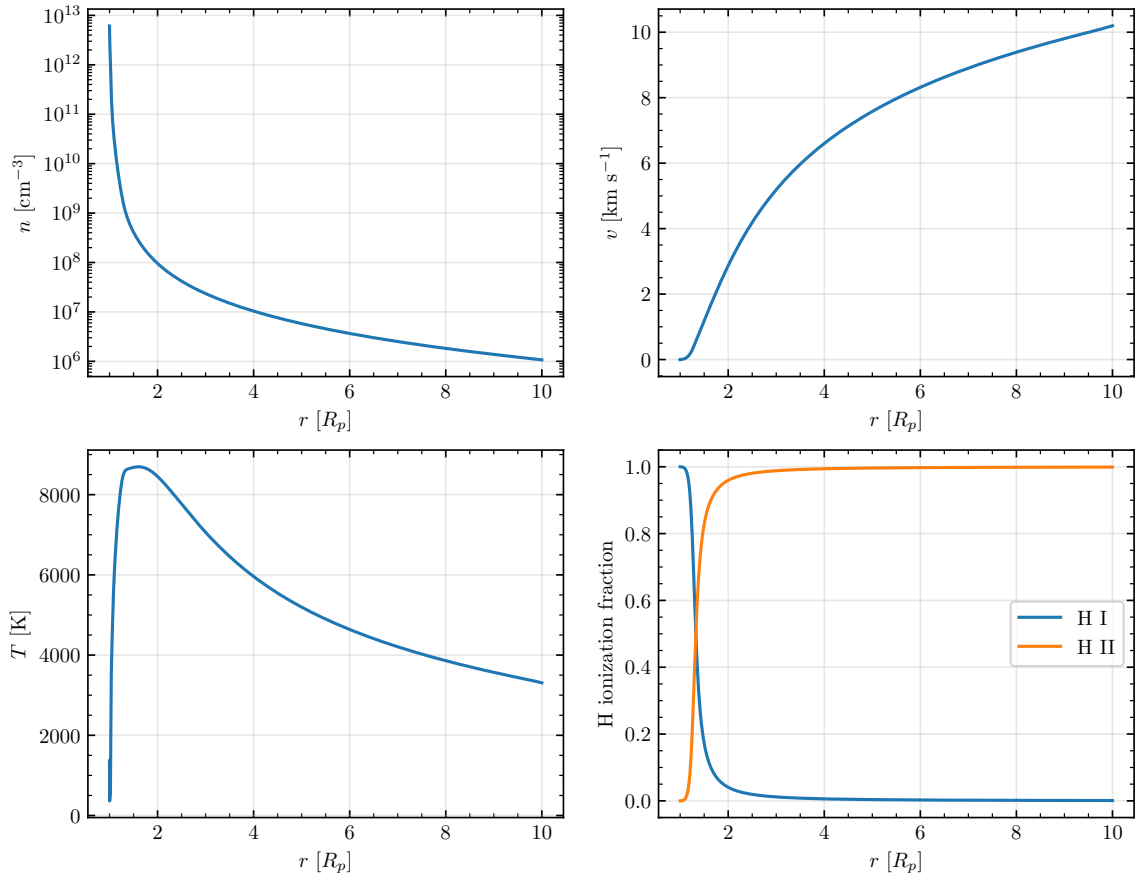


Figure 3: First-run output of the generic hot-Jupiter tutorial: density, velocity, temperature, and the H ionization fraction vs. radius.

### 6.3 Ready-made example configurations (examples/)

For every solver/physics combination there is a ready-to-run folder under `examples/` (paths relative to the EXHALE root). Folders 01–11 are the *same* planet (HD189733b) so that the effect of each option can be isolated; 12 and 14–17 use HD209458b (where the Wind-AE warm start, diffusion, molecular chemistry with and without metals, and the lower-atmosphere profile handoff are validated), and 13 spans four planets. Each folder is self-contained: `cd` into it and run the repo-root binary (`././EXHALE.x`); outputs land in the folder’s own `output/`. The “Added / changed lines” column of Table 6 is relative to each example’s natural base: the legacy `01_legacy_marching/input.inp` for the solver examples 02–04 (and Wind-AE 11/12), and the recommended-solver `03_newton/input.inp` (two-stage PLM+WENO3 + Solver: Newton) for the physics examples 05–10, which all inherit that solver and add only their signature option on top. A `diff` against the appropriate base shows exactly what an option adds; all configurations have been smoke-tested. The metastable helium triplet is on by default in the code, but the ladder holds `Include He23S? False` on purpose: in both bases and in every rung whose signature option is something else (02, 04, 05, 09–12), so that atomic helium is the baseline and `06_he23s` remains a one-line difference. Outside the ladder (tutorial/, 13–16, the planet folders) the triplet is on. `11_windae_ic` (HD189733b) exercises the self-consistent-BC continuation: its Wind-AE ramp converges and writes the IC, though EXHALE’s own time-integration of HD189733b is limited by a separate base-breathing instability (§2.10). `12_windae_ic_hd209` is the seed-adjacent HD209458b case. `14_diffusion` and `15_molecular` turn on the He/metal diffusive separation (§3.6) and the full molecular network (§3.7) respectively, on HD209458b, and `16_molecular_metals` adds a `metals.inp` to the latter. `17_lower_profile` replaces the scalar `base.inp` handoff by the profile of §3.8.

Table 6: Ready-made example configurations under `examples/` (01–11 are HD189733b; 12, 14–17 are HD209458b; 13 spans four planets). The Wind-AE IC ramp converges for both of the planets that use it (HD 189733 b (11) and HD 209458 b (12)), and 11, being far from the seed, additionally exercises the self-consistent-BC continuation and the molecular-layer turn-off. Metals on/off is a runtime switch: any folder becomes metals-on by copying a `metals.inp` into it, molecular folders included: that is what `16_molecular_metals` is.

| examples/...       | Demonstrates                                                                           | Added / changed lines                                                        |
|--------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 01_legacy_marching | classic ATEs v2 baseline: PLM marching, H/He only, Roche domain, <i>du</i> -based stop | (none, baseline)                                                             |
| 02_two_stage       | two-stage PLM→WENO3 marching (§2.4)                                                    | Reconstruction scheme: PLM+WENO3, <code>du_th [PLM,WENO3]: 0.5 1.0e-3</code> |
| 03_newton          | <b>recommended default:</b> two-stage warm-up + Newton finish (§2.7)                   | + Solver: Newton                                                             |

Table 6, continued

| examples/...         | Demonstrates                                                                                                                                                                                                                                                                                                               | Added / changed lines                                                                           |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 04_newton_from_state | Newton re-convergence of an existing state (no marching; needs output/*_IC.txt, see the folder README)                                                                                                                                                                                                                     | Load IC? True, Resid tol: 1.0e-5; run via the EXHALE_PTC hook                                   |
| 05_metals            | trace-metal cooling, solar C/N/O/Mg/Ca/Na/Fe (§3.)                                                                                                                                                                                                                                                                         | metals.inp present                                                                              |
| 06_he23s             | He 2 <sup>3</sup> S metastable level (HeI 10830); the one rung that leaves the ladder's atomic-helium baseline                                                                                                                                                                                                             | Include He23S? True                                                                             |
| 07_balmer_lya        | non-LTE H( <i>n</i> =2) + Ly $\alpha$ pumping (H $\alpha$ /H $\beta$ , §3.4)                                                                                                                                                                                                                                               | + Stellar Teff/radius, Deexc heat, Jlya escape-prob, Stellar Lya flux/halfwidth/boost           |
| 08_full              | everything on (= the HD189733b/planet folder + Newton); feeds EXHALE_transit.py                                                                                                                                                                                                                                            | 07 + metals.inp                                                                                 |
| 09_spherical         | spherical domain instead of the default Roche/L1 truncation (§1.3)                                                                                                                                                                                                                                                         | Domain mode: Spherical, Outer radius [R_p]: 10.0                                                |
| 10_warm_seed_ic      | warm-seed (hot Parker overlay) initial condition                                                                                                                                                                                                                                                                           | Hot Parker IC: 10000                                                                            |
| 11_windae_ic         | in-process Wind-AE IC for HD189733b (§2.10): far from the seed, so the self-consistent-BC continuation (base-BC re-convergence + molecular-layer turn-off) is exercised; the Wind-AE ramp converges and writes the IC. EXHALE's own HD189733b base-breathing instability (separate from the IC) then limits the warm start | + IC mode: windae, Solver: Newton; symlinks inputdata/ for the shipped seed                     |
| 12_windae_ic_hd209   | in-process Wind-AE warm-start IC that <i>works</i> : HD 209458b ( <b>not</b> HD189733b) is close to the shipped seed, so the ramp converges and EXHALE warm-starts cleanly (spherical 10 $R_p$ )                                                                                                                           | HD209458b params + Domain mode: Spherical, IC mode: windae, Solver: Newton; symlinks inputdata/ |

Table 6, continued

| examples/...        | Demonstrates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Added / changed lines                                                                                                                   |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 13_lower_atmosphere | lower-atmosphere connection for four planets (HD209458b, HD189733b, WASP-121b, WASP-52b; §3.7): an analytic $1\mu\text{bar}$ base column plus a base.inp hand-off (isothermal or Guillot $T(p)$ generator); see the folder README                                                                                                                                                                                                                                                                          | Lower column: <R_1bar>; driver-generated base_iso.inp / base_guillot.inp                                                                |
| 14_diffusion        | diffusive separation of He and metals (HD209458b; §3.6): the He/H ratio declines with altitude and each trace metal settles independently, re-shaping He I 10830                                                                                                                                                                                                                                                                                                                                           | HD209458b params + Include He23S? True, He_diffusion: True, He_metal_diffusion: True, He_Kzz: 1.0e9, He_alphaT: 0.0; metals.inp present |
| 15_molecular        | full molecular chemistry (HD209458b; §3.7): $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ in the coupled ionization equilibrium; a sharp $\text{H}_2 \rightarrow \text{H}$ front forms above a thin molecular base, the wind above it essentially atomic. Newton-converged from a cold start, re-run 2026-08-15 with the current code ( $\ R\  = 6.105 \times 10^{-6}$ in 89 outer iterations): the front sits at $r = 1.0097 R_p$ and the molecular layer below it collapses to $\sim 400\text{ K}$ | HD209458b params + Molecular chemistry: True, Solver: Newton 5.0e-2, Resid tol: 1.0e-7, Max steps: 150000 (no metals.inp here)          |

Table 6, continued

| examples/...        | Demonstrates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Added / changed lines                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16_molecular_metals | molecular chemistry <i>and</i> trace metals solved in one system (HD209458b): the $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ network and the metal ionization stages share the free electron density, which the metals dominate in the shielded molecular base. Converges only from a warm restart, and the seed matters: seed it with a 15 converged by <i>the same binary</i> (a state left by an older binary is a different seed and fails, <code>info = 2</code> ), then expect <code>info = 0</code> at <code>Resid tol: 1.0e-7</code> and $\log_{10} \dot{M} = 9.65 - 9.67$ . The cold path is closed: with metals on, the JFNK line search bottoms near $\ R\  \sim 10^{-3}$ (worst cell $r = 1.021$ , momentum) and falls back to marching at every Low-Mach damping coefficient tried: off, $5 \times 10^{-3}$ , $10^{-2}$ , $2 \times 10^{-2}$ (measured 2026-08-15) | as shipped, 15_molecular + metals.inp (solar C/N/O) without 15's three convergence keys. For the warm restart put the seed run's <code>Hydro_ioniz.txt</code> / <code>Ion_species.txt</code> in <code>output/*_IC.txt</code> and use 15's three keys + <code>Load IC? True</code> + Low-Mach damping: $5.0\text{e-}3$ (run $1.0\text{e-}2$ as well and compare, see the folder README) |
| 17_lower_profile    | lower-atmosphere <i>profile</i> handoff (HD209458b; §3.8): the lower atmosphere handed over as a table over an interval of pressure instead of the scalars of <code>base.inp</code> . The base state ( $T_0$ , $R_0$ , $p_{\text{base}}$ , $q_{\text{H}_2}$ ), the elemental reservoirs (He/H and solar C/N/O) and $K_{\text{zz}}(r)$ all come from the one file; <code>make_example_profile.py</code> regenerates it                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 14_diffusion + Lower atmosphere profile: <code>lower_atmosphere_profile.dat</code> , <i>minus</i> <code>He_Kzz</code> (the profile carries $K_{\text{zz}}$ ) and <i>minus</i> <code>metals.inp</code> (the profile carries C/N/O)                                                                                                                                                      |

## 7. Comparison studies

### 7.1 Effect of metal-line cooling (metals on vs. off)

Trace metals add strong line cooling (and some photoionization heating and free electrons). Figure 4 compares the tutorial planet with and without metals. `inp`: metal-line cooling lowers the peak temperature by  $\sim 1800$  K (acting as a thermostat that prevents the H/He-only over-heating), while the H ionization structure is barely changed. The effect is far stronger for hotter, more irradiated planets (e.g. WASP-121b, where metal cooling is essential to match the observed temperatures); the detailed cooling by channel budget is in Figure 5.

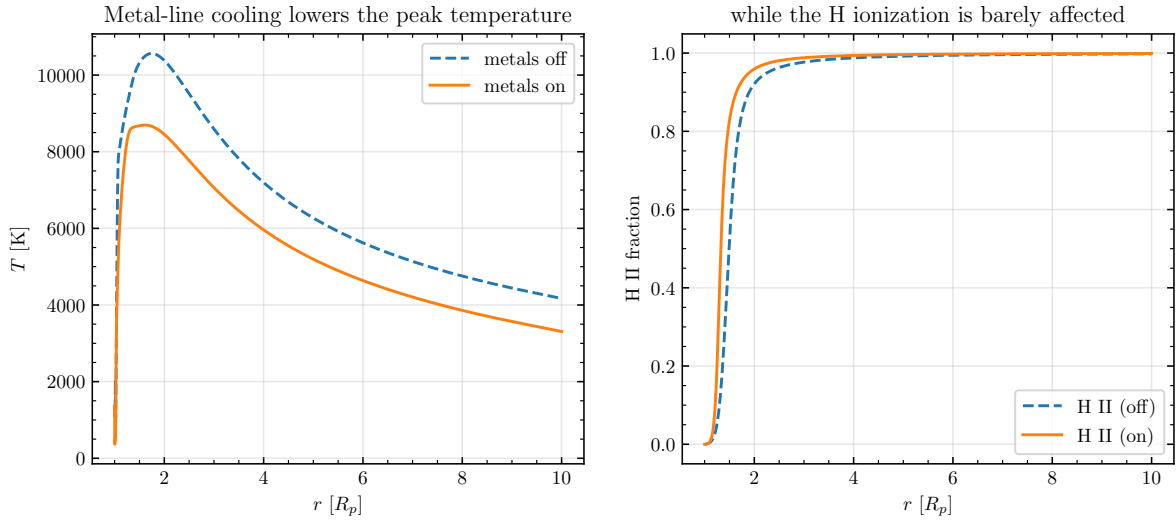


Figure 4: Metals on vs. off for the tutorial planet: metal-line cooling lowers the peak temperature (left) while the H ionization fraction is nearly unchanged (right).

### 7.2 Spherical vs. Roche (RLOF) and the mass-loss rate

Removing the `Domain` mode: `Spherical` line switches on the full Roche tidal potential, truncated at the L1 radius. Figure 6 compares a spherical model (Case A) and a Roche model (Case B) of WASP-121b: the tidal potential lowers the effective gravity toward the star, so the Roche wind accelerates far faster and the mass-loss rate rises by an order of magnitude ( $\dot{M}$  from  $\sim 0.05$  to  $\sim 0.34 M_p \text{ Gyr}^{-1}$  here), the “Roche-lobe overflow” enhancement.

### 7.3 Transmission spectra

`EXHALE_transit.py` turns a converged model into transit absorption. The metal resonance lines are the key NUV/optical diagnostics; Figure 7 compares the modeled effective transit radii  $R_{\text{eff}}/R_*$  (spherical Case A) against Huang et al. (2023). Recall the conventions of §5.2: Mg II is a  $4 \text{ \AA}$  bin, Ca II/Na are line-center, and  $R_{\text{eff}}/R_* = \sqrt{(R_p/R_*)^2 + h}$ .

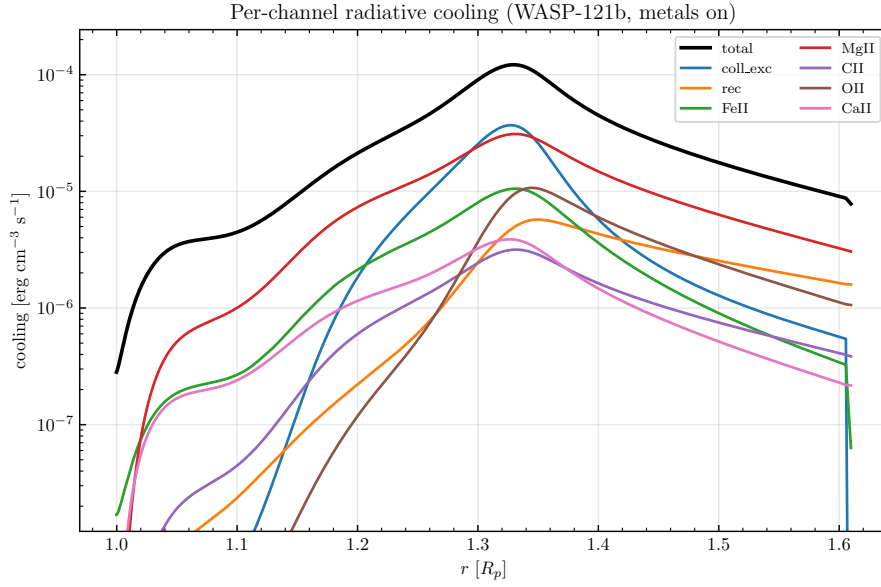


Figure 5: Radiative cooling by channel vs. radius (WASP-121b, metals on; from `Cooling_breakdown.txt`). Below  $\sim 1.4 R_p$  the metal lines (e.g. Fe II, Mg II, C/O) compete with or dominate the H/He collisional channels.

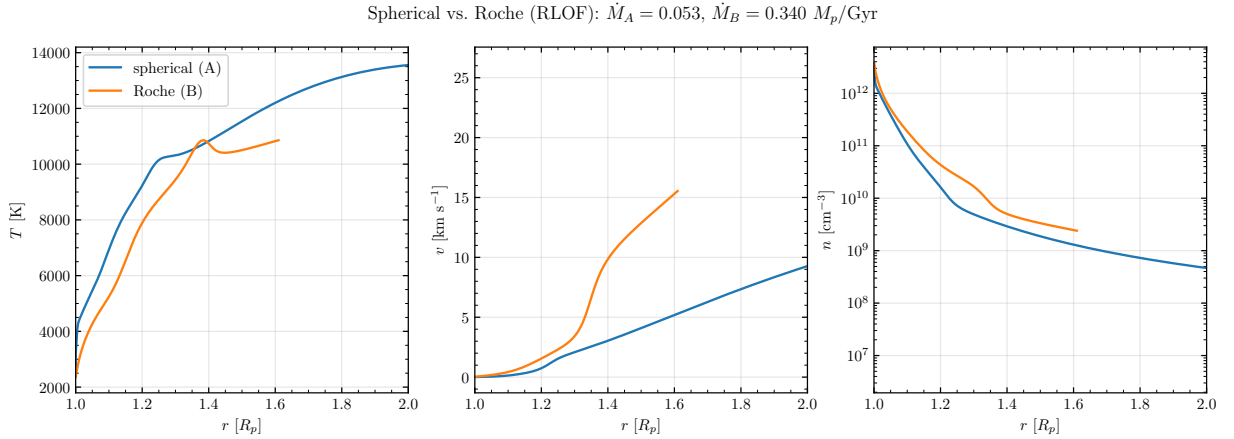


Figure 6: Spherical (Case A) vs. Roche (Case B) WASP-121b: temperature, velocity, and density. The Roche wind reaches L1 ( $\approx 1.6 R_p$ ) and accelerates much faster;  $\dot{M}$  is annotated. The temperature panel is the `_adv` profile; its base is smooth because the post-processor falls back to the equilibrium solution where the flow is not a clean outflow (see the caveat in §4.). The Roche velocity (middle) dips slightly negative near the base: the genuine, physical “breathing” inflow of a near-Roche-filling planet.

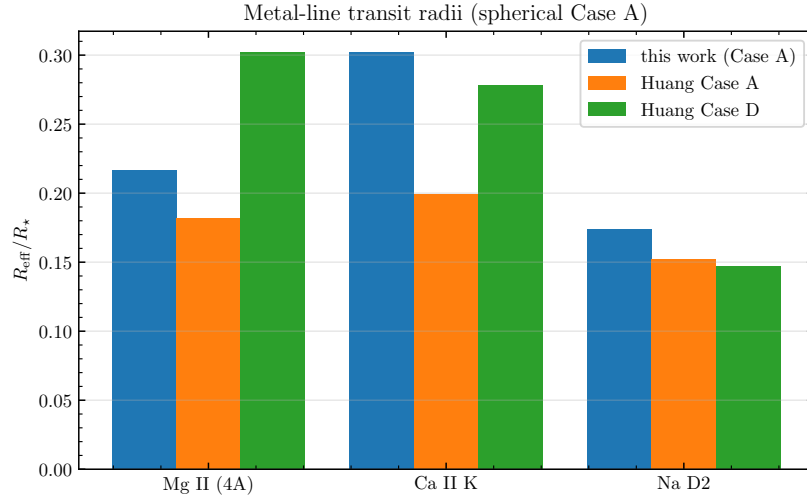


Figure 7: Metal-line transit radii  $R_{\text{eff}}/R_*$  for spherical Case A, compared with Huang et al. (2023) Case A and Case D.

## 8. Atomic and molecular data

By default EXHALE is an *atomic/ionic* code: the modeled species are H, He (including the He  $2^3\text{S}$  metastable triplet), and ten trace metals (C, N, O, Mg, Si, Ca, Na, K, S, Fe) across their first two–three ionization stages. The optional molecular network (Molecular chemistry: True, §3.7) adds  $\text{H}_2$ ,  $\text{H}_2^+$ ,  $\text{H}_3^+$  and  $\text{HeH}^+$  to the coupled equilibrium, with  $\text{H}_2$  photoionization opacity (Yan et al. 1998) and  $\text{H}_3^+$  infrared cooling (Miller et al. 2013). Beyond those four there is **no molecular chemistry and no molecular opacity** (no  $\text{H}^-$ , CO, or  $\text{H}_2\text{O}$ ), so no molecular line lists or partition functions enter the calculation. Table 7 collects the provenance of every rate and cross section, grouped by physical process. The H/He baseline is inherited essentially unchanged from the original ATES (Caldirola et al. 2021, 2022); the metal, excited-hydrogen, and charge-exchange data were added in this work.

Table 7: Atomic data sources, by process. Source files: `cross_sec.f90` (cross sections); `Cool_coeff.f90` (collisional rates, recombination, cooling); `charge_exchange.f90`; `excited_hydrogen.f90`; `EXHALE_transit.py` (transmission line data).

| Process                               | Species                         | Source                                          |
|---------------------------------------|---------------------------------|-------------------------------------------------|
| <b>Elemental abundances</b>           |                                 |                                                 |
| Solar reference                       | 10 metals (metals.inp template) | Asplund et al. (2009)                           |
| Helium                                | He/H ratio                      | input parameter (He/H number ratio)             |
| <b>Photoionization cross sections</b> |                                 |                                                 |
| Bound–free $\sigma(E)$                | H I                             | hydrogenic (Kramers) analytic form (orig. ATES) |

Table 7, continued

| Process                                                              | Species                          | Source                                                                                                                                                                                                                          |
|----------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Opacity model<br>(runtime)                                           | He I                             | Verner, Ferland, Korista & Yakovlev (1996) default; legacy ATES 2-term fit via ATES_photoionization_rate                                                                                                                        |
|                                                                      | He I $2^3S$ (metastable)         | two VFKY96 (Verner) wings + power-law bridge, fitted to TOPbase/Opacity-Project (extends the orig. Norcross 1971 broken power law past 60 eV)                                                                                   |
|                                                                      | C, N, O, Mg, Si, Na, Fe (+ ions) | Verner, Ferland, Korista & Yakovlev (1996), single-shell fits                                                                                                                                                                   |
|                                                                      | K I                              | Verner & Yakovlev (1995)                                                                                                                                                                                                        |
|                                                                      | all (via opacity.inp)            | A=analytic above (default); C=constant above threshold, $\sigma = \text{OPA\_CONST\_X}$ times the analytic value at the edge; P=that constant base times the pressure-broadening factor (Robinson & Catling); T=tabulated files |
| <b>Collisional (electron-impact) ionization</b>                      |                                  |                                                                                                                                                                                                                                 |
| Rate $C(T)$                                                          | H I, He I, He II                 | Voronov (1997, ADNDT 65, 1) default; legacy ATES fits (Abel+1997 / Hui & Gnedin 1997) via Legacy_HHe_rates                                                                                                                      |
|                                                                      | C–Fe, all stages                 | Voronov (1997, ADNDT 65, 1)                                                                                                                                                                                                     |
| <b>Recombination (radiative RR + dielectronic DR)</b>                |                                  |                                                                                                                                                                                                                                 |
| Case B (RR)                                                          | H II, He II, He III              | Badnell (2023) RR – Mao & Kaastra (2016) $\alpha_1$ (+ Badnell DR for He II) default; legacy Hui & Gnedin (1997) fits via Legacy_HHe_rates                                                                                      |
| He rec. radiation $\rightarrow$ H I<br>(def. on,<br>He_rec_coupling) | He II $\rightarrow$ He I         | Draine (2011) on-the-spot y/z: extra H I photoionization $n_{\text{HeII}}n_e[z\alpha_B + y\alpha_1]$ with photoelectron heating, He II recombination shifted to $\alpha_B + y\alpha_1$ ; False = pure case B                    |
| Metastable                                                           | He II $\rightarrow 2^3S, 1^1S$   | power-law fits (orig. ATES)                                                                                                                                                                                                     |
| Penning ionization                                                   | He $2^3S + H$                    | Taylor et al. (2025) temperature-dependent fit (replaces the const. $5 \times 10^{-10}$ ; always on)                                                                                                                            |

Table 7, continued

| Process                                                       | Species                                     | Source                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                               | He $2^3\text{S} + \text{H}_2$               | fit to García Muñoz (2025) Table A.5 (penning_HeI23S_H2, valid 500–10 000 K): $\text{He } 2^3\text{S} + \text{H}_2 \rightarrow \text{He } 1^1\text{S} + \text{H}_2^+ + e^-$ , releasing 4.4 eV to the electrons. The minor $\text{H} + \text{HeH}^+$ associative-ionization branch is not resolved: the full rate goes to the dominant Penning channel. Active only when a molecular run also tracks the triplet |
| RR + DR                                                       | C, N, O, Mg, Si, Na, K, S, Ca II (+ stages) | Badnell (2006, ApJS 167, 334) RR + Badnell adf48 DR (TAMOC clist_K)                                                                                                                                                                                                                                                                                                                                              |
| RR only                                                       | Ca I (Ca II+e)                              | Shull & Van Steenberg (1982); no DR                                                                                                                                                                                                                                                                                                                                                                              |
| RR + DR                                                       | Fe I, Fe II                                 | Huang et al. (2023) analytic fit; the Badnell DR project does not cover these isoelectronic sequences (see the note below the table)                                                                                                                                                                                                                                                                             |
| <b>Charge exchange (charge transfer)</b>                      |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                  |
| metal + $\text{H}/\text{H}^+$ (default)                       | 23 reactions (Group A)                      | Huang et al. (2023, ApJ 951, 123) Table 4                                                                                                                                                                                                                                                                                                                                                                        |
| He + $\text{H}/\text{H}^+$ (default on, He_H_charge_exchange) | 2 reactions (Group B)                       | Koskinen et al. (2013) rates via Huang et al. (2023) Table 4; applied by dedicated routines in every helium-bearing system, <i>not</i> gated by cx_full                                                                                                                                                                                                                                                          |
| full network (cx_full)                                        | + metal+He, metal+metal                     | Huang et al. (2023) Table 4 (Groups C/D)                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Radiative cooling</b>                                      |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                  |
| recomb., coll. ion./exc., free-free                           | H, He                                       | Hui & Gnedin (1997) recomb. cooling; Black (1981) collisional; free-free bremsstrahlung with the van Hoof et al. (2014) thermal Gaunt factor, weighted by ion net charge                                                                                                                                                                                                                                         |
| $\text{H}_3^+$ infrared (molecular runs)                      | $\text{H}_3^+$                              | Miller et al. (2013) LTE emission per molecule (Table 5 $z(T)$ fits) times their Table 6 non-LTE departure factor $s(T, n_{\text{H}_2})$ ; optically thin. The dominant coolant of a molecular base, where every atomic channel above is exponentially suppressed                                                                                                                                                |
| metal line cooling (analytic)                                 | C I/II, N I/II, O I/II                      | closed-form fits to CHIANTI v11.0.2 (default); legacy AIOLOS fits via cno_cool 0 (no N cooling)                                                                                                                                                                                                                                                                                                                  |

Table 7, continued

| Process                                                    | Species                                                                    | Source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metal line cooling<br>(analytic)                           | Mg I/II, Ca II H&K, Na I D, Fe II<br>coronal                               | closed-form fits to CHIANTI v11.0.2 (Burgess–Tully descaled; 1–3% accuracy); Na I D via effective $\bar{g} \approx 0.2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| metal line cooling<br>(tabulated)                          | Fe I; Fe II ( $T, n_e$ )                                                   | NIST $f$ -values + Van Regemorter; CHIANTI multilevel statistical equilibrium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ground-term fine<br>structure (statistical<br>equilibrium) | C I $2p^2\ ^3P$ , C II $2p^2\ ^3P$ , N II $2p^2\ ^3P$ ,<br>O I $2p^4\ ^3P$ | the split ground term is solved in exact statistical equilibrium at the local $(n_e, n_{\text{HI}})$ instead of the low-density coronal limit, which overstates these lines by 2.9–8.3 decades at the base. Electron collision strengths from CHIANTI v11.0.2; H-atom rates from Yan & Babb (2023) for C I/N II, Abrahamsson, Krems & Dalgarno (2007) for O I, Barinovs et al. (2005) for C II. The coronal curve is kept only for the channels leaving the ground term. The complete set: the other coolants have a single ground level, are built from permitted lines (Fe I), or already use a statistical-equilibrium table (Fe II) |
| line trapping                                              | the eight ground-term fine-structure<br>lines                              | line-center escape probability from the column above the cell, Hollenbach & McKee (1979) / de Jong, Boland & Dalgarno (1980) plane-parallel Doppler form renormalized to $\beta(0) = 1$ ; applied as $A_{ul} \rightarrow \beta A_{ul}$ inside the statistical-equilibrium solution, the only placement correct in both the low- and high-density limits. Other lines, including Mg II h&k, are treated as optically thin                                                                                                                                                                                                                |

Table 7, continued

| Process                                                                  | Species                                                                      | Source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| validity floor of the coronal fits                                       | all CHIANTI-derived coefficients                                             | the coronal part is cut off smoothly below $10^3$ K (the lower end of the fitted range), continuous in value and $dT$ -slope and bit-identical above it; only the explicitly solved ground-term fine-structure terms survive below. The roll-off is $\exp(-x^2)$ with $x = (T_{\text{floor}}/T - 1)/w$ and $w = 0.1$ by default, settable as <code>Coronal cutoff width</code> . With the C/N/O ground-term fine-structure floors saturated no coronal cooling survives below the floor for the guard to remove, and the base-cell balance temperature of all four paper planets is identical over $w = 0.02$ – $1.2$ . Not applied to the legacy AIOLOS branch ( <code>cno_cool 0</code> ) |
| <b>Photoelectric heating</b>                                             |                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| secondary ionization<br>(def. on,<br><code>Secondary_ionization</code> ) | H I, He I, H <sub>2</sub>                                                    | Shull & van Steenberg (1985) $f_{\text{ion}}(x)$ fits with the Dalgarno, Yan & Liu (1999) energy dependence, and Dalgarno et al.’s heating fraction $f_{\text{heat}}(x, E_0)$ , H <sub>2</sub> target channel and molecular heat returns; photoelectrons with $E_0 > 30$ eV drive secondary H I / He I / H <sub>2</sub> ionizations, heating scaled to $f_{\text{heat}}$ ; the He 2 <sup>3</sup> S photoelectron feeds the same heating and secondary-source budget                                                                                                                                                                                                                         |
| <b>Excited hydrogen H(<math>n=2</math>) and Ly<math>\alpha</math></b>    |                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2s/2p rate equilibrium                                                   | H( $n=2$ )                                                                   | Christie, Arras & Li (2013) Table 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| recombination cascade<br>to $n=2$                                        | H                                                                            | Draine (2011)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Ly $\alpha$ pumping field<br>(mode 0)                                    | H                                                                            | Huang et al. (2017) Eq. 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Transmission line data (EXHALE_transit.py)</b>                        |                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| $\lambda$ , $f$ -value, $A_{ki}$                                         | He 10830, Ly $\alpha$ , H $\alpha$ , H $\beta$ , Mg II,<br>Ca II H&K, Na I D | NIST ASD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| line profile                                                             | all                                                                          | Voigt (Faddeeva, <code>scipy.special.wofz</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**Note on Fe.** Iron recombination uses the Huang et al. (2023) analytic RR+DR fit while every other metal uses the

Badnell tabulation, and this is not a deferred item: no Badnell fit exists for either stage. Fe I is formed by recombining Fe II (Mn-like, 25 electrons) and Fe II by Fe III (Cr-like, 24), whereas the Badnell dielectronic-recombination project reaches the phosphorus isoelectronic sequence (15 electrons) as of its paper XVI (2022). Ca I has the same problem one sequence lower (K-like, 19 electrons) and likewise keeps a non-Badnell rate. **Note on completeness.** The metal grid carries the stages needed for the XUV-driven ionization balance and the observed NUV/optical lines; higher stages and additional elements (e.g. the full Fe forest, K I, Mg I) beyond those tabulated above are not modeled.